

Updates to P3-v5 in GEM-5.3.0-a15 for IC5

Control configuration is MoSA6 (from RPN-A)

Melissa Cholette et al. (HIWR)

Main conclusions

- P3_v5.5.0-rc9 is ready to be inserted in GEM-alpha (+ RPN-A changes)
- Tested in GEM5.3.0-a15 with MoSA6-HRDPS configuration (P3_v5.3.6):
 - 2MOM-noLF (default configuration) for H22 and E22
 - 2MOM-LF for H22 and E22
 - 3MOM-LF for H22, E22, P22 and A21 (to come)

Results:

- Improved freezing rain FBI with v5.5.0 and LF for H22 (except for small PR thresholds <0.01 mm), with mostly neutral emet scores (small degradation (within instrument noise) of the TT cold bias because of HM changes), but a plus of 2% in running time by turning off Bourgouin.
- Improved scores for P22 and E22 with both 3MOM and LF.
- Improved representations of hydrometeors properties with 3MOM (hail) and LF (wet snow) in bonus. Potential to improve feedback with the surface scheme (e.g., snow density is better with LF).
- Increased total run time of $<\sim 3\%$ (small) with LF and $\sim 7\%$ (larger) with 3MOM.

Definitions

Seasons:

- H22 = 2022010100:2022033012:36
- E22 = 2022070112:2022092912:36
- P22 = 2022040100:2022063000:36
- A21 = 2021100600:2021123012:36

P3 configurations:

- 3MOM = 3-moment ice (Milbrandt et al. 2021)
- 2MOM = 2-moment ice (default; Cholette et al. 2023)
- LF = liquid fraction (mixed-phase particles; Cholette et al. 2019)

List of experiments GEM_5.3.0-a15

P3 version	Description	H22	E22	P22	A21
5.3.6	Control of MoSA6	X	X	X	X
5.5.0-rc9 2mom-noLF	2-moment ice, no liquid fraction	X X bourge	X	X bourge	X
5.5.0-rc9 2mom-LF	2-moment ice, liquid fraction	X X bourge	X	X	X bourge
5.5.0-rc9 3mom-LF	3-moment ice, liquid fraction	X	X	X	X bourge

What has changed since P3_v5.3.6, the control in MoSA6-HRDPS?

- Changes to Hallett-Mossop, a secondary ice production process, to improve the precipitation phases scores, especially freezing rain FBI (Cholette et al. 2024, & already tested in P3v.5.3.14^{*}).
- A bunch of optimisation, especially for 3-moment ice.

Timings

Timings (H22)

Detailed Statistics:

P3-v5.3.6 2MOM-noLF: Mean = 5937.72s, Std Dev = 117.18s, Bruit = 2.0%

P3-v5.5.0-rc9 2MOM-noLF: Mean = 5797.28s, Std Dev = 84.84s, Bruit = 1.5%

P3-v5.5.0-rc9 2MOM-noLF bourge: Mean = 5752.8s, Std Dev = 86.57s

P3-v5.5.0-rc9 2MOM-LF nil: Mean = 6063.98s, Std Dev = 111.66s, Bruit = 1.8%

P3-v5.5.0-rc9 2MOM-LF bourge: Mean = 6024.56s, Std Dev = 126.19s

P3-v5.5.0-rc9 3MOM-LF: Mean = 6497.55s, Std Dev = 226.30s, Bruit = 3.5%

P3 version	Compared to CTL	Compared to 2MOM-noLF	Noise
5.3.6 2MOM-noLF	-	+2.4%	±1.75%
5.5.0-rc8 2MOM-noLF 'nil' 2MOM-noLF 'bourge'	-2.4% -3.12%	- -0.76%	±1.75%
5.5.0-rc9 2MOM-LF 'nil' 2MOM-LF 'bourge'	+2.5% +1.5%	+4.5% +3.9%	±2.65% ±2.7%
5.5.0-rc9 3MOM-LF 'nil'	+9.4%	+12%	±2.63%

Timings (E22)

Detailed Statistics:

P3-v5.3.6 2MOM-noLF: Mean = 6071.70s, Std Dev = 75.59s

P3-v5.5.0-rc9 2MOM-noLF: Mean = 5890.41s, Std Dev = 68.32s

P3-v5.5.0-rc9 2MOM-LF: Mean = 6124.29s, Std Dev = 71.93s

P3-v5.5.0-rc9 3MOM-LF: Mean = 6543.88s, Std Dev = 121.14s

P3 version	Compared to CTL	Compared to 2MOM-noLF
5.3.6 2MOM-noLF	-	+2.99%
5.5.0-rc8 2MOM-noLF	-2.99%	-
5.5.0-rc9 2MOM-LF	+0.87%	+3.96%
5.5.0-rc9 3MOM-LF	+7.78%	+11.1%

Timings (P22)

Detailed Statistics:

P3-v5.3.6 2MOM-noLF: Mean = 6047.14s, Std Dev = 83.78s
P3-v5.5.0-rc9 2MOM-noLF: Mean = 5902.32s, Std Dev = 102.13s
P3-v5.5.0-rc9 2MOM-LF: Mean = 6173.17s, Std Dev = 103.57s
P3-v5.5.0-rc9 3MOM-LF: Mean = 6713.96s, Std Dev = 217.40s

P3 version	Compared to CTL	Compared to 2MOM-noLF
5.3.6 2MOM-noLF	-	+2.4%
5.5.0-rc9 2MOM-noLF	-2.4% (note: not bourge)	-
5.5.0-rc9 2MOM-LF	+2.1%	+4.6%
5.5.0-rc9 3MOM-LF	+11%	+13.7%

Timings (A21)

Detailed Statistics:

P3-v5.3.6 2MOM-noLF: Mean = 5965.3s, Std Dev = 87.27s
P3-v5.5.0-rc9 2MOM-noLF: Mean = 5806.02s, Std Dev = 82.51s
P3-v5.5.0-rc9 2MOM-LF: Mean = 6070.23s, Std Dev = 82.98s
P3-v5.5.0-rc9 3MOM-LF: Mean = 6542.87s, Std Dev = 174.45s

P3 version	Compared to CTL	Compared to 2MOM-noLF
5.3.6 2MOM-noLF	-	+2.67%
5.5.0-rc9 2MOM-noLF	-2.67%	-
5.5.0-rc9 2MOM-LF	+0.2%	+1.76%
5.5.0-rc9 3MOM-LF	+9.68%	+12.7%

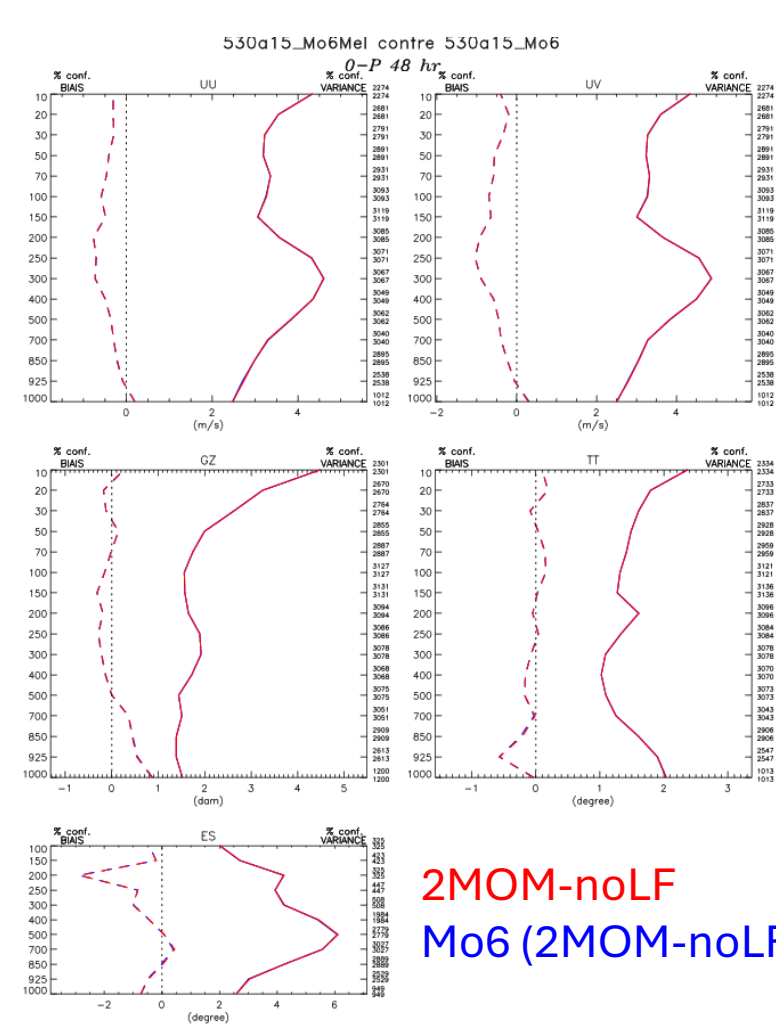
Arcad

'nil'

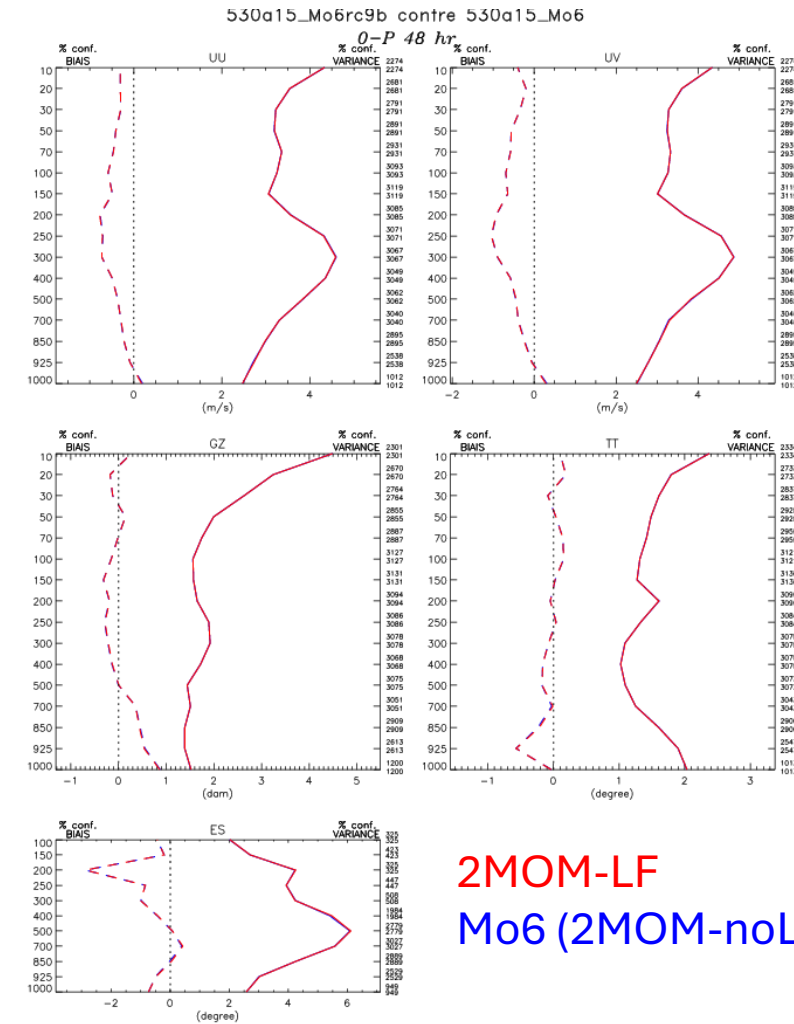
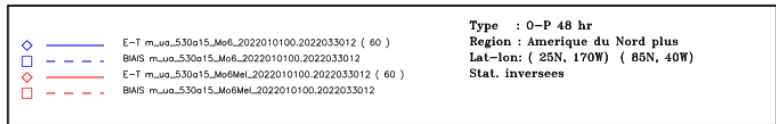
Arcad (H22)

P3-v5.5.0-rc9

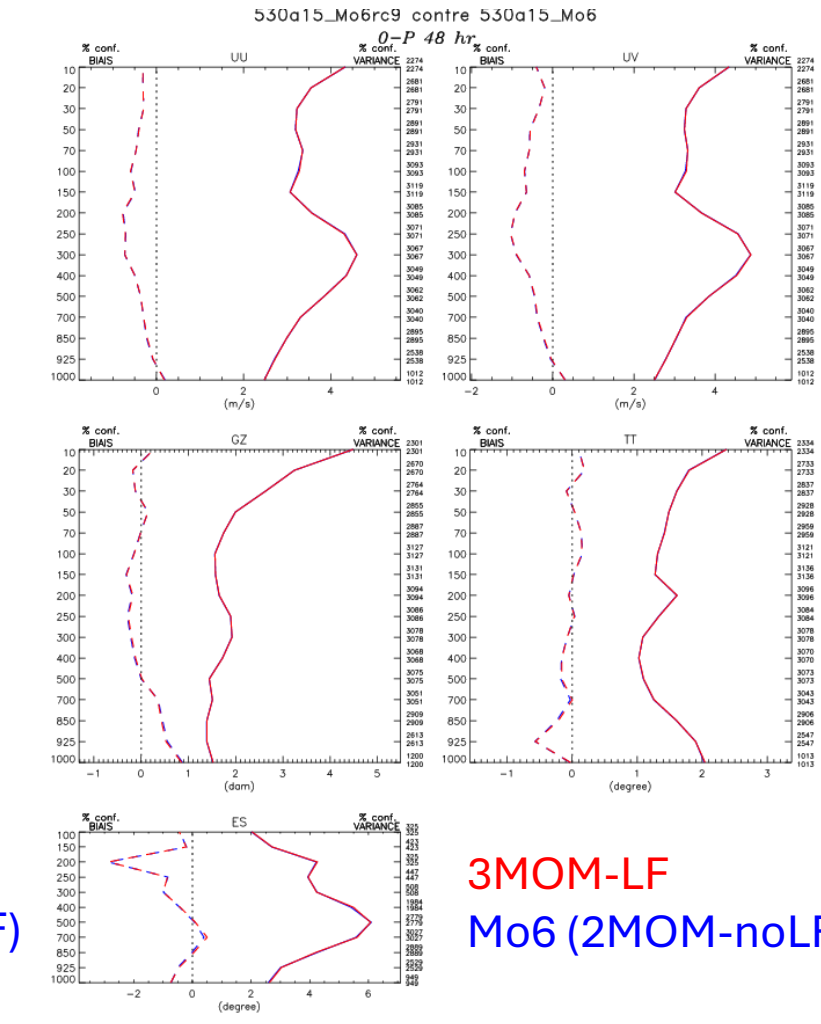
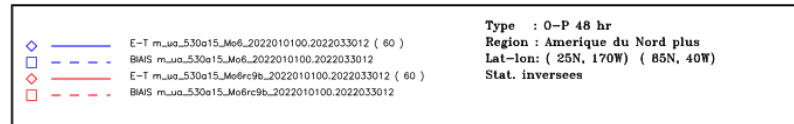
P3_v5.3.6



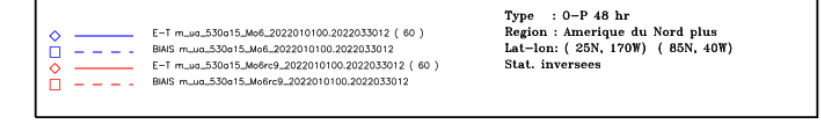
2MOM-noLF
Mo6 (2MOM-noLF)



2MOM-LF
Mo6 (2MOM-noLF)



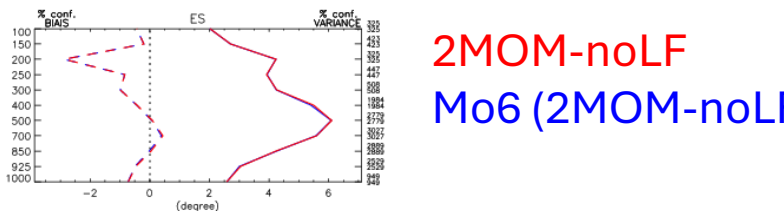
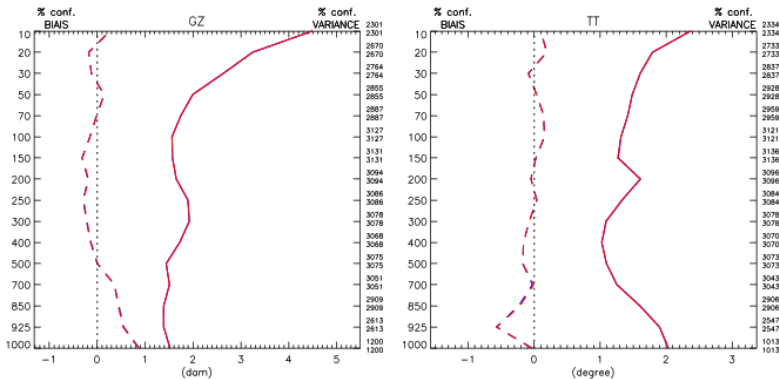
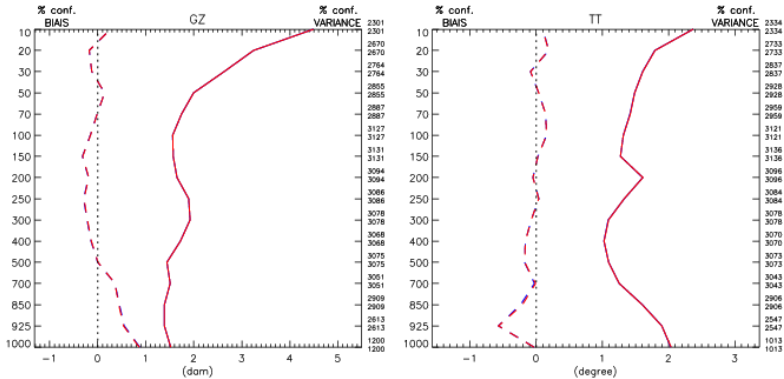
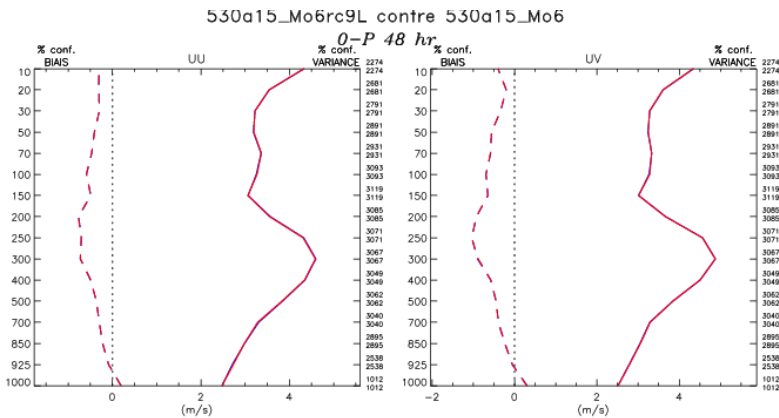
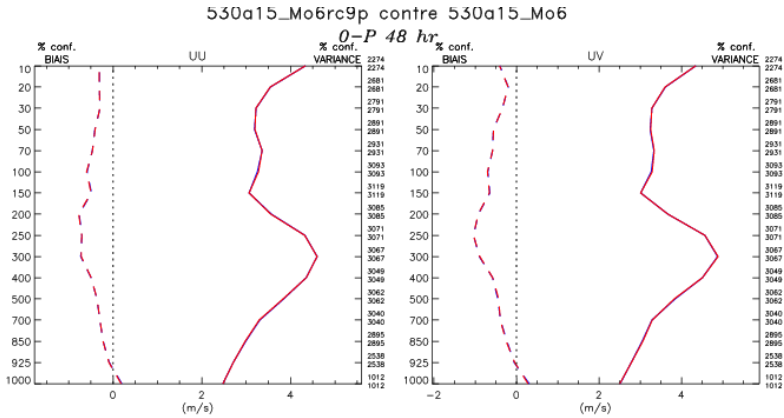
3MOM-LF
Mo6 (2MOM-noLF)



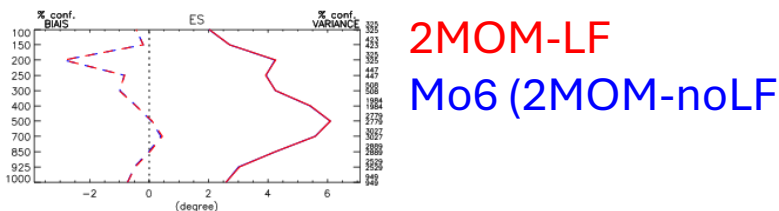
‘bourge’

Arcad (H22)

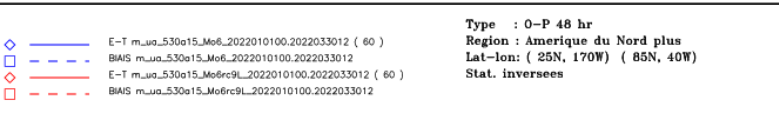
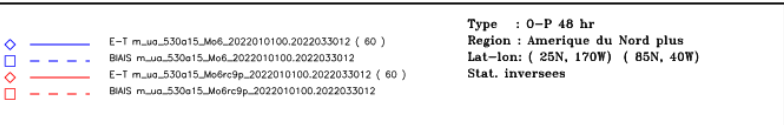
P3-v5.5.0-rc9
P3_v5.3.6



2MOM-noLF
Mo6 (2MOM-noLF)



2MOM-LF
Mo6 (2MOM-noLF)



'nil'

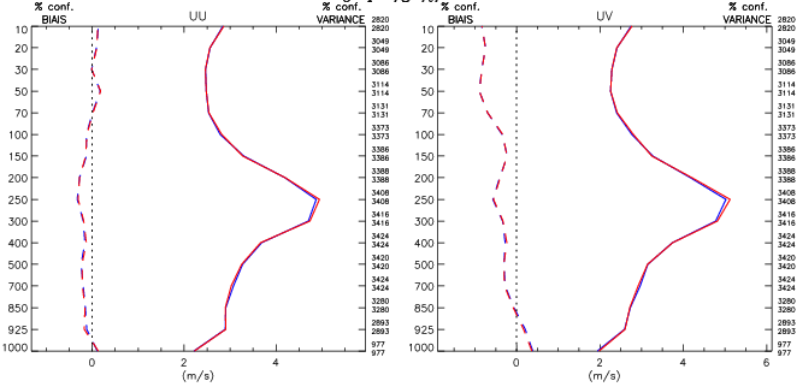
Arcad (E22)

P3-v5.5.0-rc9

P3_v5.3.6

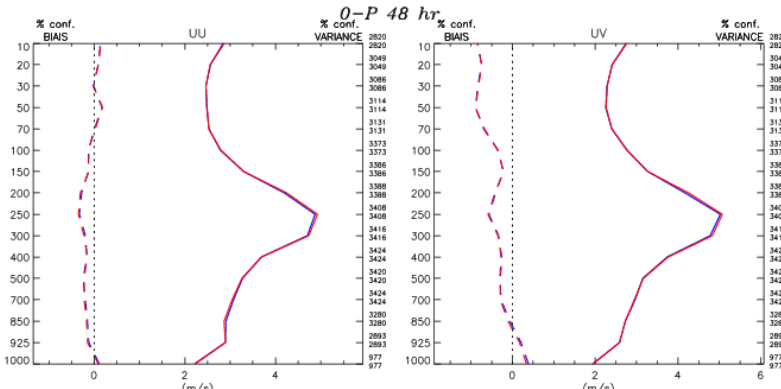
530a15_Mo6rc9p contre 530a15_Mo6

0-P 48 hr



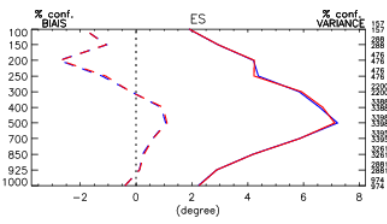
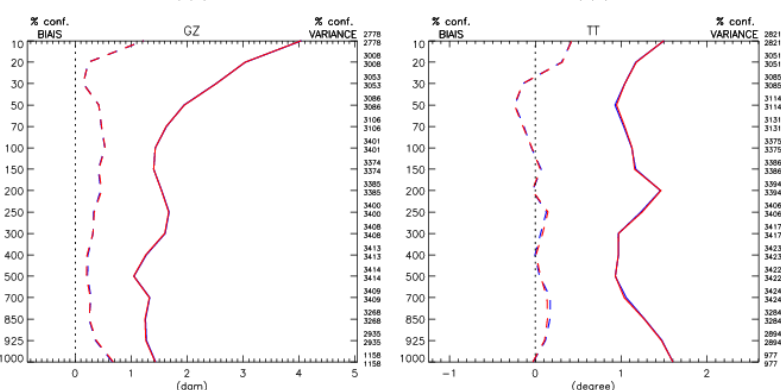
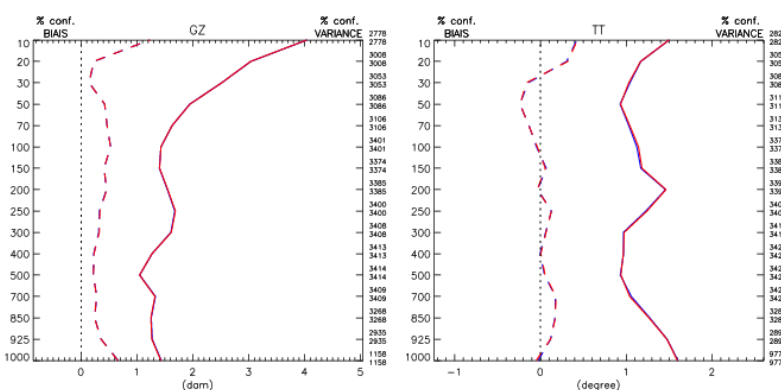
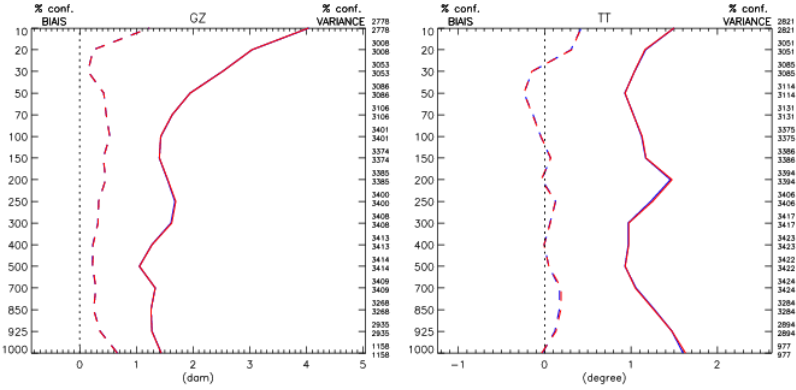
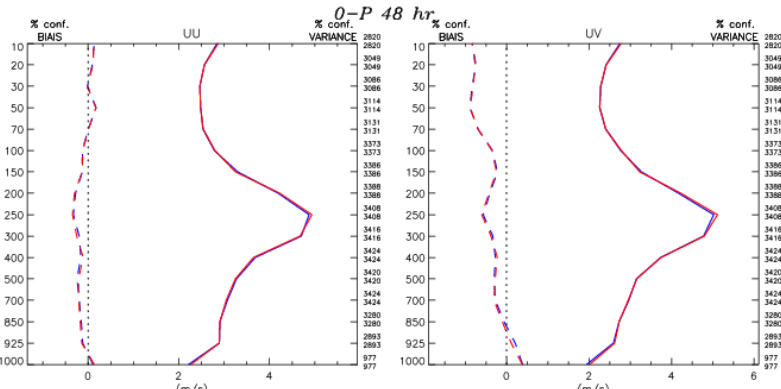
530a15_Mo6rc9L contre 530a15_Mo6

0-P 48 hr

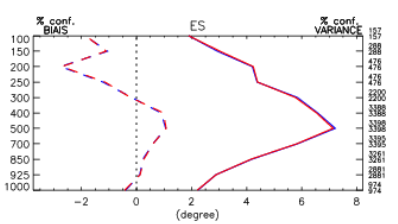


530a15_Mo6rc9b contre 530a15_Mo6

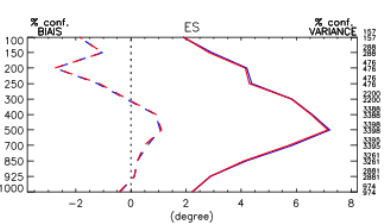
0-P 48 hr



2MOM-noLF
Mo6 (2MOM-noLF)



2MOM-LF
Mo6 (2MOM-noLF)



3MOM-LF
Mo6 (2MOM-noLF)

Type : 0-P 48 hr

Region : Amerique du Nord plus

Lat-lon: (25N, 170W) (85N, 40W)

Stat. inversees

Type : 0-P 48 hr

Region : Amerique du Nord plus

Lat-lon: (25N, 170W) (85N, 40W)

Stat. inversees

Type : 0-P 48 hr

Region : Amerique du Nord plus

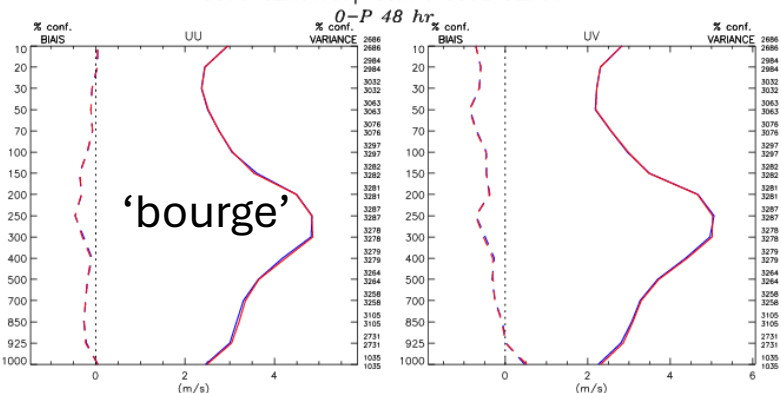
Lat-lon: (25N, 170W) (85N, 40W)

Stat. inversees

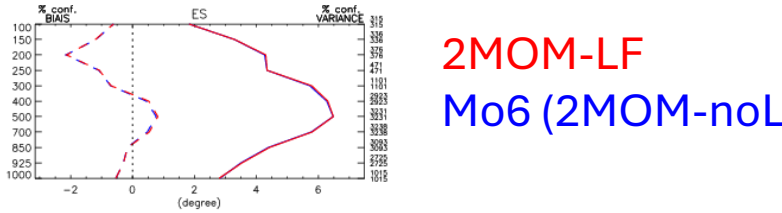
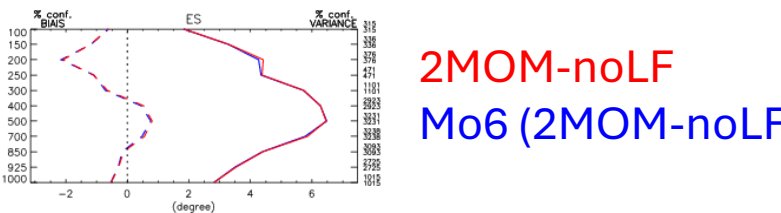
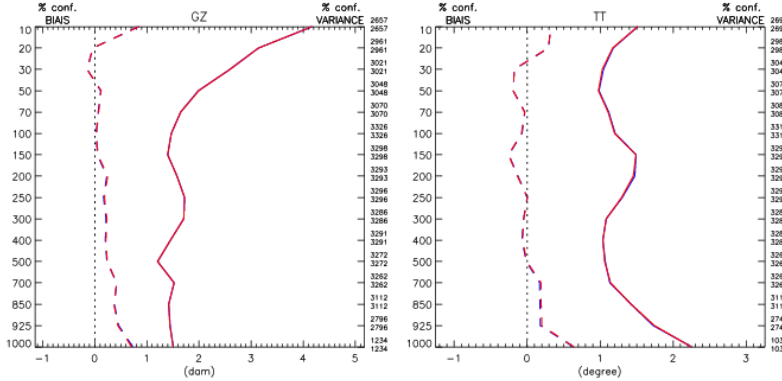
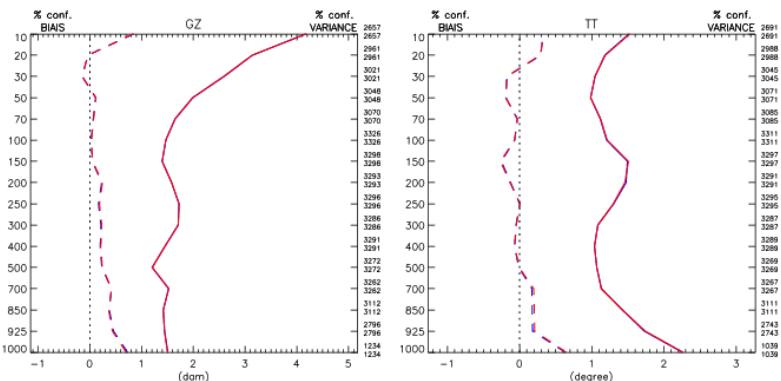
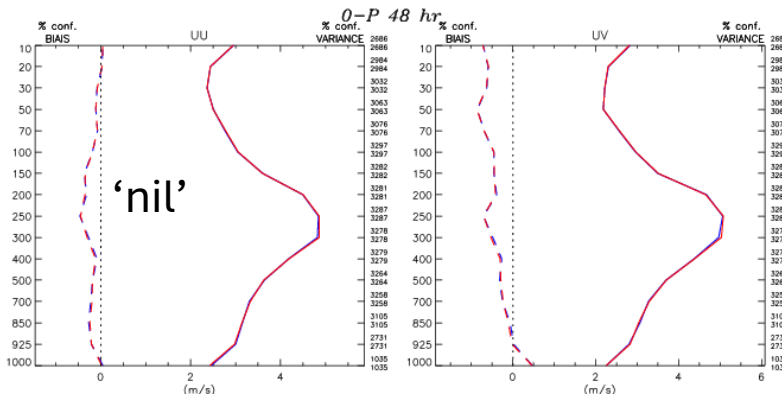
Arcad (P22)

P3-v5.5.0-rc9
P3_v5.3.6

530a15_Mo6rc9p contre 530a15_Mo6



530a15_Mo6rc9L contre 530a15_Mo6

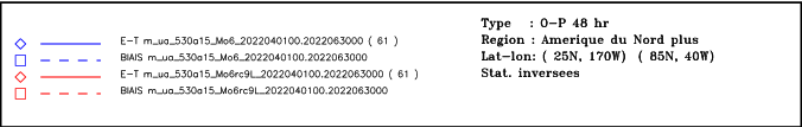
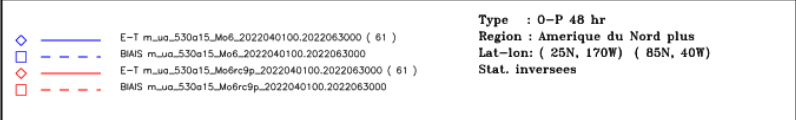


2MOM-noLF
Mo6 (2MOM-noLF)

2MOM-LF
Mo6 (2MOM-noLF)

3MOM-LF
Mo6 (2MOM-noLF)

The graph
did not get
produced



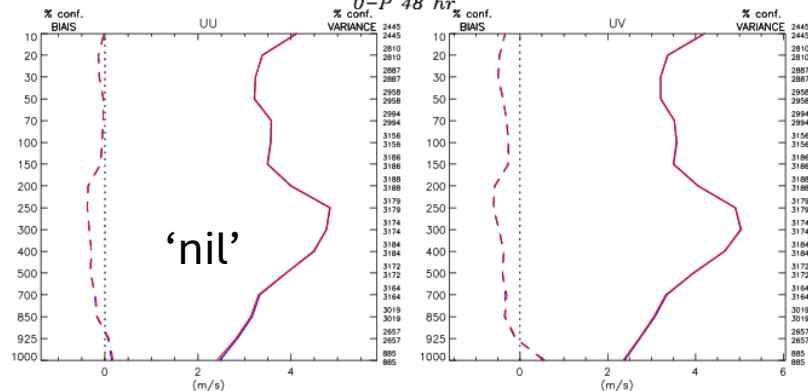
Arcad (A21)

P3-v5.5.0-rc9

P3_v5.3.6

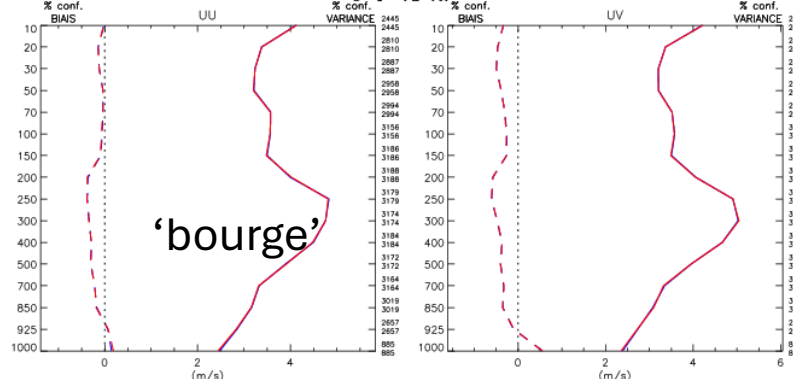
530a15_Mo6rc9p contre 530a15_Mo6

0-P 48 hr



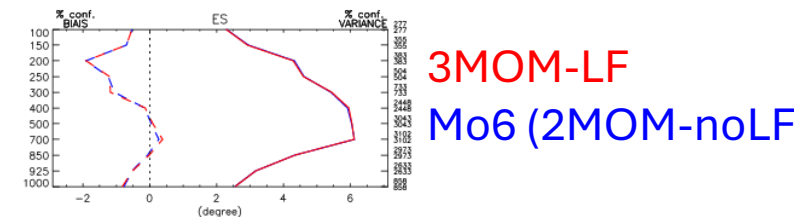
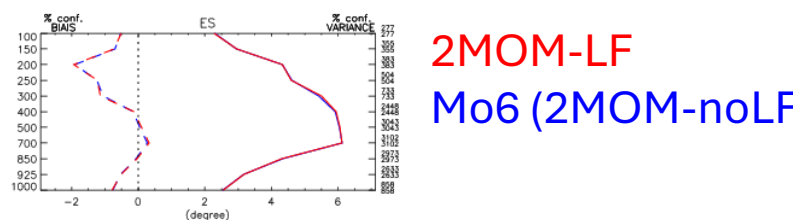
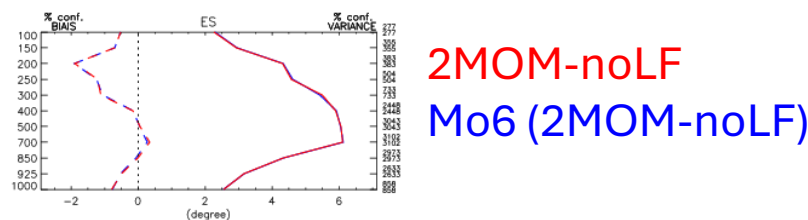
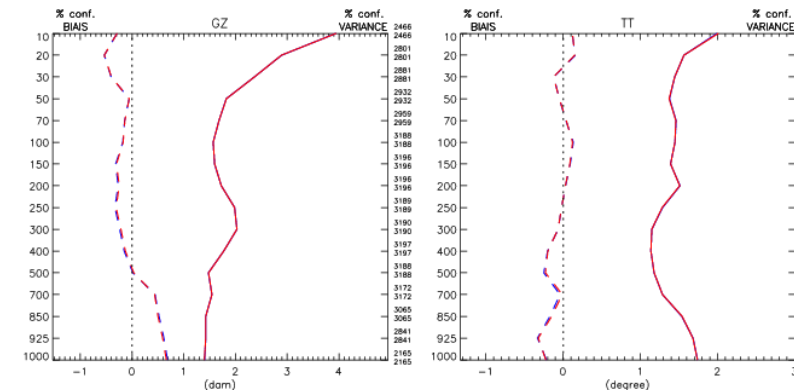
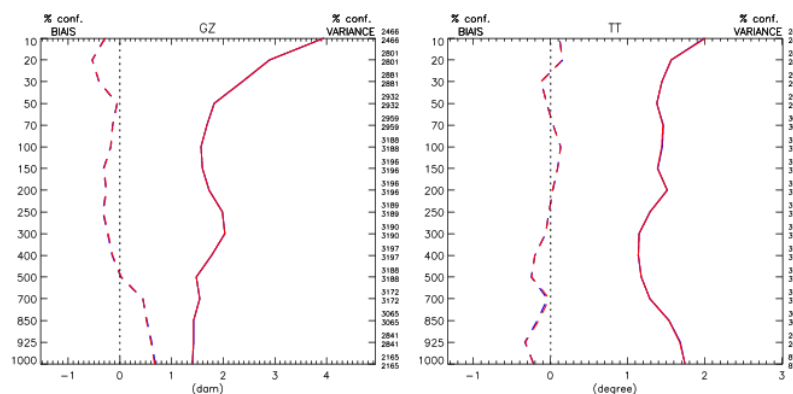
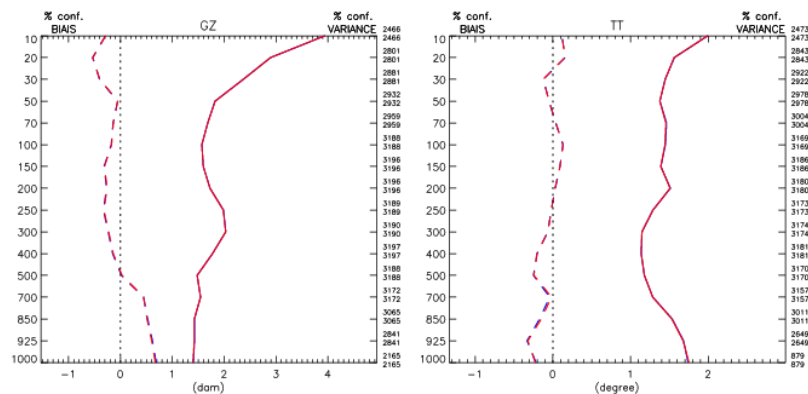
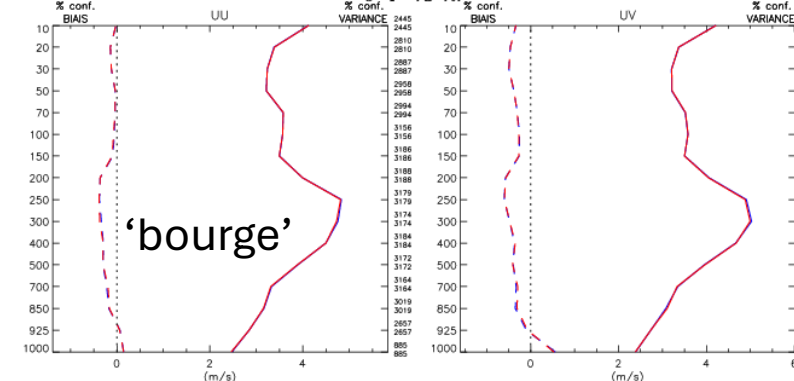
530a15_Mo6rc9L contre 530a15_Mo6

0-P 48 hr



530a15_Mo6rc9b contre 530a15_Mo6

0-P 48 hr



Type : 0-P 48 hr
Region : Amerique du Nord plus
Lat-lon: (25N, 170W) (85N, 40W)
Stat. inversees

◇ E-T m_uu_530a15_Mo6_2021100600.2021123012 (58)
□ BIAS m_uu_530a15_Mo6_2021100600.2021123012
◇ E-T m_uu_530a15_Mo6rc9p_2021100600.2021123012 (58)
□ BIAS m_uu_530a15_Mo6rc9p_2021100600.2021123012

Type : 0-P 48 hr
Region : Amerique du Nord plus
Lat-lon: (25N, 170W) (85N, 40W)
Stat. inversees

◇ E-T m_uu_530a15_Mo6_2021100600.2021123012 (58)
□ BIAS m_uu_530a15_Mo6_2021100600.2021123012
◇ E-T m_uu_530a15_Mo6rc9L_2021100600.2021123012 (58)
□ BIAS m_uu_530a15_Mo6rc9L_2021100600.2021123012

Type : 0-P 48 hr
Region : Amerique du Nord plus
Lat-lon: (25N, 170W) (85N, 40W)
Stat. inversees

◇ E-T m_uu_530a15_Mo6_2021100600.2021123012 (58)
□ BIAS m_uu_530a15_Mo6_2021100600.2021123012
◇ E-T m_uu_530a15_Mo6rc9b_2021100600.2021123012 (58)
□ BIAS m_uu_530a15_Mo6rc9b_2021100600.2021123012

EMET

Autumn 2021 (A21)

1.2MOM-noLF

2.2MOM-LF

3.3MOM-LF

‘nil’

P3_v5.5.0-rc9

P3 v5.3.6 2MOM-noLF

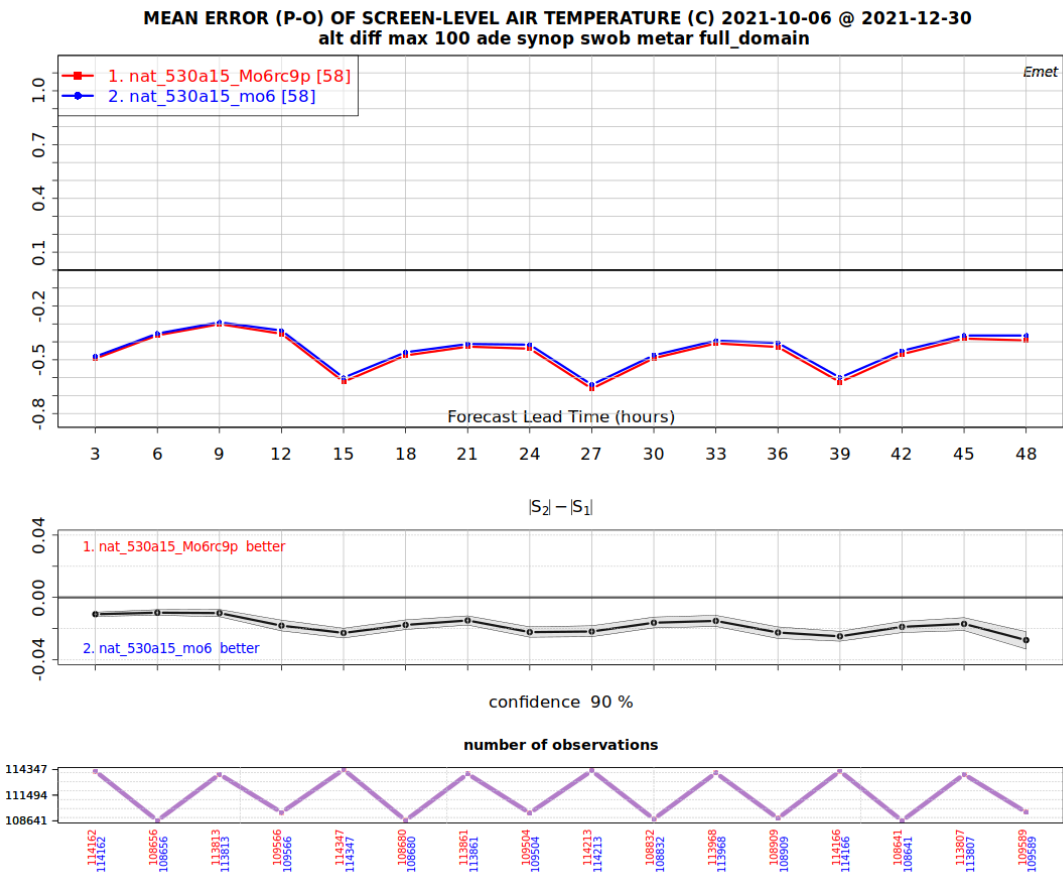
EMET (A21), TT & TD 2MOM-noLF

Pretty neutral

bias

bias		20211006 / 20211230		
nat_530a15_Mo6rc9p / nat_530a15_mo6		00z	12z	All
Arctic All CLIM	TD	-0.032	-0.026	-0.031
	TT	-0.039	-0.036	-0.038
Boreal CLIM	TD	-0.0086	-0.015	-0.012
	TT	-0.02	-0.026	-0.023
Canada	TD	-0.015	-0.013	-0.011
	TT	-0.023	-0.025	-0.026
East CLIM	TD	-0.001	-0.00063	-0.00034
	TT	-0.01	-0.0097	-0.0092
Great Lakes CLIM	TD	-0.00019	-0.0012	0.0021
	TT	-0.017	-0.016	-0.017
Mt West CLIM	TD	-0.029	-0.027	-0.033
	TT	-0.033	-0.024	-0.029
North Plains CLIM	TD	0.0033	0.0074	0.016
	TT	-0.02	-0.012	-0.012
Pacific North West CLIM	TD	0.0	0.0	0.0
	TT	-0.016	-0.017	-0.015
Prairie CLIM	TD	0.0	0.0	-0.00039
	TT	-0.0078	-0.0058	-0.0082
United States of America East	TD	-0.0024	-0.001	-2.4e-05
	TT	-0.01	-0.0088	-0.0096
United States of America West	TD	-0.012	-0.01	-0.0092
	TT	-0.021	-0.013	-0.017
full_domain	TD	-0.0074	-0.0083	-0.0073
	TT	-0.019	-0.017	-0.018

TT-bias (full domain)



'nil'

EMET (A21), P0 & UV 2MOM-noLF

P3_v5.5.0-rc9

P3 v5.3.6 2MOM-noLF

bias		20211006 / 20211230		
bias nat_530a15_Mo6rc9p / nat_530a15_mo6		00z	12z	All
Arctic All CLIM	P0	0.018	0.017	0.017
Boreal CLIM	P0	0.016	0.018	0.017
Canada	P0	0.014	0.016	0.015
East CLIM	P0	0.0076	0.016	0.014
Great Lakes CLIM	P0	0.0041	0.01	0.01
Mt West CLIM	P0	0.0061	-0.0086	0.0012
North Plains CLIM	P0	0.012	0.016	0.014
Pacific North West CLIM	P0	0.0045	0.002	0.0045
Prairie CLIM	P0	0.00021	0.0066	0.0047
United States of America East	P0	-0.00034	0.0018	0.0033
United States of America West	P0	-0.0076	-0.011	-0.011
full_domain	P0	0.013	0.015	0.014

rmse		20211006 / 20211230		
rmse nat_530a15_Mo6rc9p / nat_530a15_mo6		00z	12z	All
Arctic All CLIM	P0	0.0039	0.0065	0.0064
Boreal CLIM	P0	0.0026	0.0014	0.0032
Canada	P0	0.0034	0.0022	0.0031
East CLIM	P0	0.0014	0.0017	0.0011
Great Lakes CLIM	P0	0.0025	0.0021	0.003
Mt West CLIM	P0	0.0014	0.0	0.00028
North Plains CLIM	P0	0.0019	0.00077	0.0016
Pacific North West CLIM	P0	0.00072	0.0	-0.00062
Prairie CLIM	P0	0.00077	0.0023	0.0019
United States of America East	P0	0.0028	7.8e-05	0.0018
United States of America West	P0	0.0	0.00012	-0.0013
full_domain	P0	0.0034	0.0019	0.003

bias		20211006 / 20211230		
bias nat_530a15_Mo6rc9p / nat_530a15_mo6		00z	12z	All
Arctic All CLIM	UV	0.0049	0.0032	0.005
Boreal CLIM	UV	0.0023	0.0	0.0023
Canada	UV	0.0041	0.0019	0.0037
East CLIM	UV	0.0	0.0	0.0
Great Lakes CLIM	UV	0.0	0.0015	0.0
Mt West CLIM	UV	-0.0024	-0.0056	-0.0026
North Plains CLIM	UV	-0.0033	-0.00035	-0.0031
Pacific North West CLIM	UV	0.0	-2.3e-05	-0.003
Prairie CLIM	UV	0.0	0.00059	0.0
United States of America East	UV	0.00026	0.00012	0.00057
United States of America West	UV	-0.0027	0.0	-0.0018
full_domain	UV	0.0018	0.00038	0.002

rmse		20211006 / 20211230		
rmse nat_530a15_Mo6rc9p / nat_530a15_mo6		00z	12z	All
Arctic All CLIM	UV	0.0	0.0012	-0.00038
Boreal CLIM	UV	0.0015	0.00054	0.0019
Canada	UV	-0.00085	0.0011	0.00033
East CLIM	UV	0.00075	-0.00025	0.0
Great Lakes CLIM	UV	0.00013	-0.00072	-8.1e-05
Mt West CLIM	UV	-0.0018	0.0	-0.0026
North Plains CLIM	UV	0.0	0.0	0.0
Pacific North West CLIM	UV	-0.00088	0.0	0.0
Prairie CLIM	UV	-0.00038	0.0	0.00067
United States of America East	UV	0.0	-6.1e-05	0.0
United States of America West	UV	-0.0018	-0.0032	-0.0029
full_domain	UV	0.0	0.0	0.0

'nil'

EMET (A21), PR6 & PR24 2MOM-noLF

P3_v5.5.0-rc9

P3 v5.3.6 2MOM-noLF

dpmean

dpmean	nat_530a15_Mo6rc9p / nat_530a15_m06	00z	12z	All
Arctic All CLIM	PR24	-0.021	-0.0068	-0.013
	PR6	-0.0029	-0.0033	-0.0023
Boreal CLIM	PR24	-0.004	-0.0029	-0.0031
	PR6	-0.0035	-0.0042	-0.0045
Canada	PR24	-0.0029	-0.0073	-0.0069
	PR6	-0.0031	-0.0054	-0.0044
East CLIM	PR24	-0.0042	-0.0098	-0.0035
	PR6	0.00087	0.003	0.00098
Great Lakes CLIM	PR24	0.0	-0.0024	-0.00058
	PR6	0.0	-0.002	0.00093
Mt West CLIM	PR24	-0.011	-0.026	-0.022
	PR6	-0.00049	-0.0077	-0.00024
North Plains CLIM	PR24	0.0	0.0015	0.0029
	PR6	-0.00029	0.0	0.00015
Pacific North West CLIM	PR24	-0.0013	-0.08	-0.064
	PR6	-0.017	-0.024	-0.021
Prairie CLIM	PR24	-0.00031	0.0	0.0
	PR6	0.00077	0.0	0.00032
United States of America East	PR24	-0.0026	-0.005	-0.00033
	PR6	0.0	0.0017	0.00022
United States of America West	PR24	0.014	-0.027	0.0021
	PR6	0.0013	-0.0045	-0.0058
full_domain	PR24	-0.012	-0.015	-0.017

ets

ets	nat_530a15_Mo6rc9p / nat_530a15_m06	00z	12z	All
Arctic All CLIM	PR24	-0.0055	-0.019	-0.019
	PR6	0.0	0.0031	0.0
Boreal CLIM	PR24	0.0	0.0016	0.0
	PR6	0.0053	0.0046	0.005
Canada	PR24	0.0021	-0.0028	-0.003
	PR6	0.003	0.0	0.0039
East CLIM	PR24	0.002	0.001	0.0015
	PR6	0.0061	-0.0034	0.0013
Great Lakes CLIM	PR24	0.014	0.0049	0.019
	PR6	0.0016	0.0	0.0016
Mt West CLIM	PR24	0.0	0.002	0.004
	PR6	0.016	0.0017	0.0039
North Plains CLIM	PR24	-0.0037	-0.0065	-0.0041
	PR6	-0.0077	0.0066	-0.0012
Pacific North West CLIM	PR24	0.00029	-0.0039	-0.0028
	PR6	0.0008	-0.0017	0.0
Prairie CLIM	PR24	-0.0025	-2.3e-05	0.00054
	PR6	-0.0065	0.0	-0.002
United States of America East	PR24	0.0018	0.0029	0.0033
	PR6	0.0	-0.0029	0.0
United States of America West	PR24	0.0012	-0.0062	0.00019
	PR6	0.0	-0.0023	0.0
full_domain	PR24	0.00062	0.0018	0.00088

fbi

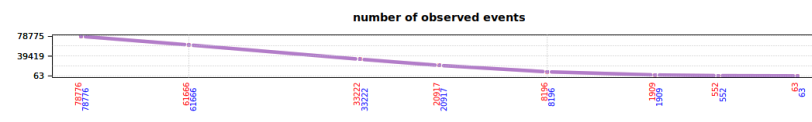
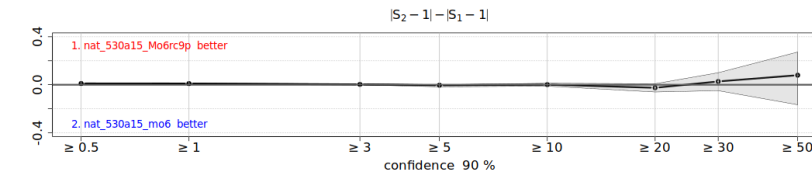
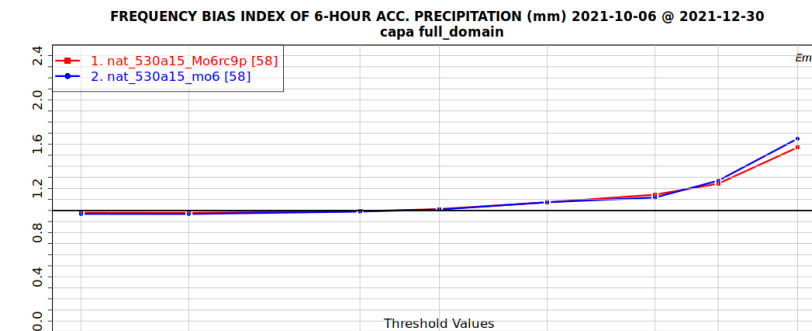
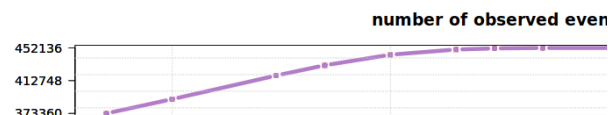
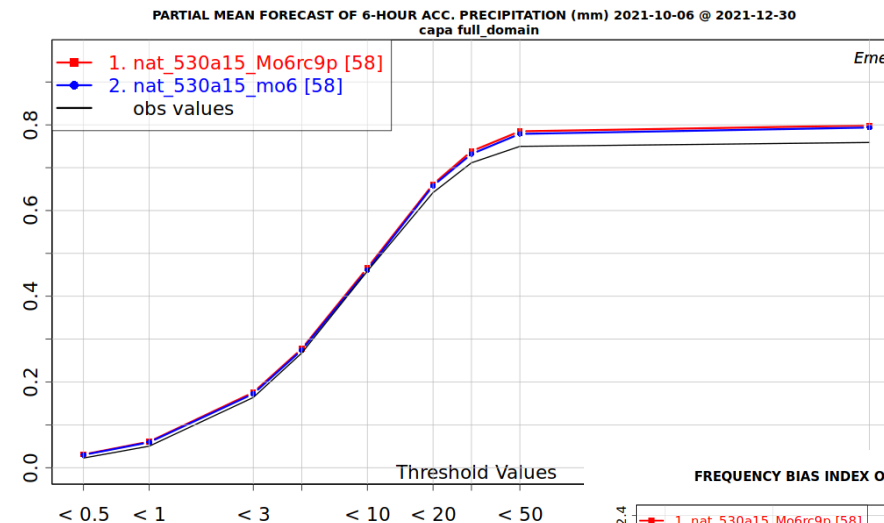
fbi	nat_530a15_Mo6rc9p / nat_530a15_m06	00z	12z	All
Arctic All CLIM	PR24	-0.057	-0.044	-0.059
	PR6	0.0	0.016	0.0
Boreal CLIM	PR24	-0.011	-0.034	-0.039
	PR6	-0.0076	-0.1	-0.029
Canada	PR24	-0.00044	-0.0061	-0.006
	PR6	0.0015	-0.018	-0.0067
East CLIM	PR24	0.0052	-0.0082	-0.006
	PR6	-3.4e-05	0.0037	0.0024
Great Lakes CLIM	PR24	0.0061	0.055	0.0
	PR6	0.0	0.025	0.0
Mt West CLIM	PR24	-0.008	-0.022	-0.0091
	PR6	0.0	-0.008	-0.0018
North Plains CLIM	PR24	0.0	0.039	-0.0022
	PR6	-0.0055	0.0	0.0078
Pacific North West CLIM	PR24	0.0015	-0.019	-0.0073
	PR6	-0.005	0.011	-0.015
Prairie CLIM	PR24	-0.008	-0.023	0.0
	PR6	0.0	0.0015	0.0038
United States of America East	PR24	-0.0061	-0.0059	-0.0032
	PR6	0.0	0.0028	0.0
United States of America West	PR24	0.00089	-0.013	0.016
	PR6	0.0024	-0.0043	0.0039
full_domain	PR24	-0.00075	-0.0058	-0.0052
	PR6	0.0016	-0.00057	0.0026

pod

pod	nat_530a15_Mo6rc9p / nat_530a15_m06	00z	12z	All
Arctic All CLIM	PR24	0.0	-0.00098	-0.031
	PR6	0.0	0.0	0.0
Boreal CLIM	PR24	-0.0026	0.009	0.0017
	PR6	0.01	0.04	0.0075
Canada	PR24	0.0	-0.0022	-0.00031
	PR6	0.0029	0.0063	0.0058
East CLIM	PR24	0.00073	0.0091	0.0048
	PR6	0.0086	0.0019	0.0027
Great Lakes CLIM	PR24	0.0091	-0.0013	0.017
	PR6	0.0035	0.0	0.0
Mt West CLIM	PR24	-0.0021	0.0031	0.0081
	PR6	0.015	0.0017	0.0055
North Plains CLIM	PR24	-0.0044	0.0	0.0013
	PR6	-0.017	0.0015	-0.01
Pacific North West CLIM	PR24	0.0049	0.0084	0.0078
	PR6	0.0024	0.0062	0.0054
Prairie CLIM	PR24	-0.0023	-0.0093	0.00059
	PR6	-0.012	0.0012	-0.0022
United States of America East	PR24	0.0035	0.0066	0.0066
	PR6	0.0	0.0011	0.0
United States of America West	PR24	0.0066	0.0095	0.0091
	PR6	0.0014	0.0017	0.0025
full_domain	PR24	0.0032	0.0084	0.0061
	PR6	0.0021	0.0025	0.0021

far

far	nat_530a15_Mo6rc9p / nat_530a15_m06	00z	12z	All
Arctic All CLIM	PR24	-0.0043	-0.032	-0.039
	PR6	0.0	0.0059	0.0
Boreal CLIM	PR24	-0.0017	-0.006	-0.0069
	PR6	0.0044	0.0	0.0032
Canada	PR24	0.0001	-0.004	-0.0032
	PR6	0.0022	-0.0013	0.0037
East CLIM	PR24	0.0038	-0.00075	-0.00054
	PR6	0.0085	-0.0071	0.00093
Great Lakes CLIM	PR24	0.0059	0.015	0.022
	PR6	-0.0006	0.0073	0.0032
Mt West CLIM	PR24	-0.002	0.0	-0.0013
	PR6	0.02	0.0	0.0042
North Plains CLIM	PR24	0.0	-0.016	-0.0031
	PR6	0.0	0.0091	0.0052
Pacific North West CLIM	PR24	-0.0034	-0.0072	-0.0068
	PR6	-0.00067	-0.0039	-0.0021
Prairie CLIM	PR24	-0.0022	8.9e-05	-0.00013
	PR6	-0.0053	0.0	0.0
United States of America East	PR24	0.0	-0.00055	-0.00035
	PR6	0.0	-0.0053	0.0
United States of America West	PR24	-0.0017	-0.0087	-0.0075
	PR6	0.0	0.0	0.0
full_domain	PR24	0.0	-0.00086	-0.00095
	PR6	0.0012	-0.0015	-9.1e-05



Autumn 2021 (A21)

1.2MOM-noLF

2.2MOM-LF

3.3MOM-LF

‘bourgé’

EMET (A21), TT & TD

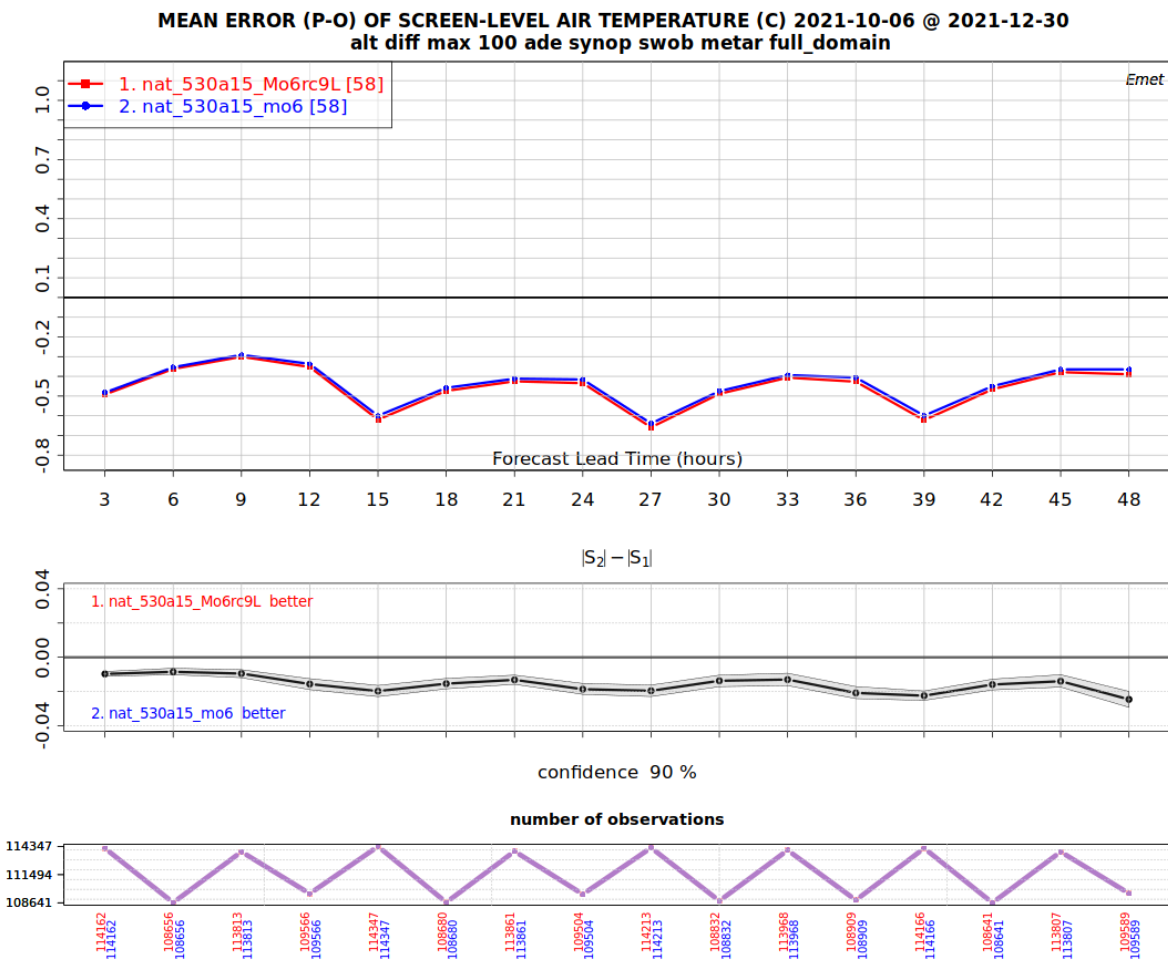
2MOM-LF

P3_v5.5.0-rc9
P3 v5.3.6 2MOM-noLF

Pretty neutral

bias		20211006 / 20211230		
nat_530a15_Mo6rc9L / nat_530a15_mo6		00z	12z	All
Arctic All CLIM	TD	-0.017	-0.012	-0.018
	TT	-0.025	-0.021	-0.024
Boreal CLIM	TD	-0.0069	-0.014	-0.01
	TT	-0.017	-0.022	-0.02
Canada	TD	-0.013	-0.011	-0.0092
	TT	-0.019	-0.02	-0.021
East CLIM	TD	0.0	-0.003	-0.0025
	TT	-0.011	-0.013	-0.011
Great Lakes CLIM	TD	-0.0014	0.00019	0.0018
	TT	-0.017	-0.014	-0.016
Mt West CLIM	TD	-0.025	-0.027	-0.028
	TT	-0.029	-0.023	-0.025
North Plains CLIM	TD	-0.00071	0.0048	0.012
	TT	-0.015	-0.0071	-0.0065
Pacific North West CLIM	TD	0.0	0.0	0.0
	TT	-0.012	-0.018	-0.013
Prairie CLIM	TD	-0.00077	0.00054	0.0007
	TT	-0.0095	-0.0053	-0.0078
United States of America East	TD	-0.0021	-0.0017	-0.00095
	TT	-0.01	-0.011	-0.01
United States of America West	TD	-0.0095	-0.01	-0.0073
	TT	-0.019	-0.013	-0.016
full_domain	TD	-0.0063	-0.0074	-0.0064
	TT	-0.017	-0.015	-0.016

TT-bias (full domain)



'bourge'

EMET (A21), P0 & UV 2MOM-LF

P3_v5.5.0-rc9

P3 v5.3.6 2MOM-noLF



bias < >

bias nat_530a15_Mo6rc9L / nat_530a15_mo6		20211006 / 20211230		
		00z	12z	All
Arctic All CLIM	P0	0.016	0.016	0.016
Boreal CLIM	P0	0.014	0.017	0.016
Canada	P0	0.013	0.017	0.015
East CLIM	P0	0.007	0.017	0.014
Great Lakes CLIM	P0	0.004	0.013	0.011
Mt West CLIM	P0	0.0056	-0.0096	0.0019
North Plains CLIM	P0	0.01	0.015	0.013
Pacific North West CLIM	P0	0.0049	-0.00069	0.0026
Prairie CLIM	P0	0.0021	0.0072	0.0055
United States of America East	P0	-0.00027	0.002	0.0034
United States of America West	P0	-0.0095	-0.013	-0.013
full_domain	P0	0.012	0.016	0.014

stdev < >

stdev nat_530a15_Mo6rc9L / nat_530a15_mo6		20211006 / 20211230		
		00z	12z	All
Arctic All CLIM	P0	0.0	0.0015	0.00069
Boreal CLIM	P0	-0.0013	-0.00088	0.0
Canada	P0	0.0	0.0	0.0
East CLIM	P0	5.1e-05	0.0	0.0
Great Lakes CLIM	P0	0.0016	0.0	0.0
Mt West CLIM	P0	0.00088	0.0	0.0
North Plains CLIM	P0	0.0	0.00046	0.0
Pacific North West CLIM	P0	0.00081	0.0	0.00077
Prairie CLIM	P0	0.0037	0.0041	0.0031
United States of America East	P0	0.0031	-0.00053	0.0011
United States of America West	P0	0.00097	0.0	0.0
full_domain	P0	0.00024	0.0	0.0

bias < >

bias nat_530a15_Mo6rc9L / nat_530a15_mo6		20211006 / 20211230		
		00z	12z	All
Arctic All CLIM	UV	0.0053	0.0028	0.0047
Boreal CLIM	UV	0.0029	0.00099	0.0015
Canada	UV	0.0029	0.0013	0.0022
East CLIM	UV	0.0022	-9.7e-05	0.001
Great Lakes CLIM	UV	0.0	0.00053	0.0
Mt West CLIM	UV	-0.0032	0.0	-0.0023
North Plains CLIM	UV	-0.0032	0.0	-0.0024
Pacific North West CLIM	UV	0.00036	0.0026	0.0021
Prairie CLIM	UV	0.0	0.0	0.0
United States of America East	UV	0.00039	0.0	8.3e-05
United States of America West	UV	-0.0029	-0.0014	-0.0019
full_domain	UV	0.0015	0.0001	0.0022

rmse < >

rmse nat_530a15_Mo6rc9L / nat_530a15_mo6		20211006 / 20211230		
		00z	12z	All
Arctic All CLIM	UV	0.00037	0.0017	0.0018
Boreal CLIM	UV	0.0016	0.0014	0.0015
Canada	UV	-0.00052	0.0	-0.0006
East CLIM	UV	1.2e-05	0.0	-0.00057
Great Lakes CLIM	UV	0.0	0.0	-0.0011
Mt West CLIM	UV	-0.0036	-0.00017	-0.0034
North Plains CLIM	UV	-0.00013	0.00021	0.00034
Pacific North West CLIM	UV	-0.00094	0.0	0.0
Prairie CLIM	UV	0.0013	0.0	0.001
United States of America East	UV	0.00023	-0.00037	-6.9e-05
United States of America West	UV	-0.0036	-0.0016	-0.0027
full_domain	UV	0.00034	0.0	-0.00061

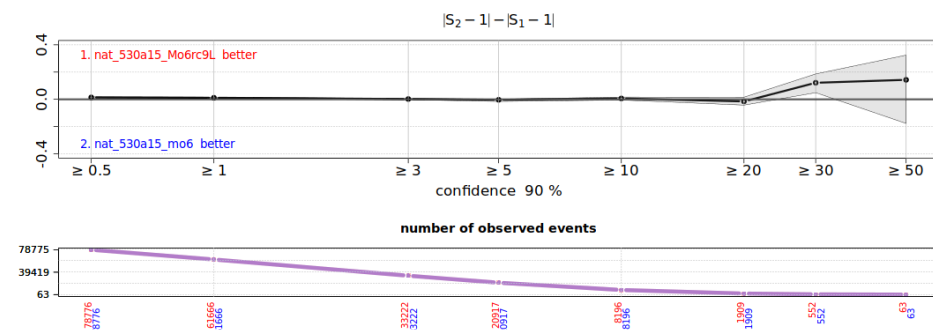
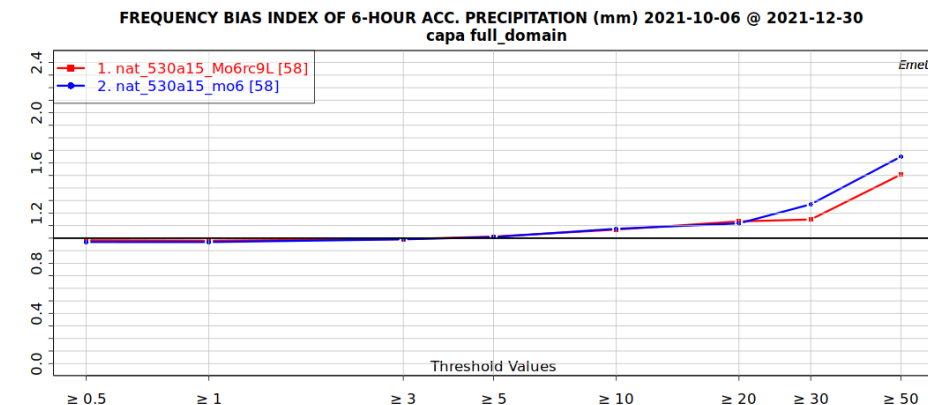
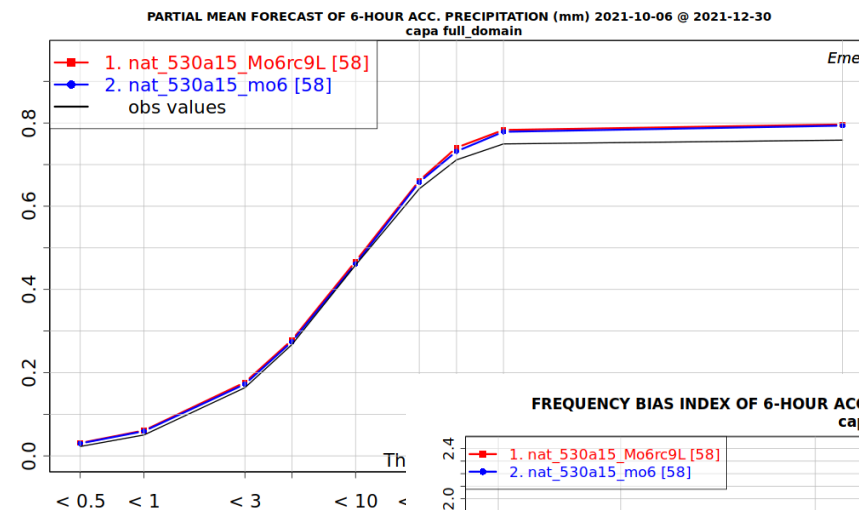
'bourne'

EMET (A21), PR6 & PR24 2MOM-LF

P3_v5.5.0-rc9

P3 v5.3.6 2MOM-noLF

dpmean									
ets									
fbi									
pod									
far									



Autumn 2021 (A21)

1.2MOM-noLF

2.2MOM-LF

3.3MOM-LF

‘bourge’

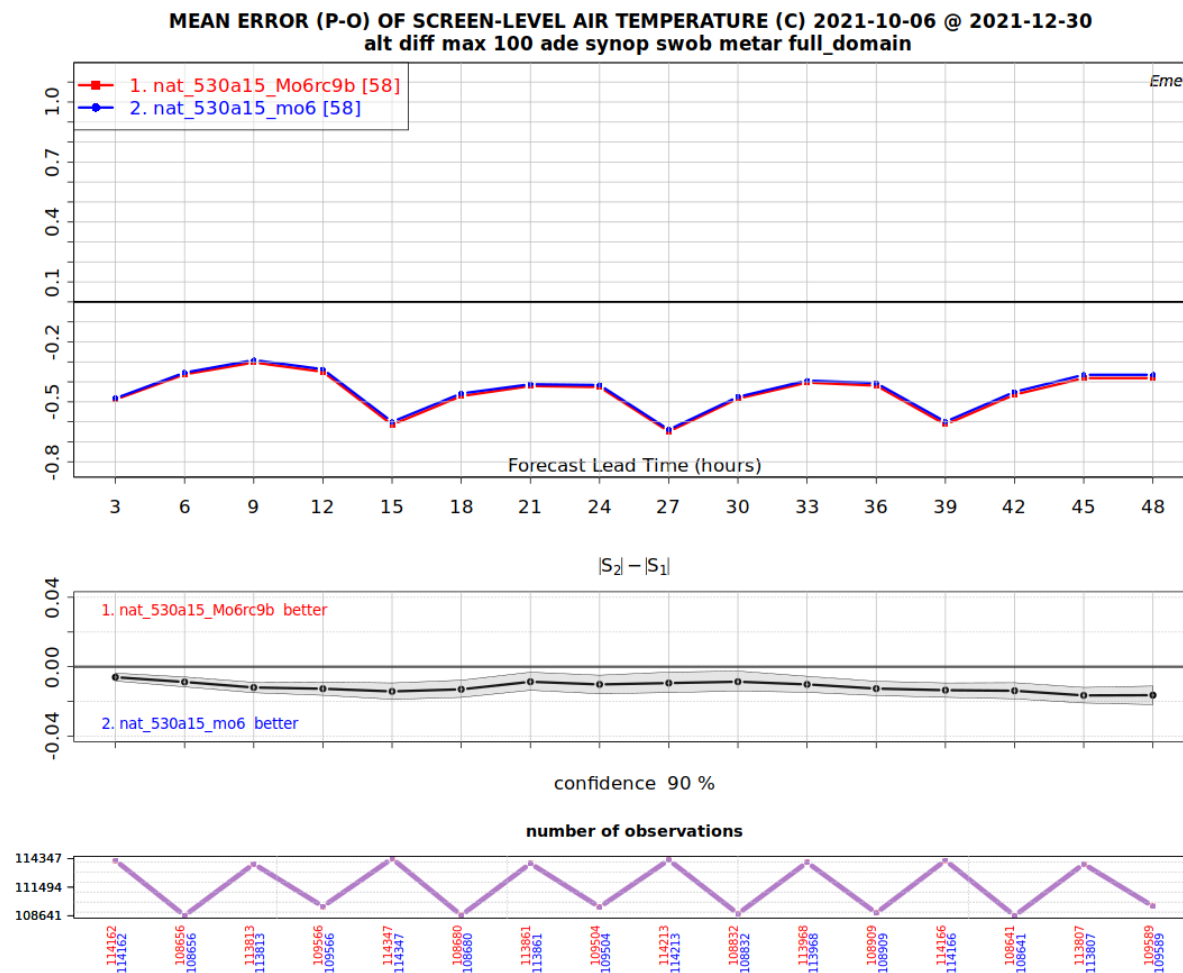
EMET (A21), TT & TD 3MOM-LF

P3_v5.5.0-rc9
P3 v5.3.6 2MOM-noLF

Pretty neutral

bias		20211006 / 20211230		
		00z	12z	All
Arctic All CLIM	TD	-0.013	0.0	-0.01
	TT	-0.037	-0.034	-0.036
Boreal CLIM	TD	0.0064	0.0032	0.0063
	TT	-0.0064	-0.011	-0.012
Canada	TD	0.0025	0.0072	0.0047
	TT	-0.014	-0.0099	-0.016
East CLIM	TD	0.00096	-0.00057	0.00034
	TT	-0.0064	-0.0073	-0.0092
Great Lakes CLIM	TD	-0.00085	-0.0025	0.00043
	TT	-0.013	-0.0089	-0.014
Mt West CLIM	TD	0.0081	0.0051	0.0082
	TT	-0.0081	-0.011	-0.0087
North Plains CLIM	TD	-0.003	-0.012	-0.015
	TT	0.0	0.013	0.0049
Pacific North West CLIM	TD	0.045	0.044	0.045
	TT	-0.015	-0.012	-0.016
Prairie CLIM	TD	-0.0014	0.0	0.0
	TT	-0.0034	0.0	-0.0028
United States of America East	TD	0.0012	0.0	0.00038
	TT	-0.006	-0.0048	-0.0072
United States of America West	TD	0.0046	0.0033	0.0091
	TT	-0.0062	-0.0018	-0.0072
full_domain	TD	0.002	0.0035	0.0052
	TT	-0.013	-0.0096	-0.012

TT-bias (full domain)



'bourge'

EMET (A21), P0 & UV 3MOM-LF

P3_v5.5.0-rc9

P3 v5.3.6 2MOM-noLF

bias < >

bias nat_530a15_Mo6rc9b / nat_530a15_mo6		20211006 / 20211230		
		00z	12z	All
Arctic All CLIM	P0	0.027	0.03	0.029
Boreal CLIM	P0	0.026	0.031	0.029
Canada	P0	0.022	0.027	0.024
East CLIM	P0	0.01	0.022	0.019
Great Lakes CLIM	P0	0.0088	0.017	0.016
Mt West CLIM	P0	0.015	-0.012	0.0047
North Plains CLIM	P0	0.019	0.027	0.024
Pacific North West CLIM	P0	0.011	0.0016	0.0051
Prairie CLIM	P0	0.0058	0.0093	0.0067
United States of America East	P0	0.0015	0.0042	0.0061
United States of America West	P0	-0.015	-0.015	-0.016
full_domain	P0	0.02	0.025	0.023

rmse < >

rmse nat_530a15_Mo6rc9b / nat_530a15_mo6		20211006 / 20211230		
		00z	12z	All
Arctic All CLIM	P0	0.005	0.013	0.0095
Boreal CLIM	P0	0.0074	0.0052	0.0072
Canada	P0	0.0072	0.0056	0.0063
East CLIM	P0	0.0024	0.006	0.0051
Great Lakes CLIM	P0	0.0053	0.0034	0.0049
Mt West CLIM	P0	0.00012	-0.0011	0.00013
North Plains CLIM	P0	0.0047	0.0027	0.0046
Pacific North West CLIM	P0	0.0032	-0.00091	0.0017
Prairie CLIM	P0	0.0022	0.0036	0.0025
United States of America East	P0	0.0018	0.0021	0.002
United States of America West	P0	0.0011	-0.0045	-0.0023
full_domain	P0	0.0065	0.005	0.0063

bias < >

bias nat_530a15_Mo6rc9b / nat_530a15_mo6		20211006 / 20211230		
		00z	12z	All
Arctic All CLIM	UV	0.0041	0.0029	0.0038
Boreal CLIM	UV	0.0027	0.0	0.0
Canada	UV	0.0011	-0.0011	0.0
East CLIM	UV	0.001	0.0	0.00028
Great Lakes CLIM	UV	-0.0016	-0.0029	-0.0016
Mt West CLIM	UV	0.0	0.0051	0.002
North Plains CLIM	UV	-0.0046	-0.00077	-0.0016
Pacific North West CLIM	UV	0.0	-0.0018	-0.0053
Prairie CLIM	UV	0.0	-0.00028	0.0
United States of America East	UV	0.0	0.0	0.0
United States of America West	UV	0.0014	0.0016	0.0019
full_domain	UV	0.00063	-0.0011	0.0

rmse < >

rmse nat_530a15_Mo6rc9b / nat_530a15_mo6		20211006 / 20211230		
		00z	12z	All
Arctic All CLIM	UV	0.0	0.00022	0.0
Boreal CLIM	UV	0.0016	0.0	0.00074
Canada	UV	-0.00067	-0.0031	-0.0016
East CLIM	UV	-0.00026	0.0	0.0
Great Lakes CLIM	UV	0.0	-0.005	-0.0027
Mt West CLIM	UV	0.0	-0.0022	0.0
North Plains CLIM	UV	0.0	0.0	-0.0009
Pacific North West CLIM	UV	-0.0048	-0.011	-0.006
Prairie CLIM	UV	-0.00075	0.0	-0.00085
United States of America East	UV	-0.0012	-0.0005	-0.00068
United States of America West	UV	-0.00053	-0.001	-0.0022
full_domain	UV	-0.0011	-0.002	-0.0012

'bourne'

EMET (A21), PR6 & PR24 3MOM-LF

P3_v5.5.0-rc9

P3 v5.3.6 2MOM-noLF

dpmean

<>

dpmean

20211006 / 20211230

	00z	12z	All
Arctic All CLIM	PR24 -0.003	-0.0042	-0.0046
	PR6 0.00037	-0.00078	0.0
Boreal CLIM	PR24 0.0	-0.013	0.00033
	PR6 0.00019	-0.0013	-0.00082
Canada	PR24 0.0	-0.0094	-0.0031
	PR6 -0.0012	-0.0032	-0.002
East CLIM	PR24 0.001	-0.0015	0.0011
	PR6 0.00016	0.0012	0.0006
Great Lakes CLIM	PR24 0.00038	0.0029	0.0016
	PR6 0.0002	-0.0012	0.0
Mt West CLIM	PR24 -0.00066	-0.0033	-0.0019
	PR6 0.01	0.0044	0.0098
North Plains CLIM	PR24 0.0	0.0006	0.0049
	PR6 0.0	0.00016	0.0022
Pacific North West CLIM	PR24 0.0	-0.06	-0.001
	PR6 -0.0064	-0.023	-0.017
Prairie CLIM	PR24 0.0041	-0.01	-0.0067
	PR6 0.0026	-0.0018	-0.0014
United States of America East	PR24 0.0013	0.005	0.00075
	PR6 0.00084	-0.0043	0.0007
United States of America West	PR24 0.00076	-0.012	0.02
	PR6 0.00068	0.0	-0.0026
full_domain	PR24 0.0011	-0.0048	-0.0068
	PR6 -0.00014	-0.0021	0.00077

ets

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ETS

20211006 / 20211230

	00z	12z	All
Arctic All CLIM	PR24 -0.0063	-0.024	0.0
	PR6 0.0	0.0	0.0
Boreal CLIM	PR24 0.00034	-0.0058	0.003
	PR6 0.0046	0.0	0.0041
Canada	PR24 0.0	-0.0041	0.0023
	PR6 0.0	0.0014	0.0043
East CLIM	PR24 -0.00039	0.0019	0.006
	PR6 0.0056	0.002	0.0017
Great Lakes CLIM	PR24 0.0067	0.005	0.0037
	PR6 0.0074	0.0	0.0
Mt West CLIM	PR24 -0.02	0.0073	-0.0023
	PR6 0.028	0.0	0.022
North Plains CLIM	PR24 0.01	-0.00024	0.0077
	PR6 -0.0034	0.0	0.01
Pacific North West CLIM	PR24 0.0	0.0015	0.0012
	PR6 0.002	-0.0006	0.00027
Prairie CLIM	PR24 0.0039	-0.015	0.0
	PR6 -0.0013	0.0	0.0
United States of America East	PR24 0.0056	0.0015	0.0064
	PR6 0.003	0.0015	0.0025
United States of America West	PR24 0.0	0.0017	0.0015
	PR6 -0.0017	-0.0024	0.0
full_domain	PR24 0.0019	0.0018	0.0019
	PR6 0.00085	0.0013	0.001

fbi

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FBI

20211006 / 20211230

	00z	12z	All
Arctic All CLIM	PR24 0.0	0.0	-0.0028
	PR6 0.0	-0.012	-0.0049
Boreal CLIM	PR24 0.0	0.025	0.0
	PR6 0.003	-0.062	0.0015
Canada	PR24 0.0	0.013	-0.003
	PR6 0.015	-0.016	0.0066
East CLIM	PR24 -0.0011	0.0	0.0
	PR6 -0.0015	0.0	0.0
Great Lakes CLIM	PR24 -0.003	-0.0043	0.00097
	PR6 -0.026	0.0021	-0.0055
Mt West CLIM	PR24 -0.024	-0.017	0.0
	PR6 -0.012	-0.0041	-0.01
North Plains CLIM	PR24 0.0072	0.0	0.0041
	PR6 -0.012	0.0	0.0058
Pacific North West CLIM	PR24 0.0	-0.016	-0.007
	PR6 0.0	0.02	-0.0058
Prairie CLIM	PR24 -0.028	-0.005	-0.0012
	PR6 0.0	-0.02	0.0075
United States of America East	PR24 -0.0075	0.0	0.0
	PR6 -0.003	0.0	0.093
United States of America West	PR24 0.0018	-0.0024	0.012
	PR6 0.0092	-0.016	-0.0053
full_domain	PR24 0.0038	0.00081	-0.00024
	PR6 0.013	0.12	0.07

pod

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POD

20211006 / 20211230

	00z	12z	All
Arctic All CLIM	PR24 0.0	-0.022	0.0
	PR6 0.0	0.0032	0.0019
Boreal CLIM	PR24 0.0021	-0.023	0.0025
	PR6 0.0033	0.042	0.0054
Canada	PR24 -0.0023	-0.0059	-0.0024
	PR6 0.0	0.005	0.0021
East CLIM	PR24 0.00055	0.0	0.0026
	PR6 0.0013	0.0023	0.0018
Great Lakes CLIM	PR24 0.013	0.0062	0.0063
	PR6 0.03	0.0026	0.007
Mt West CLIM	PR24 -0.022	-0.0099	-0.0064
	PR6 0.015	-0.0019	0.0049
North Plains CLIM	PR24 0.016	-0.013	0.01
	PR6 -0.0052	0.0093	0.0061
Pacific North West CLIM	PR24 0.0	0.0029	0.0025
	PR6 0.0	0.0067	0.0011
Prairie CLIM	PR24 0.0076	-0.024	0.0
	PR6 -0.015	0.0	-0.012
United States of America East	PR24 0.008	0.00062	0.0034
	PR6 0.0012	0.0017	0.001
United States of America West	PR24 0.0013	0.0037	0.0031
	PR6 0.0017	0.011	0.0017
full_domain	PR24 0.0031	0.0027	0.0033
	PR6 0.002	0.0021	0.0019

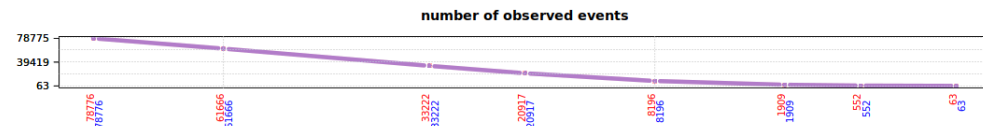
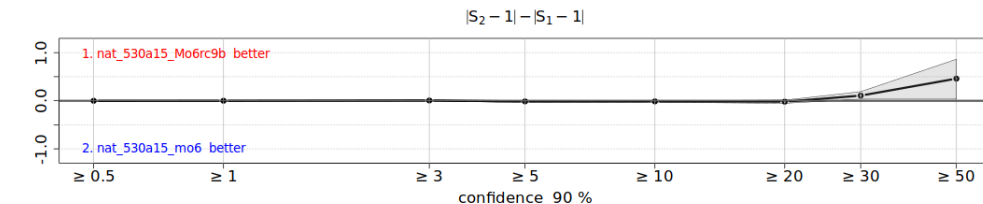
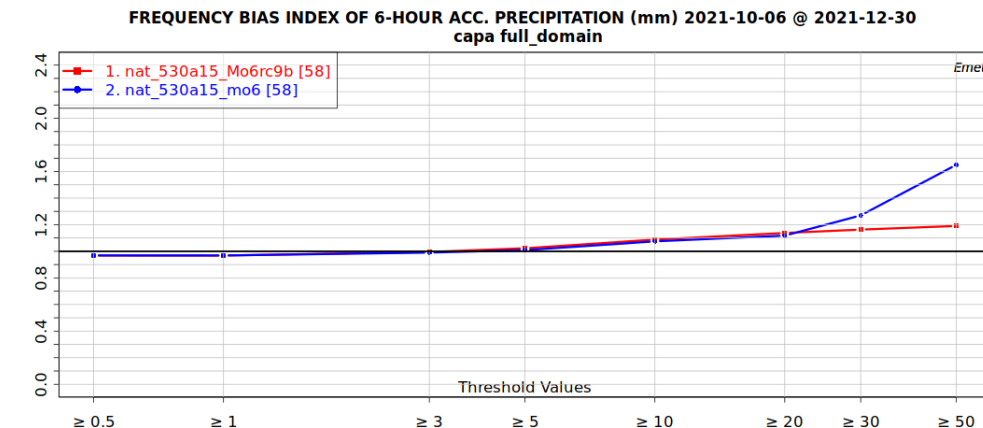
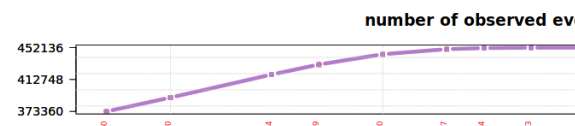
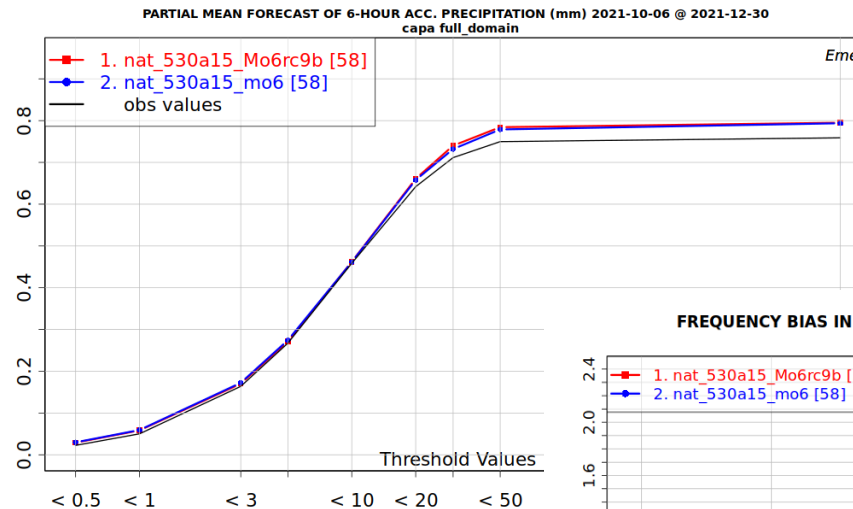
far

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FAR

20211006 / 20211230

	00z	12z	All
Arctic All CLIM	PR24 -0.005	0.0	-0.0057
	PR6 0.0	0.0	0.0
Boreal CLIM	PR24 0.0	0.0	0.0018
	PR6 0.0024	0.0	0.0029
Canada	PR24 0.0	-0.0031	-0.0025
	PR6 0.0	0.0	0.0056
East CLIM	PR24 0.016	0.011	0.011
	PR6 0.01	0.0014	0.0013
Great Lakes CLIM	PR24 0.0024	0.0	0.0
	PR6 0.01	0.0	0.0
Mt West CLIM	PR24 0.012	0.0	0.0018
	PR6 0.043	0.0021	0.027
North Plains CLIM	PR24 0.01	-0.0016	0.0051
	PR6 -0.0042	0.0	0.0094
Pacific North West CLIM	PR24 -0.00056	-0.0063	-0.0044
	PR6 -0.00071	-0.0027	-0.0046
Prairie CLIM	PR24 0.0	0.00095	0.00082
	PR6 0.0025	0.0	-0.0012
United States of America East	PR24 0.00043	0.0033	0.0095
	PR6 0.0048	0.0012	0.0028
United States of America West	PR24 0.0	-0.0038	-0.0033
	PR6 -0.0015	-0.0073	-0.0019
full_domain	PR24 0.0021	0.0	0.0011
	PR6 0.0042	0.0	0.0



H22: winter 2022

1.2MOM-noLF

2.2MOM-LF

3.3MOM-LF

‘nil’

P3_v5.5.0-rc9

P3 v5.3.6 2MOM-noLF

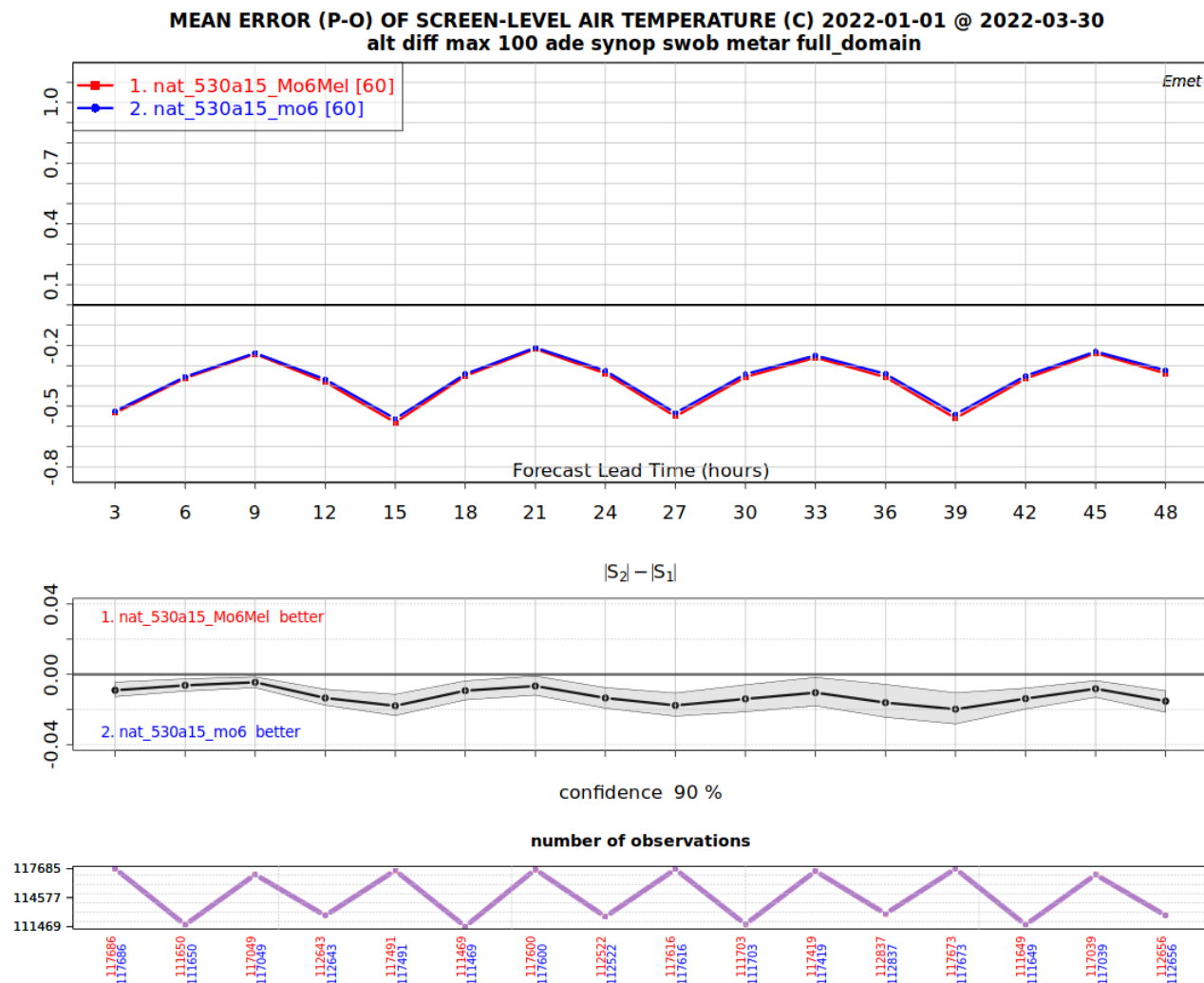
EMET (H22), TT & TD 2MOM-noLF*

Small cooling everywhere caused by the change to secondary ice production to improve freezing rain forecasts. This is consistent with initial results in P3_v5.3.14

More results* and interpretation:
Cholette et al. (2024)*

bias		< >		
bias		20220101 / 20220330		
nat_530a15_Mo6Mel / nat_530a15_mo6		00z	12z	All
Arctic All CLIM	TD	0.0072	0.0098	0.01
	TT	0.0083	0.013	0.011
Boreal CLIM	TD	0.0092	-0.0019	0.00092
	TT	0.0076	0.0042	-0.0012
Canada	TD	0.00041	0.01	-0.00036
	TT	-0.017	-0.0022	-0.0043
East CLIM	TD	0.0017	0.0027	0.0014
	TT	-0.0065	-0.0042	-0.0026
Great Lakes CLIM	TD	0.004	0.0022	0.0011
	TT	-0.0042	-0.0025	-0.0028
Mt West CLIM	TD	-0.065	-0.056	-0.07
	TT	-0.042	-0.043	-0.041
North Plains CLIM	TD	0.016	0.02	0.027
	TT	-0.014	0.0018	0.0024
Pacific North West CLIM	TD	-0.0072	-0.012	-0.009
	TT	-0.027	-0.031	-0.032
Prairie CLIM	TD	0.0011	0.0	0.0
	TT	-0.0029	-0.0035	0.0
United States of America East	TD	0.0013	0.0019	0.0018
	TT	-0.0022	-0.0056	-0.0029
United States of America West	TD	-0.024	-0.013	-0.015
	TT	-0.03	-0.027	-0.02
full_domain	TD	-0.0091	-0.0064	-0.0052
	TT	-0.015	-0.013	-0.012

TT-bias (full domain)



‘nil’

P3_v5.5.0-rc9

P3 v5.3.6 2MOM-noLF

EMET (H22), TT & TD 2MOM-noLF*

Small cooling everywhere caused by the change to secondary ice production to improve freezing rain forecasts. This is consistent with initial results in P3_v5.3.14

More results* and interpretation:
Cholette et al. (2024)*

bias				
nat_530a15_Mo6Mel / nat_530a15_mo6		20220101 / 20220330		
		00z	12z	All
Arctic All CLIM	TD	0.0072	0.0098	0.01
	TT	0.0083	0.013	0.011
Boreal CLIM	TD	0.0092	-0.0019	0.00092
	TT	0.0076	0.0042	-0.0012
Canada	TD	0.00041	0.01	-0.00036
	TT	-0.017	-0.0022	-0.0043
East CLIM	TD	0.0017	0.0027	0.0014
	TT	-0.0065	-0.0042	-0.0026
Great Lakes CLIM	TD	0.004	0.0022	0.0011
	TT	-0.0042	-0.0025	-0.0028
Mt West CLIM	TD	-0.065	-0.056	-0.07
	TT	-0.042	-0.043	-0.041
North Plains CLIM	TD	0.016	0.02	0.027
	TT	-0.014	0.0018	0.0024
Pacific North West CLIM	TD	-0.0072	-0.012	-0.009
	TT	-0.027	-0.031	-0.032
Prairie CLIM	TD	0.0011	0.0	0.0
	TT	-0.0029	-0.0035	0.0
United States of America East	TD	0.0013	0.0019	0.0018
	TT	-0.0022	-0.0056	-0.0029
United States of America West	TD	-0.024	-0.013	-0.015
	TT	-0.03	-0.027	-0.02
full_domain	TD	-0.0091	-0.0064	-0.0052
	TT	-0.015	-0.013	-0.012

rmse				
nat_530a15_Mo6Mel / nat_530a15_mo6		20220101 / 20220330		
		00z	12z	All
Arctic All CLIM	TD	-0.0065	-0.0044	-0.005
	TT	-0.0062	-0.0066	-0.0074
Boreal CLIM	TD	-0.0031	-0.014	-0.01
	TT	-0.0066	-0.017	-0.014
Canada	TD	-0.0086	-0.013	-0.012
	TT	-0.0096	-0.016	-0.014
East CLIM	TD	-0.00091	0.0	-0.00016
	TT	-0.0088	-0.0081	-0.011
Great Lakes CLIM	TD	0.0	-0.0052	-0.0054
	TT	-0.013	-0.01	-0.013
Mt West CLIM	TD	-0.057	-0.047	-0.054
	TT	-0.047	-0.044	-0.05
North Plains CLIM	TD	-0.0011	-0.0039	-0.0041
	TT	-0.0016	-0.0099	-0.0065
Pacific North West CLIM	TD	-0.011	-0.017	-0.014
	TT	-0.013	-0.016	-0.018
Prairie CLIM	TD	0.0	-0.0019	-0.0023
	TT	-0.011	-0.0078	-0.011
United States of America East	TD	-0.0012	-0.0024	-0.0017
	TT	-0.015	-0.0098	-0.013
United States of America West	TD	-0.024	-0.017	-0.022
	TT	-0.024	-0.021	-0.025
full_domain	TD	-0.009	-0.0098	-0.01
	TT	-0.014	-0.015	-0.015

stdev				
nat_530a15_Mo6Mel / nat_530a15_mo6		20220101 / 20220330		
		00z	12z	All
Arctic All CLIM	TD	-0.0077	-0.0048	-0.0073
	TT	-0.0072	-0.01	-0.009
Boreal CLIM	TD	-0.0056	-0.016	-0.011
	TT	-0.0087	-0.017	-0.014
Canada	TD	-0.0072	-0.014	-0.012
	TT	-0.0087	-0.017	-0.014
East CLIM	TD	-0.0016	0.0	-0.0021
	TT	-0.0064	-0.0063	-0.011
Great Lakes CLIM	TD	0.0	-0.006	-0.0058
	TT	-0.011	-0.0079	-0.013
Mt West CLIM	TD	-0.035	-0.033	-0.04
	TT	-0.023	-0.024	-0.034
North Plains CLIM	TD	0.0	-0.01	-0.01
	TT	0.0	-0.011	-0.0075
Pacific North West CLIM	TD	-0.0083	-0.0085	-0.0094
	TT	-0.0087	-0.0074	-0.01
Prairie CLIM	TD	-0.002	-0.0027	-0.0035
	TT	-0.0078	-0.0053	-0.011
United States of America East	TD	-0.0034	-0.0027	-0.0035
	TT	-0.012	-0.0087	-0.012
United States of America West	TD	-0.018	-0.016	-0.02
	TT	-0.013	-0.013	-0.02
full_domain	TD	-0.0074	-0.0092	-0.0096
	TT	-0.011	-0.013	-0.013

'nil'

P3_v5.5.0-rc9

P3 v5.3.6 2MOM-noLF

EMET (H22), P0 & UV 2MOM-noLF*

bias		20220101 / 20220330		
nat_530a15_Mo6Mel / nat_530a15_mo6		00z	12z	All
Arctic All CLIM	P0	0.0016	0.0029	0.0023
Boreal CLIM	P0	0.012	0.015	0.014
Canada	P0	0.013	0.016	0.015
East CLIM	P0	0.018	0.025	0.022
Great Lakes CLIM	P0	0.0055	0.015	0.012
Mt West CLIM	P0	0.0032	0.012	0.008
North Plains CLIM	P0	0.015	0.018	0.016
Pacific North West CLIM	P0	-0.01	-0.0094	-0.008
Prairie CLIM	P0	0.012	0.01	0.012
United States of America East	P0	0.0074	0.0098	0.012
United States of America West	P0	-0.019	-0.018	-0.02
full_domain	P0	0.013	0.016	0.015

rmse		20220101 / 20220330		
nat_530a15_Mo6Mel / nat_530a15_mo6		00z	12z	All
Arctic All CLIM	P0	0.00022	0.0015	0.00089
Boreal CLIM	P0	0.0037	0.005	0.0055
Canada	P0	0.0027	0.0045	0.0045
East CLIM	P0	0.0021	0.0023	0.0027
Great Lakes CLIM	P0	0.0	0.0015	0.0014
Mt West CLIM	P0	-0.0011	0.0	0.0
North Plains CLIM	P0	0.0029	0.0033	0.0036
Pacific North West CLIM	P0	0.0018	-0.0021	-0.00061
Prairie CLIM	P0	0.0047	0.004	0.0054
United States of America East	P0	0.001	0.0013	0.0014
United States of America West	P0	-0.002	-0.0038	-0.0034
full_domain	P0	0.0022	0.0037	0.004

bias		20220101 / 20220330		
nat_530a15_Mo6Mel / nat_530a15_mo6		00z	12z	All
Arctic All CLIM	UV	0.0	0.0	0.0
Boreal CLIM	UV	0.00099	0.0022	0.002
Canada	UV	0.001	0.00094	0.0019
East CLIM	UV	-4.6e-05	-0.00011	0.00039
Great Lakes CLIM	UV	0.00082	-0.001	8.4e-05
Mt West CLIM	UV	-0.0015	0.0	-0.0019
North Plains CLIM	UV	-0.0063	-0.0022	-0.0059
Pacific North West CLIM	UV	-0.0018	0.0	0.0
Prairie CLIM	UV	-6.9e-06	0.0	-0.00042
United States of America East	UV	0.00028	0.0009	0.001
United States of America West	UV	0.00025	-0.0016	-0.0012
full_domain	UV	0.00052	0.0007	0.0011

rmse		20220101 / 20220330		
nat_530a15_Mo6Mel / nat_530a15_mo6		00z	12z	All
Arctic All CLIM	UV	0.0	0.00049	-0.00032
Boreal CLIM	UV	0.0012	0.0	0.001
Canada	UV	0.0	-0.00085	-0.00057
East CLIM	UV	0.0	-0.00087	0.0
Great Lakes CLIM	UV	0.0	0.0	0.0
Mt West CLIM	UV	-0.0012	0.0	-0.00091
North Plains CLIM	UV	-0.0024	0.0	-0.0017
Pacific North West CLIM	UV	0.0	-0.00016	0.00026
Prairie CLIM	UV	-0.00095	0.0	0.0
United States of America East	UV	0.0	0.0	0.0
United States of America West	UV	-0.0014	0.0	-0.00083
full_domain	UV	-0.0005	-0.0003	-0.00034

'nil'

P3_v5.5.0-rc9

P3 v5.3.6 2MOM-noLF

EMET (H22), TCC 2MOM-noLF*

fbi_clrfew 0 - 2				
		20220101 / 20220330		
		00z	12z	All
Arctic All CLIM	NT	-0.0055	-0.0095	-0.0069
	TCC	0.00023	0.0	0.00011
Boreal CLIM	NT	0.0	-0.01	-0.0088
	TCC	0.0	0.0	0.0029
Canada	NT	-0.014	-0.019	-0.02
	TCC	0.0058	0.0048	0.0066
East CLIM	NT	-0.0029	-0.0042	-0.0052
	TCC	0.0046	0.0016	0.0052
Great Lakes CLIM	NT	-0.004	-0.0053	-0.0081
	TCC	0.0	0.0	0.0015
Mt West CLIM	NT	-0.024	-0.013	-0.02
	TCC	0.017	0.0	0.012
North Plains CLIM	NT	0.0	-0.0097	-0.0028
	TCC	0.0	0.0013	0.0043
Pacific North West CLIM	NT	-0.011	-0.011	-0.02
	TCC	0.0	0.0081	0.0
Prairie CLIM	NT	0.0	0.0023	-0.0025
	TCC	0.002	-0.0013	0.00089
United States of America East	NT	-0.006	-0.0028	-0.0067
	TCC	0.005	0.0012	0.0052
United States of America West	NT	-0.0068	-0.01	-0.0086
	TCC	0.0075	0.0	0.0061
full_domain	NT	-0.0087	-0.01	-0.011
	TCC	0.0054	0.0046	0.0069

fbi_sct 3 - 4				
		20220101 / 20220330		
		00z	12z	All
Arctic All CLIM	NT	0.0	0.013	-0.0069
	TCC	0.0	0.0	0.0
Boreal CLIM	NT	0.0	0.0	0.0
	TCC	0.0	0.0	0.0
Canada	NT	-0.023	-0.01	-0.024
	TCC	0.0034	0.0	0.0
East CLIM	NT	-0.019	-0.015	-0.014
	TCC	0.0	0.0	-0.0053
Great Lakes CLIM	NT	0.0	0.0	0.0
	TCC	0.018	0.0	0.0018
Mt West CLIM	NT	0.0	0.0	-0.043
	TCC	0.0	0.0	0.0
North Plains CLIM	NT	0.0	0.0	0.0
	TCC	0.0	0.0067	0.0
Pacific North West CLIM	NT	0.0098	-0.041	-0.068
	TCC	0.0	0.0	0.0
Prairie CLIM	NT	0.0	0.0	0.0
	TCC	0.0	0.0	0.0
United States of America East	NT	-0.017	-0.012	-0.016
	TCC	0.0	-0.006	-0.0045
United States of America West	NT	-0.088	0.0	-0.022
	TCC	-0.0083	0.0	0.0
full_domain	NT	-0.019	-0.014	-0.021
	TCC	0.0	-0.004	-0.0011

fbi_bkn_ovc 5 - 8				
		20220101 / 20220330		
		00z	12z	All
Arctic All CLIM	NT	0.0	-0.0029	-0.0026
	TCC	0.0	0.0	0.0
Boreal CLIM	NT	-0.0029	-0.0059	-0.0058
	TCC	0.0	0.0019	0.00091
Canada	NT	-0.0097	-0.009	-0.011
	TCC	0.0025	0.0022	0.0027
East CLIM	NT	-0.0086	-0.0073	-0.01
	TCC	0.0028	0.0	0.0034
Great Lakes CLIM	NT	-0.0042	-0.0034	-0.0056
	TCC	0.0022	0.0	0.00098
Mt West CLIM	NT	-0.029	-0.035	-0.04
	TCC	0.01	0.0041	0.01
North Plains CLIM	NT	-0.0065	-0.0045	-0.011
	TCC	0.0	0.0	0.0014
Pacific North West CLIM	NT	-0.0046	-0.0073	-0.0078
	TCC	0.0022	-5.1e-05	0.0
Prairie CLIM	NT	-0.0064	-0.0049	-0.0087
	TCC	0.0022	-0.004	0.00059
United States of America East	NT	-0.0075	-0.0084	-0.011
	TCC	0.0038	0.0016	0.0048
United States of America West	NT	-0.013	-0.016	-0.019
	TCC	0.0038	0.0052	0.0047
full_domain	NT	-0.012	-0.011	-0.012
	TCC	0.0041	0.0031	0.0042

Not sure how to interpret those scores.

'nil'

EMET (H22), PR6 & PR24 2MOM-noLF*

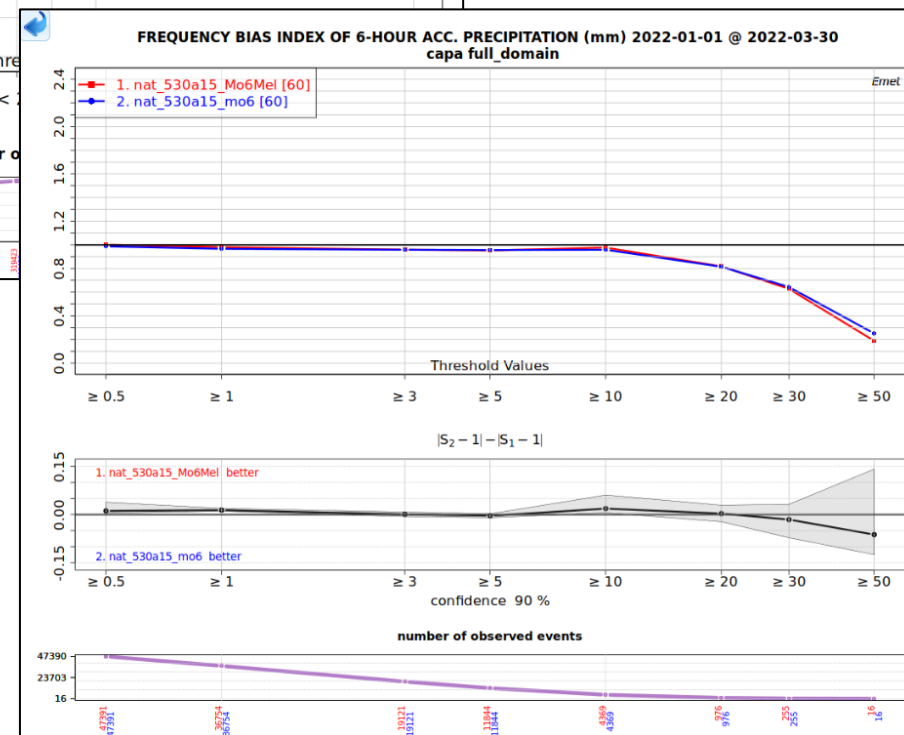
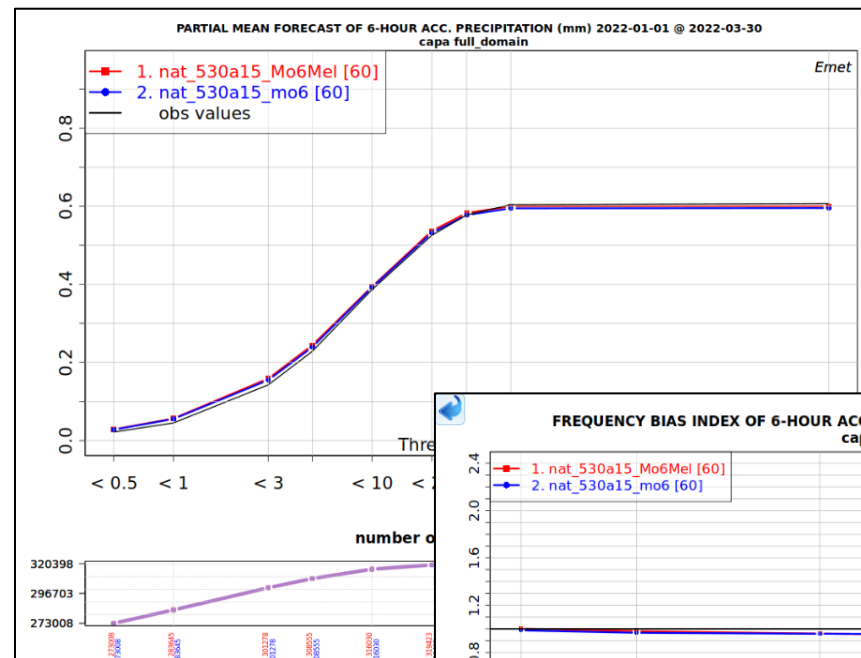
P3_v5.5.0-rc9

P3 v5.3.6 2MOM-noLF

dpmean			
dpmean			
nat_530a15_Mo6Mel / nat_530a15_mo6			
	00z	12z	All
Arctic All CLIM	PR24 0.0	-0.0019	-0.00042
	PR6 -0.00058	-0.00064	-0.00044
Boreal CLIM	PR24 -0.0014	0.0	-0.00094
	PR6 -0.00043	0.0	-7.9e-05
Canada	PR24 -0.012	-0.0074	-0.011
	PR6 -0.0049	-0.0038	-0.0044
East CLIM	PR24 0.0036	0.0063	0.0079
	PR6 0.00036	0.00088	0.0009
Great Lakes CLIM	PR24 0.021	-0.026	0.0093
	PR6 -0.00034	-0.0031	-0.0017
Mt West CLIM	PR24 0.0045	-0.00031	-0.00035
	PR6 0.00095	-0.00041	-0.00017
North Plains CLIM	PR24 0.0	0.0	-0.00083
	PR6 0.0	0.0	0.0
Pacific North West CLIM	PR24 -0.06	0.025	-0.011
	PR6 -0.022	-0.017	-0.02
Prairie CLIM	PR24 -0.0015	0.0	-0.00064
	PR6 0.00047	0.00062	0.0
United States of America East	PR24 0.0028	0.0027	0.0049
	PR6 -0.0006	-0.00034	-0.00012
United States of America West	PR24 -0.052	0.022	-0.0082
	PR6 -0.0055	-0.0064	-0.0056
full_domain	PR24 0.012	0.0098	0.016
	PR6 -0.0014	-0.00084	-0.0014

ETS			
ETS			
nat_530a15_Mo6Mel / nat_530a15_mo6			
	00z	12z	All
Arctic All CLIM	PR24 0.0	0.0	0.00012
	PR6 0.0	0.0	-0.00093
Boreal CLIM	PR24 -0.0014	0.0	-0.00094
	PR6 -0.00043	0.0	-7.9e-05
Canada	PR24 -0.012	-0.0074	-0.011
	PR6 -0.0049	-0.0038	-0.0044
East CLIM	PR24 0.0036	0.0063	0.0079
	PR6 0.00036	0.00088	0.0009
Great Lakes CLIM	PR24 0.021	-0.026	0.0093
	PR6 -0.00034	-0.0031	-0.0017
Mt West CLIM	PR24 0.0045	-0.00031	-0.00035
	PR6 0.00095	-0.00041	-0.00017
North Plains CLIM	PR24 0.0	0.0	-0.00083
	PR6 0.0	0.0	0.0
Pacific North West CLIM	PR24 -0.06	0.025	-0.011
	PR6 -0.022	-0.017	-0.02
Prairie CLIM	PR24 -0.0015	0.0	-0.00064
	PR6 0.00047	0.00062	0.0
United States of America East	PR24 0.0028	0.0027	0.0049
	PR6 -0.0006	-0.00034	-0.00012
United States of America West	PR24 -0.052	0.022	-0.0082
	PR6 -0.0055	-0.0064	-0.0056
full_domain	PR24 0.012	0.0098	0.016
	PR6 -0.0014	-0.00084	-0.0014

FBI			
FBI			
nat_530a15_Mo6Mel / nat_530a15_mo6			
	00z	12z	All
Arctic All CLIM	PR24 0.0	0.01	0.0062
	PR6 0.0	-0.031	-0.003
Boreal CLIM	PR24 -0.0057	0.0	-0.004
	PR6 -0.01	0.0	0.0
Canada	PR24 0.022	-0.027	-0.011
	PR6 0.0098	-0.00057	0.014
East CLIM	PR24 -0.0016	-0.00014	-0.00071
	PR6 0.0031	-0.014	-0.011
Great Lakes CLIM	PR24 -0.11	0.0032	-0.00038
	PR6 -0.0049	-0.0085	-0.0073
Mt West CLIM	PR24 0.0059	-0.019	-0.012
	PR6 0.0065	0.0	-0.018
North Plains CLIM	PR24 0.0	0.0	0.0
	PR6 0.0	0.0	0.0
Pacific North West CLIM	PR24 0.0088	0.02	0.015
	PR6 -0.024	-0.0062	-0.02
Prairie CLIM	PR24 -0.00031	0.0	0.0
	PR6 0.0037	-0.0016	0.0022
United States of America East	PR24 -0.0013	0.0009	-0.0068
	PR6 0.0028	-0.011	-0.011
United States of America West	PR24 0.0029	0.003	0.01
	PR6 -0.013	-0.0051	-0.0046
full_domain	PR24 0.0067	0.0073	0.0074
	PR6 0.0039	0.0057	0.0055



H22: winter 2022

1.2MOM-noLF

2.2MOM-LF

3.3MOM-LF

‘nil’

P3_v5.5.0-rc9

P3 v5.3.6 2MOM-noLF

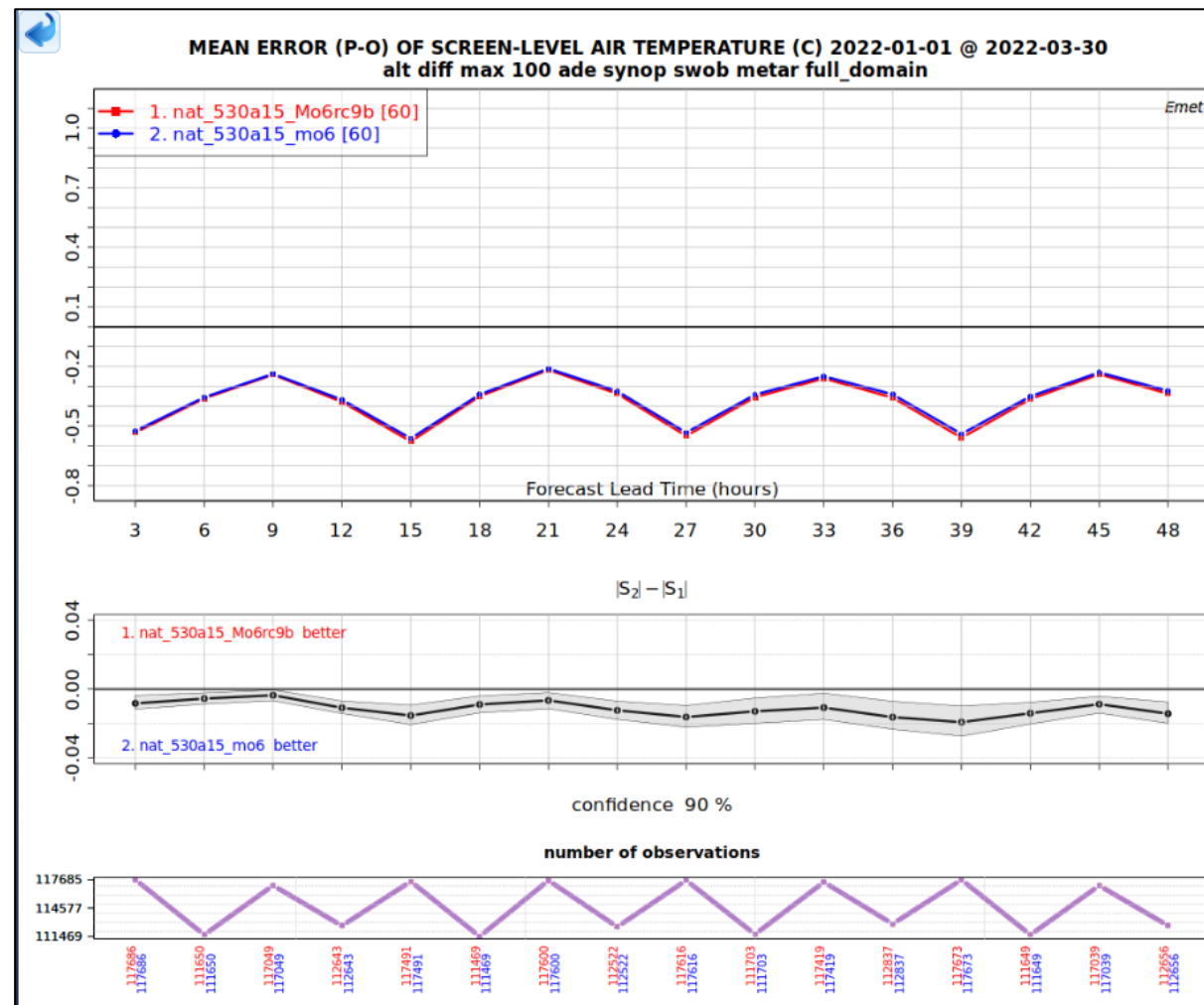
EMET (H22), TT & TD 2MOM-LF*

Expected results!
Same signal as for
2MOM-noLF

Very similar to 2MOM-
noLF also for RMSE and
STDEV (not shown)

bias		20220101 / 20220330		
nat_530a15_Mo6rc9b / nat_530a15_mo6		00Z	12Z	All
Arctic All CLIM	TD	0.0027	0.0033	0.0036
	TT	0.0028	0.0053	0.006
Boreal CLIM	TD	0.0073	-0.0025	0.0006
	TT	0.0061	0.0034	-0.001
Canada	TD	0.00037	0.0086	0.00029
	TT	-0.015	-0.0018	-0.0037
East CLIM	TD	0.0	0.00086	0.0012
	TT	-0.0054	-0.005	-0.0028
Great Lakes CLIM	TD	0.0038	-5.4e-06	0.0
	TT	-0.0071	-0.0016	-0.0032
Mt West CLIM	TD	-0.063	-0.056	-0.071
	TT	-0.043	-0.039	-0.04
North Plains CLIM	TD	0.017	0.022	0.026
	TT	-0.0096	0.0048	0.0021
Pacific North West CLIM	TD	-0.0099	-0.011	-0.0055
	TT	-0.026	-0.028	-0.028
Prairie CLIM	TD	0.0	0.0	0.0
	TT	-0.0015	-0.0042	0.0
United States of America East	TD	0.0	0.0011	0.00065
	TT	-0.0049	-0.0059	-0.0026
United States of America West	TD	-0.026	-0.011	-0.016
	TT	-0.028	-0.026	-0.02
full_domain	TD	-0.0083	-0.0072	-0.0056
	TT	-0.014	-0.012	-0.011

TT-bias (full domain)



'nil'

P3_v5.5.0-rc9

P3 v5.3.6 2MOM-noLF

EMET (H22), P0 & UV 2MOM-LF*

bias nat_530a15_Mo6rc9b / nat_530a15_mo6		20220101 / 20220330		
		00z	12z	All
Arctic All CLIM	P0	0.00082	0.00088	0.0015
Boreal CLIM	P0	0.012	0.013	0.012
Canada	P0	0.012	0.015	0.013
East CLIM	P0	0.017	0.022	0.02
Great Lakes CLIM	P0	0.0043	0.014	0.011
Mt West CLIM	P0	0.0024	0.012	0.0072
North Plains CLIM	P0	0.013	0.016	0.014
Pacific North West CLIM	P0	-0.011	-0.011	-0.009
Prairie CLIM	P0	0.011	0.0095	0.01
United States of America East	P0	0.0068	0.0082	0.011
United States of America West	P0	-0.017	-0.018	-0.018
full_domain	P0	0.012	0.015	0.013

rmse nat_530a15_Mo6rc9b / nat_530a15_mo6		20220101 / 20220330		
		00z	12z	All
Arctic All CLIM	P0	-0.0001	0.0013	0.00094
Boreal CLIM	P0	0.0037	0.0049	0.0045
Canada	P0	0.0027	0.0047	0.0047
East CLIM	P0	0.0022	0.0022	0.0029
Great Lakes CLIM	P0	0.0	0.0017	0.0015
Mt West CLIM	P0	-0.0003	0.0	0.0
North Plains CLIM	P0	0.0026	0.0044	0.0035
Pacific North West CLIM	P0	-0.00011	0.0	-0.0002
Prairie CLIM	P0	0.0041	0.0035	0.0054
United States of America East	P0	0.0013	0.0012	0.0015
United States of America West	P0	-0.0013	-0.0013	-0.0015
full_domain	P0	0.0024	0.004	0.0039

bias nat_530a15_Mo6rc9b / nat_530a15_mo6		20220101 / 20220330		
		00z	12z	All
Arctic All CLIM	UV	0.0	0.0	0.0
Boreal CLIM	UV	0.00099	0.00052	0.00067
Canada	UV	0.0006	0.00098	0.0013
East CLIM	UV	-0.0005	-0.0001	-0.00017
Great Lakes CLIM	UV	0.0	-0.0016	-0.00078
Mt West CLIM	UV	-0.00071	-0.0016	-0.0024
North Plains CLIM	UV	-0.0056	-0.0026	-0.0051
Pacific North West CLIM	UV	-0.0025	0.0014	-0.00031
Prairie CLIM	UV	0.0004	0.00097	0.00054
United States of America East	UV	5.9e-05	0.00076	0.0005
United States of America West	UV	0.00086	-0.0014	-0.00047
full_domain	UV	0.0011	0.00071	0.0011

rmse nat_530a15_Mo6rc9b / nat_530a15_mo6		20220101 / 20220330		
		00z	12z	All
Arctic All CLIM	UV	0.0	0.00075	-0.00091
Boreal CLIM	UV	0.001	0.0	0.00078
Canada	UV	0.0	-0.0017	-0.00044
East CLIM	UV	0.0	-0.00051	0.0
Great Lakes CLIM	UV	0.00058	0.0	-4.4e-06
Mt West CLIM	UV	0.0018	0.0	0.0
North Plains CLIM	UV	-0.0013	0.0	-0.0015
Pacific North West CLIM	UV	0.0	0.0017	0.0014
Prairie CLIM	UV	0.0	0.0	0.0
United States of America East	UV	0.0	-0.00032	0.0
United States of America West	UV	-0.0011	0.0	-0.00083
full_domain	UV	6.8e-05	-0.00066	0.0

'nil'

P3_v5.5.0-rc9

P3 v5.3.6 2MOM-noLF

EMET (H22), TCC 2MOM-LF*

fbi_clrfew 0 - 2		20220101 / 20220330		
		00z	12z	All
Arctic All CLIM	NT	-0.011	-0.0068	-0.0059
	TCC	0.00043	-0.00076	0.0
Boreal CLIM	NT	-0.013	0.0	-0.011
	TCC	0.0	0.0019	-0.00051
Canada	NT	-0.021	-0.0084	-0.021
	TCC	0.0013	0.0016	-0.00031
East CLIM	NT	-0.0048	-0.0017	-0.0055
	TCC	0.0	0.0	0.0
Great Lakes CLIM	NT	-0.0096	-0.0053	-0.011
	TCC	0.0031	0.0	-0.0016
Mt West CLIM	NT	-0.018	-0.015	-0.022
	TCC	0.0	0.0	0.0045
North Plains CLIM	NT	-0.013	0.0	-0.0078
	TCC	0.00099	0.0014	0.00067
Pacific North West CLIM	NT	-0.01	0.0	-0.015
	TCC	0.0	-0.002	0.00038
Prairie CLIM	NT	-0.0056	-0.0015	-0.0058
	TCC	0.0017	0.0	0.00087
United States of America East	NT	-0.0078	-0.0064	-0.0088
	TCC	0.0	0.0	0.0
United States of America West	NT	-0.011	-0.0078	-0.011
	TCC	0.0059	0.0	0.0
full_domain	NT	-0.012	-0.011	-0.013
	TCC	0.0	0.00078	0.0

fbi_sct 3 - 4		20220101 / 20220330		
		00z	12z	All
Arctic All CLIM	NT	0.015	0.015	0.011
	TCC	0.0	0.0	0.0
Boreal CLIM	NT	6.2e-05	0.0	-0.014
	TCC	0.0	0.0091	0.0064
Canada	NT	0.0031	-0.0045	-0.0096
	TCC	0.0	0.0002	0.0
East CLIM	NT	0.0	-0.015	-0.024
	TCC	0.0	-0.0077	0.00014
Great Lakes CLIM	NT	-0.017	0.0	0.0
	TCC	0.0	0.0	0.0072
Mt West CLIM	NT	-0.13	0.0	-0.081
	TCC	0.0	0.012	0.0
North Plains CLIM	NT	-0.02	0.0	0.0
	TCC	0.0	-0.012	0.0023
Pacific North West CLIM	NT	0.0	0.0	0.0
	TCC	0.0	0.0	0.0
Prairie CLIM	NT	0.0	0.0	0.0
	TCC	0.0	0.0	0.0
United States of America East	NT	-0.022	-0.014	-0.022
	TCC	0.0	-0.0076	-0.0032
United States of America West	NT	-0.066	-0.035	-0.03
	TCC	0.0	0.0	0.0
full_domain	NT	-0.014	-0.013	-0.015
	TCC	0.0	-0.0046	0.0

fbi_bkn ovc 5 - 8		20220101 / 20220330		
		00z	12z	All
Arctic All CLIM	NT	-0.0032	0.0	-0.00055
	TCC	-0.00043	-0.0008	-5.4e-05
Boreal CLIM	NT	0.0	0.0	0.0
	TCC	-0.00038	0.0	0.0
Canada	NT	0.00041	0.00045	0.0
	TCC	-0.00045	0.0	0.0
East CLIM	NT	0.0	0.0	0.0
	TCC	-0.00061	-0.0014	-0.001
Great Lakes CLIM	NT	0.0	-0.00023	0.00013
	TCC	-0.00059	-0.00038	-0.0005
Mt West CLIM	NT	-0.0021	0.0018	-0.0017
	TCC	-0.00081	0.0	-0.0013
North Plains CLIM	NT	-0.00032	0.0	0.0
	TCC	-0.0015	-0.00089	-0.00069
Pacific North West CLIM	NT	-0.00063	0.0	-0.0015
	TCC	0.0	0.0	0.0
Prairie CLIM	NT	0.0	0.0	-0.00043
	TCC	0.0	-0.00074	-0.00091
United States of America East	NT	0.0	-0.00057	0.0
	TCC	-0.00086	-0.0015	-0.0015
United States of America West	NT	0.0003	0.0	0.0
	TCC	-0.0011	-0.00078	-0.00097
full_domain	NT	0.0	0.0	0.0
	TCC	-0.00067	-0.00085	-0.0011

Not sure how to interpret those scores.

Seems more neutral than 2MOM-noLF.

'nil'

EMET (H22), PR6 & PR24 2MOM-LF

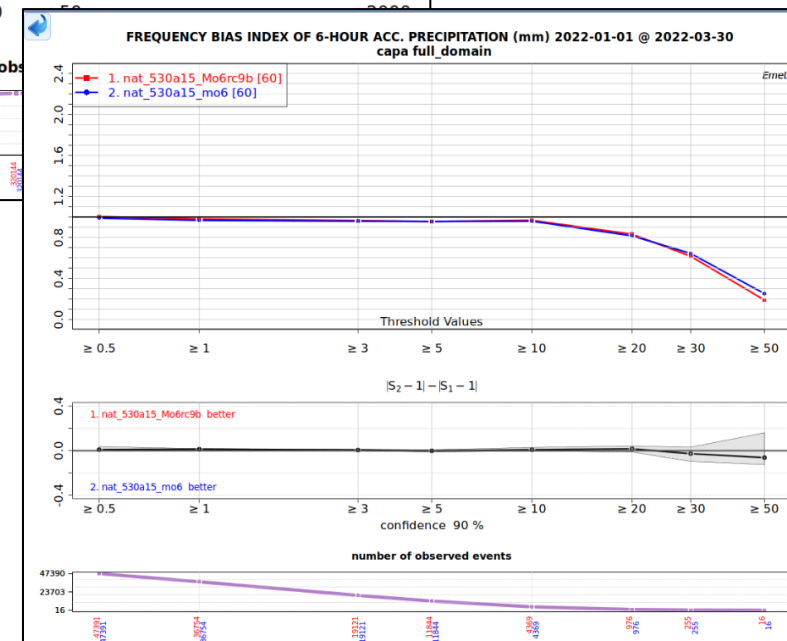
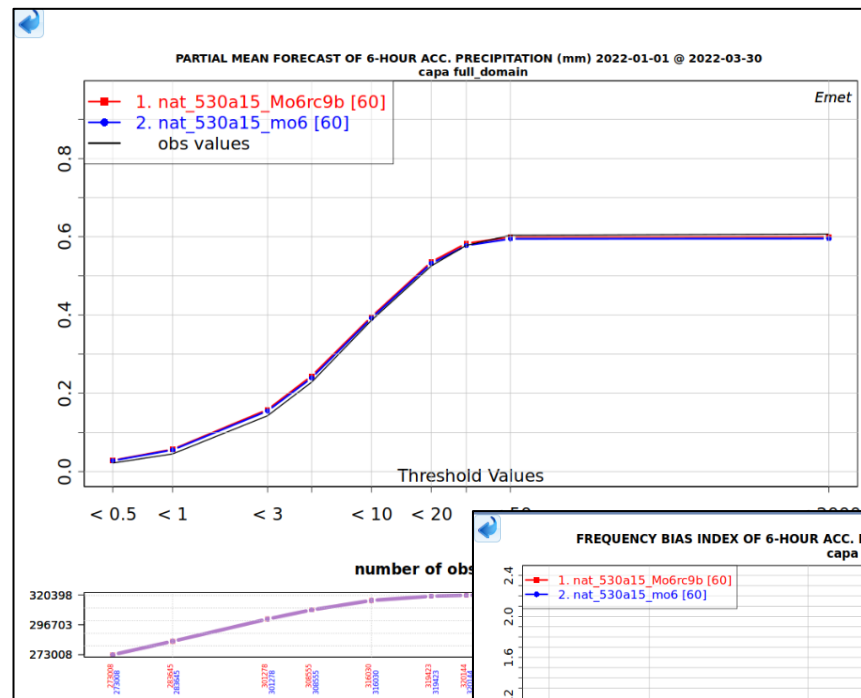
P3_v5.5.0-rc9

P3 v5.3.6 2MOM-noLF

dpmean				
dpmean				
nat_530a15_Mo6rc9b / nat_530a15_mo6				
		00z	12z	All
Arctic All CLIM	PR24	0.0	0.0	0.0
	PR6	-0.00042	-0.00073	-0.00085
Boreal CLIM	PR24	0.0	0.0	0.0
	PR6	-0.00026	0.0	0.0
Canada	PR24	-0.014	-0.013	-0.013
	PR6	-0.0041	-0.0031	-0.004
East CLIM	PR24	0.0	0.0052	0.0022
	PR6	0.0011	0.0011	0.0011
Great Lakes CLIM	PR24	0.015	-0.0054	-0.0056
	PR6	-0.00033	0.0	-9.8e-05
Mt West CLIM	PR24	-0.00048	-0.012	-0.0058
	PR6	-0.00082	-0.00043	-0.00059
North Plains CLIM	PR24	-0.00039	0.0	-0.00025
	PR6	-6.9e-05	0.0	0.0
Pacific North West CLIM	PR24	-0.06	0.017	-0.014
	PR6	-0.021	-0.018	-0.019
Prairie CLIM	PR24	0.0	-0.01	-0.0059
	PR6	0.00045	-6.7e-05	0.00032
United States of America East	PR24	0.0	0.0039	0.0044
	PR6	-0.00036	-0.00026	-0.00062
United States of America West	PR24	-0.051	0.017	-0.011
	PR6	-0.0053	-0.0076	-0.0067
full_domain	PR24	0.016	0.01	0.016
	PR6	-0.002	-0.00049	-0.0016

ETS				
ETS				
nat_530a15_Mo6rc9b / nat_530a15_mo6				
		00z	12z	All
Arctic All CLIM	PR24	0.0	0.0	0.0
	PR6	-0.0021	0.0	-0.00097
Boreal CLIM	PR24	0.02	0.0	0.0
	PR6	-0.0032	0.0	0.0091
Canada	PR24	0.0	0.0	0.0
	PR6	0.0	0.0058	-0.0017
East CLIM	PR24	0.0	0.00075	0.0
	PR6	-0.00078	-0.0061	-0.0033
Great Lakes CLIM	PR24	0.0053	-0.0032	-0.0046
	PR6	0.0	0.0	0.0
Mt West CLIM	PR24	0.0	-0.0018	0.0036
	PR6	-0.0023	-0.0042	-0.0024
North Plains CLIM	PR24	0.0	0.0	0.0
	PR6	0.0	0.0	0.0039
Pacific North West CLIM	PR24	0.0	0.0	0.0
	PR6	-0.0032	0.0	0.0
Prairie CLIM	PR24	0.002	-0.0015	-3.3e-05
	PR6	-0.0023	-0.001	-0.0048
United States of America East	PR24	0.001	0.0	0.0
	PR6	-0.00051	-0.0033	-0.0011
United States of America West	PR24	0.0	0.0	0.0
	PR6	0.0	0.0	0.0
full_domain	PR24	0.0012	0.0	0.00056
	PR6	-0.0007	-0.0035	-0.0027

FBI				
FBI				
nat_530a15_Mo6rc9b / nat_530a15_mo6				
		00z	12z	All
Arctic All CLIM	PR24	0.0	0.0	0.0
	PR6	0.0049	-0.033	-0.024
Boreal CLIM	PR24	-0.0048	0.0	-0.0014
	PR6	0.0	0.0	0.007
Canada	PR24	0.0028	-0.033	-0.0041
	PR6	-0.0042	0.0087	0.0085
East CLIM	PR24	0.0	-0.0036	-0.0092
	PR6	0.0036	-0.021	-0.013
Great Lakes CLIM	PR24	-0.17	-0.0025	-0.0033
	PR6	-0.011	-0.0078	-0.0064
Mt West CLIM	PR24	-0.021	-0.014	0.0
	PR6	0.0	-0.0074	-0.018
North Plains CLIM	PR24	0.0	0.0	0.0
	PR6	0.0	0.0	0.0
Pacific North West CLIM	PR24	-0.0056	0.02	0.013
	PR6	-0.018	-0.0084	-0.017
Prairie CLIM	PR24	0.0013	0.0036	0.0
	PR6	0.0075	0.0	0.00089
United States of America East	PR24	0.0	-0.01	-0.011
	PR6	0.0022	-0.012	-0.0098
United States of America West	PR24	-0.0016	0.0027	0.0086
	PR6	-0.0042	-0.0053	-0.0044
full_domain	PR24	0.0044	0.002	0.004
	PR6	0.0038	0.0032	0.0017



POD				
POD				
nat_530a15_Mo6rc9b / nat_530a15_mo6				
		00z	12z	All
Arctic All CLIM	PR24	0.0	0.0	0.0
	PR6	0.0	0.0	0.0016
Boreal CLIM	PR24	0.016	0.0	0.0
	PR6	-0.0055	0.0	0.01
Canada	PR24	0.0057	-0.014	0.0021
	PR6	0.0023	0.0097	0.0066
East CLIM	PR24	0.0	-0.00085	-0.00069
	PR6	-0.0003	-0.0092	-0.0051
Great Lakes CLIM	PR24	0.0027	0.0014	0.0015
	PR6	-0.011	0.0018	0.0013
Mt West CLIM	PR24	0.0	0.0	0.0
	PR6	0.0	-0.0068	-0.0037
North Plains CLIM	PR24	0.0	0.0	0.0
	PR6	0.0	0.0	0.0
Pacific North West CLIM	PR24	0.0049	0.0099	0.011
	PR6	0.0036	0.0053	0.0049
Prairie CLIM	PR24	0.0	0.00075	0.0
	PR6	0.0022	0.0	-0.0064
United States of America East	PR24	0.0	-0.00046	-0.003
	PR6	-0.00025	-0.0055	0.0
United States of America West	PR24	7.4e-05	0.011	0.012
	PR6	0.0	0.0095	0.0015
full_domain	PR24	0.0017	0.00022	0.002
	PR6	0.0011	0.0024	-0.0014

FAR				
FAR				
nat_530a15_Mo6rc9b / nat_530a15_mo6				
		00z	12z	All
Arctic All CLIM	PR24	0.0	0.0	0.0
	PR6	-0.0017	0.0	-0.0014
Boreal CLIM	PR24	0.077	0.0017	0.0
	PR6	0.0	0.0	-0.013
Canada	PR24	0.0	-0.00072	-0.001
	PR6	-0.0039	-0.02	-0.0029
East CLIM	PR24	0.0	0.0	0.0
	PR6	0.0031	0.0	-0.00082
Great Lakes CLIM	PR24	0.0	-0.0021	-0.0077
	PR6	-0.0012	-0.002	-0.0017
Mt West CLIM	PR24	0.0	-0.0059	-0.0019
	PR6	0.0025	0.0045	0.0059
North Plains CLIM	PR24	0.0	0.025	0.0
	PR6	0.0	0.0	0.008
Pacific North West CLIM	PR24	0.0094	-0.008	-0.011
	PR6	-0.0069	-0.0041	-0.0077
Prairie CLIM	PR24	0.0	-0.0027	-0.0022
	PR6	0.0	0.0	0.00095
United States of America East	PR24	0.0	0.0	0.0
	PR6	0.0018	-0.00072	-0.0034
United States of America West	PR24	0.0093	-0.011	-0.0098
	PR6	-0.0014	-0.005	-0.0038
full_domain	PR24	-0.0014	-0.0023	-0.003
	PR6	-0.0026	-0.001	-0.0054

H22: winter 2022

1.2MOM-noLF

2.2MOM-LF

3.3MOM-LF

No big difference between 'bourge'
and 'nil'

‘bourge’

EMET (H22), TT & TD 2MOM-LF*

P3_v5.5.0-rc9

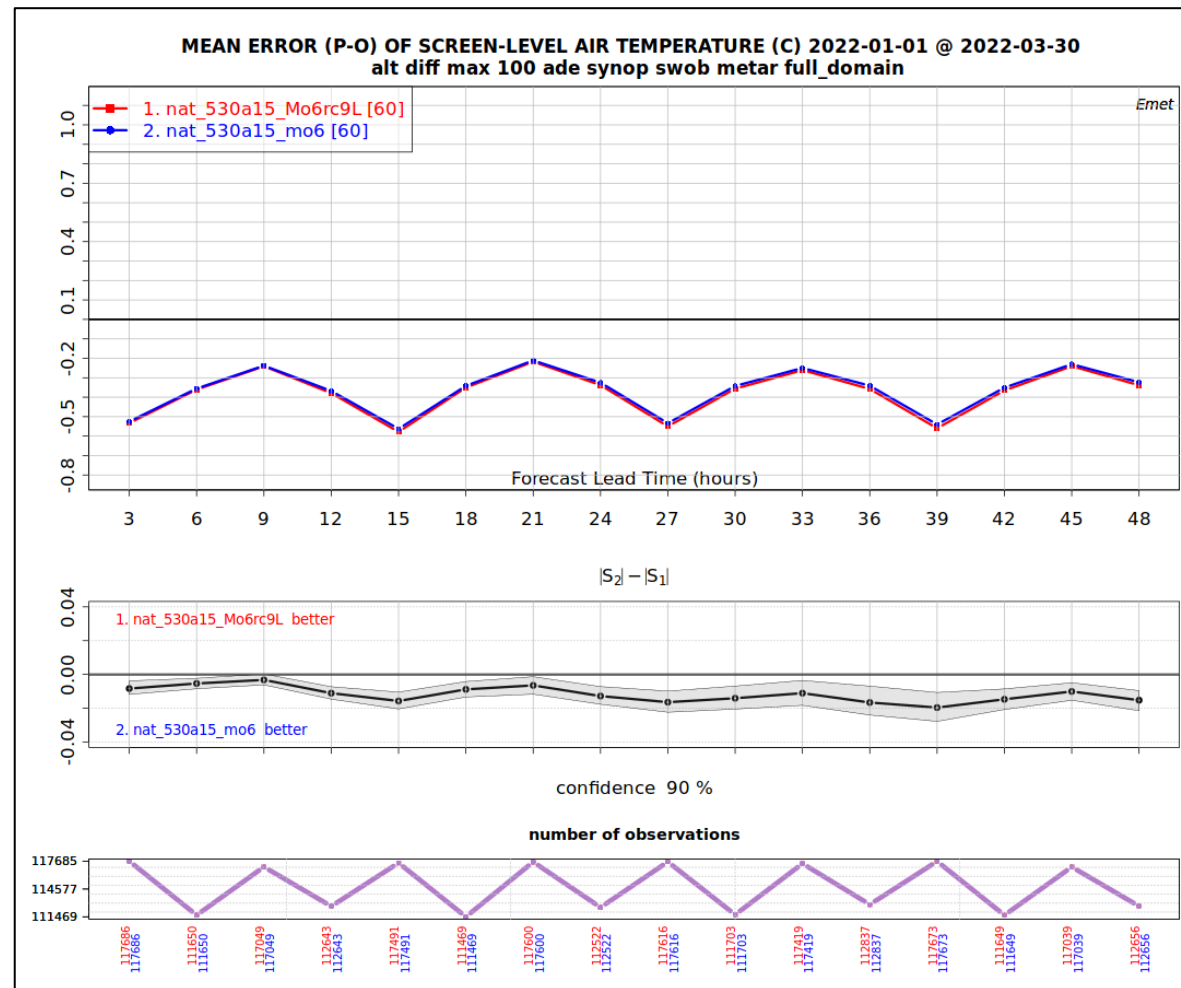
P3 v5.3.6 2MOM-noLF

Expected results!
Same signal as for
2MOM-noLF

Very similar to 2MOM-
noLF also for RMSE and
STDEV (not shown)

bias				
bias		20220101 / 20220330		
nat_530a15_Mo6rc9L / nat_530a15_mo6		00z	12z	All
Arctic All CLIM	TD	0.0041	0.0052	0.0048
	TT	0.0035	0.0064	0.0068
Boreal CLIM	TD	0.0088	-0.0025	0.0015
	TT	0.0071	0.0032	-0.00076
Canada	TD	0.00032	0.0097	0.00028
	TT	-0.014	-0.0015	-0.0036
East CLIM	TD	0.00093	0.0013	0.0013
	TT	-0.0057	-0.0045	-0.0024
Great Lakes CLIM	TD	0.0028	0.0	0.0
	TT	-0.0061	-0.00026	-0.0015
Mt West CLIM	TD	-0.065	-0.055	-0.07
	TT	-0.044	-0.041	-0.04
North Plains CLIM	TD	0.017	0.023	0.027
	TT	-0.0096	0.0049	0.0043
Pacific North West CLIM	TD	-0.0078	-0.0078	-0.0061
	TT	-0.027	-0.03	-0.03
Prairie CLIM	TD	0.0	0.0	0.0
	TT	-0.0011	-0.0041	0.0
United States of America East	TD	0.0	0.00071	0.00069
	TT	-0.0055	-0.0064	-0.0026
United States of America West	TD	-0.027	-0.013	-0.016
	TT	-0.028	-0.027	-0.021
full_domain	TD	-0.0092	-0.0074	-0.0055
	TT	-0.014	-0.012	-0.012

TT-bias (full domain)

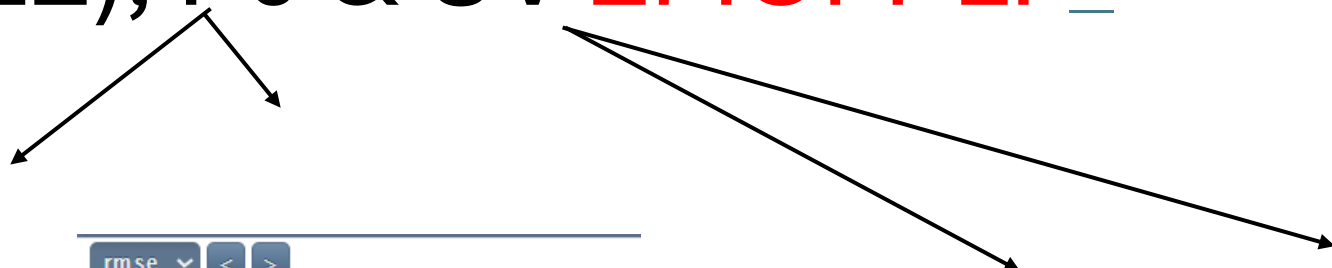


'bourge'

P3_v5.5.0-rc9

P3 v5.3.6 2MOM-noLF

EMET (H22), P0 & UV 2MOM-LF*



bias		20220101 / 20220330		
nat_530a15_Mo6rc9L / nat_530a15_mo6		00z	12z	All
Arctic All CLIM	P0	0.0013	0.0016	0.00099
Boreal CLIM	P0	0.011	0.013	0.012
Canada	P0	0.012	0.015	0.013
East CLIM	P0	0.017	0.023	0.02
Great Lakes CLIM	P0	0.004	0.014	0.011
Mt West CLIM	P0	0.0028	0.012	0.0066
North Plains CLIM	P0	0.013	0.016	0.015
Pacific North West CLIM	P0	-0.011	-0.01	-0.0089
Prairie CLIM	P0	0.0078	0.009	0.01
United States of America East	P0	0.0073	0.0086	0.011
United States of America West	P0	-0.02	-0.017	-0.019
full_domain	P0	0.012	0.015	0.013

rmse		20220101 / 20220330		
nat_530a15_Mo6rc9L / nat_530a15_mo6		00z	12z	All
Arctic All CLIM	P0	0.00064	0.00083	0.00085
Boreal CLIM	P0	0.0026	0.0041	0.0038
Canada	P0	0.0021	0.0046	0.0037
East CLIM	P0	0.0013	0.0016	0.0028
Great Lakes CLIM	P0	0.00067	0.001	0.0019
Mt West CLIM	P0	-0.0009	-6.5e-05	6.4e-05
North Plains CLIM	P0	0.0016	0.004	0.0033
Pacific North West CLIM	P0	-0.00044	0.0	-0.00055
Prairie CLIM	P0	0.0015	0.002	0.0029
United States of America East	P0	0.00082	0.0024	0.0015
United States of America West	P0	-0.0027	-0.0015	-0.0019
full_domain	P0	0.0019	0.0039	0.0033

20220101 / 20220330		
00z	12z	All
-0.00062	0.00072	-0.00012
0.00078	0.0019	0.0011
0.00011	8.4e-05	0.00092
-0.00071	-0.00024	-0.00031
0.0	-0.0027	-0.001
-0.0017	-0.0018	-0.0013
-0.005	-0.0033	-0.0051
-0.0022	-0.00061	-3.9e-05
2.4e-05	0.00098	-3.9e-05
0.00019	0.00067	0.00056
0.00045	-0.0027	-0.0018
0.00033	0.0	0.00093

20220101 / 20220330		
00z	12z	All
0.0	0.001	-4.2e-05
0.0013	-0.00078	6.7e-05
0.0	-0.0017	0.00011
0.0	0.0	0.0
0.00064	0.00012	0.0
0.00036	0.0	0.0012
-0.001	0.0	-0.001
-0.0017	0.0	-0.0013
0.0	0.0	0.0
0.0	0.0	0.0
-0.00085	0.0	0.00034
0.0005	-0.00066	0.00012

'bourne'

EMET (H22), PR6 & PR24 2MOM-LF*

P3_v5.5.0-rc9

P3 v5.3.6 2MOM-noLF

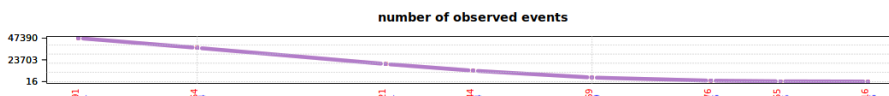
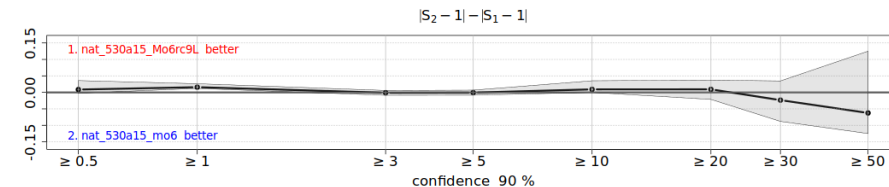
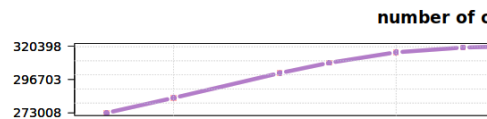
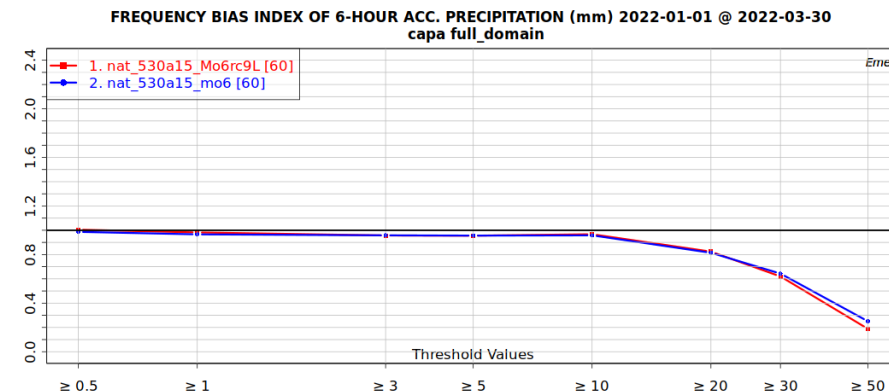
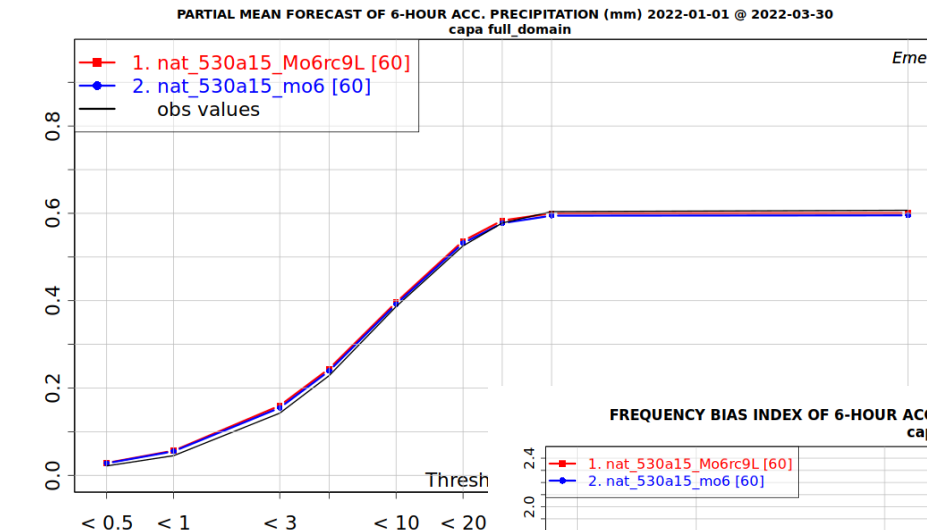
dpmean				
pmean		20220101 / 20220330		
nat_530a15_Mo6rc9L / nat_530a15_mo6		00z	12z	All
Arctic All CLIM	PR24	0.0	0.0	0.0
	PR6	0.0076	-0.012	-0.026
Boreal CLIM	PR24	0.0	0.0	-0.0029
	PR6	-0.0063	0.0017	-0.0025
Canada	PR24	0.018	-0.0029	-0.004
	PR6	-0.0041	-0.00022	8.3e-05
East CLIM	PR24	-0.0045	0.0046	0.0022
	PR6	0.0099	-0.021	-0.0042
Great Lakes CLIM	PR24	-0.21	-0.0029	-0.0034
	PR6	-0.0035	-0.0069	-0.0064
Mt West CLIM	PR24	-0.0046	-0.012	0.0
	PR6	0.005	0.004	-0.018
North Plains CLIM	PR24	0.0	0.0	0.0
	PR6	0.0	0.0	0.0
Pacific North West CLIM	PR24	0.0081	0.02	0.014
	PR6	-0.011	-0.0067	-0.017
Prairie CLIM	PR24	0.0	-0.0016	0.0
	PR6	0.012	0.0	0.0042
United States of America East	PR24	-0.0029	-0.0055	-0.0095
	PR6	0.0021	-0.015	-0.01
United States of America West	PR24	0.011	0.0082	0.0064
	PR6	0.0	-0.0069	-0.037
full_domain	PR24	0.0036	0.011	0.0085
	PR6	0.0025	0.0021	0.002

ets				
nat_530a15_Mo6rc9L / nat_530a15_mo6		20220101 / 20220330		
		00z	12z	All
Arctic All CLIM	PR24	0.0	0.0	0.0
	PR6	0.0	0.0	0.0
Boreal CLIM	PR24	0.0	-0.0021	-0.0012
	PR6	0.0013	0.0	0.0
Canada	PR24	0.0016	0.0	0.0013
	PR6	-0.0076	0.0054	0.0
East CLIM	PR24	-0.00089	-0.0027	-0.0014
	PR6	0.0	-0.0043	0.0
Great Lakes CLIM	PR24	0.0	-0.0023	-0.011
	PR6	0.0	0.0022	-0.0013
Mt West CLIM	PR24	0.0	0.0	0.0
	PR6	0.0029	0.0	0.002
North Plains CLIM	PR24	0.0	0.02	-0.0022
	PR6	0.014	0.0	0.0053
Pacific North West CLIM	PR24	0.0	0.0	0.0
	PR6	-0.0034	0.0039	-0.0026
Prairie CLIM	PR24	0.0	0.0011	0.0012
	PR6	0.0028	0.0014	0.0014
United States of America East	PR24	-0.00053	-0.0013	-0.002
	PR6	0.0023	0.00017	0.0
United States of America West	PR24	0.0	-0.001	0.0
	PR6	0.007	0.0	0.0038
full_domain	PR24	0.0015	-0.001	-0.0016

fbi				
nat_530a15_Mo6rc9L / nat_530a15_mo6		20220101 / 20220330		
		00z	12z	All
Arctic All CLIM	PR24	0.0	0.0	0.0
	PR6	0.0076	-0.012	-0.026
Boreal CLIM	PR24	0.0	0.0	-0.0029
	PR6	-0.0063	0.0017	-0.0025
Canada	PR24	0.018	-0.0029	-0.004
	PR6	-0.0041	-0.00022	8.3e-05
East CLIM	PR24	-0.0045	0.0046	0.0022
	PR6	0.0099	-0.021	-0.0042
Great Lakes CLIM	PR24	-0.21	-0.0029	-0.0034
	PR6	-0.0035	-0.0069	-0.0064
Mt West CLIM	PR24	-0.0046	-0.012	0.0
	PR6	0.005	0.004	-0.018
North Plains CLIM	PR24	0.0	0.0	0.0
	PR6	0.0	0.0	0.0
Pacific North West CLIM	PR24	0.0081	0.02	0.014
	PR6	-0.011	-0.0067	-0.017
Prairie CLIM	PR24	0.0	-0.0016	0.0
	PR6	0.012	0.0	0.0042
United States of America East	PR24	-0.0029	-0.0055	-0.0095
	PR6	0.0021	-0.015	-0.01
United States of America West	PR24	0.011	0.0082	0.0064
	PR6	0.0	-0.0069	-0.037
full_domain	PR24	0.0036	0.011	0.0085
	PR6	0.0025	0.0021	0.002

pod				
nat_530a15_Mo6rc9L / nat_530a15_mo6		20220101 / 20220330		
		00z	12z	All
Arctic All CLIM	PR24	0.0	0.0	0.0
	PR6	-0.0018	-0.016	-0.0094
Boreal CLIM	PR24	0.0	-0.0013	0.0
	PR6	0.0053	0.0015	0.0021
Canada	PR24	0.0025	0.00084	0.001
	PR6	0.0023	0.011	0.0024
East CLIM	PR24	-0.0013	0.0005	-0.0016
	PR6	0.0022	-0.0079	-0.003
Great Lakes CLIM	PR24	-0.0013	-0.0005	-0.00018
	PR6	0.0	0.0039	0.0
Mt West CLIM	PR24	0.0	0.0045	0.0043
	PR6	0.0	0.0	-0.004
North Plains CLIM	PR24	0.0	0.0	-0.0039
	PR6	0.026	0.0	0.013
Pacific North West CLIM	PR24	0.013	0.013	0.012
	PR6	0.0042	0.0041	0.0041
Prairie CLIM	PR24	0.00064	0.0014	0.00056
	PR6	0.0041	0.0	0.0026
United States of America East	PR24	-0.0014	-0.00026	-0.0016
	PR6	0.00016	-0.0033	0.00068
United States of America West	PR24	0.00086	0.012	0.014
	PR6	0.0019	0.012	0.0039
full_domain	PR24	0.0023	0.0037	0.0045
	PR6	0.0022	0.0018	0.0018

far				
nat_530a15_Mo6rc9L / nat_530a15_mo6		20220101 / 20220330		
		00z	12z	All
Arctic All CLIM	PR24	0.0	0.0	0.0
	PR6	0.0	0.0	0.0
Boreal CLIM	PR24	0.006	-0.0098	-0.0091
	PR6	0.0038	0.0067	0.0
Canada	PR24	0.0	-0.00064	0.0
	PR6	-0.013	-0.0016	-0.0037
East CLIM	PR24	0.0018	0.0	-0.00065
	PR6	0.0041	0.0	-0.00047
Great Lakes CLIM	PR24	0.0	0.0	-0.016
	PR6	-0.0019	-0.002	-0.0029
Mt West CLIM	PR24	0.0	-0.0052	0.0
	PR6	0.012	-0.0023	0.0
North Plains CLIM	PR24	0.0	0.033	0.017
	PR6	0.019	0.0	0.007
Pacific North West CLIM	PR24	0.0051	-0.0079	-0.008
	PR6	-0.0083	-0.0048	-0.0084
Prairie CLIM	PR24	0.0	0.0	0.0
	PR6	0.0	0.0077	0.0
United States of America East	PR24	0.0	-0.0017	-0.0014
	PR6	0.0045	0.0	0.00046
United States of America West	PR24	-0.0053	-0.006	-0.0082
	PR6	-0.0031	-0.0046	-0.0042
full_domain	PR24	-0.0017	-0.0046	-0.0047
	PR6	-0.0034	-0.0015	-0.0023



H22: winter 2022

1.2MOM-noLF

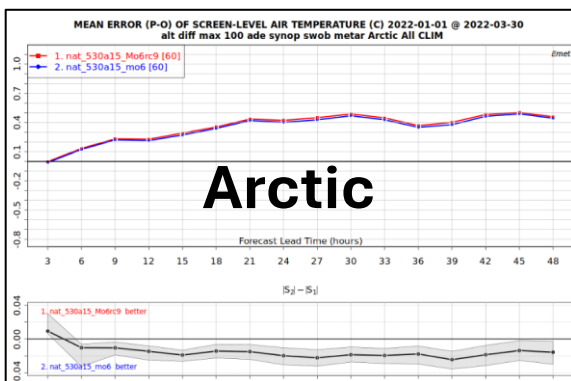
2.2MOM-LF

3.3MOM-LF

EMET (H22), TT & TD 3MOM-LF*

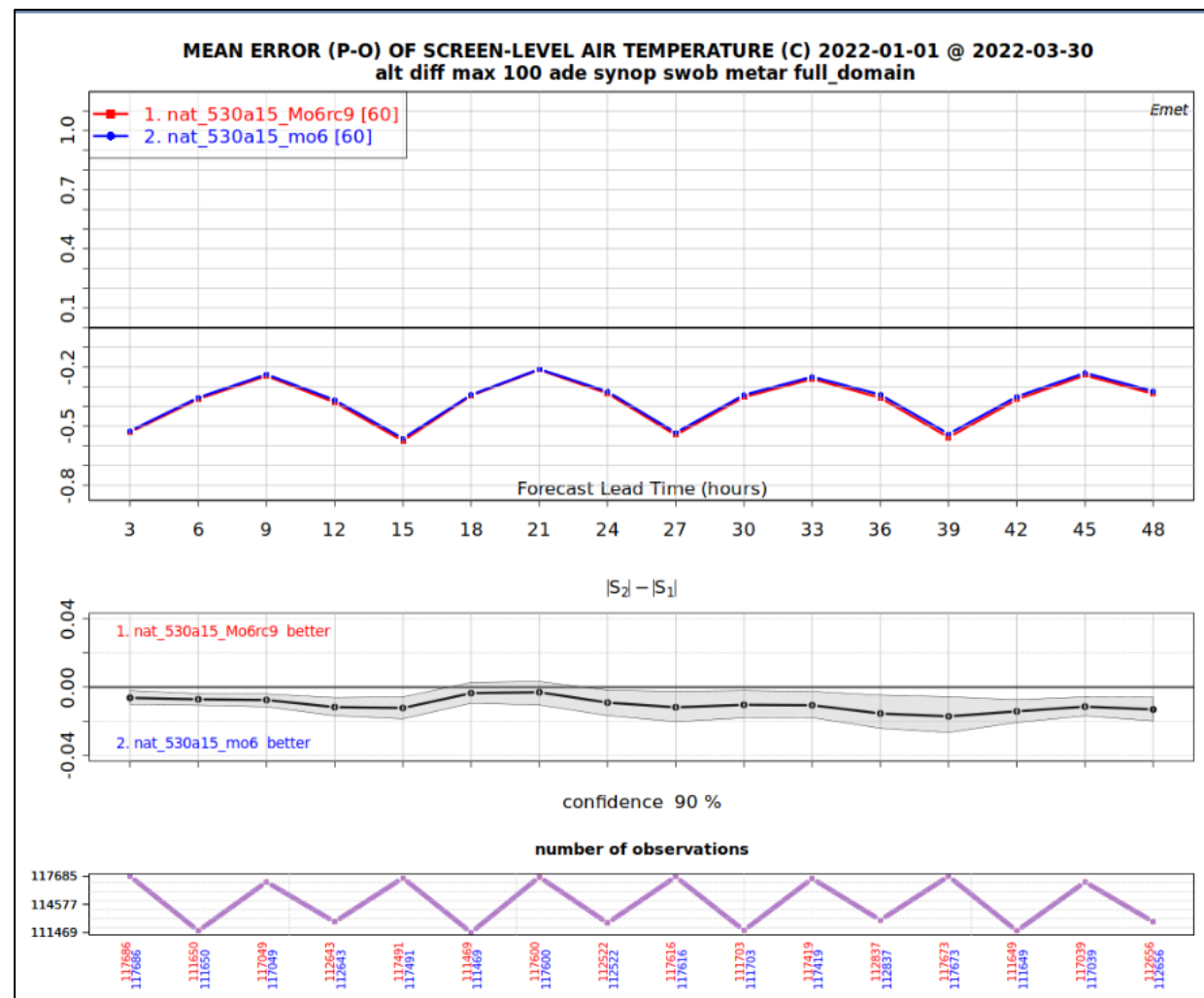
Same signal as for
2MOM-noLF and 2MOM-
LF, except in Arctic:
warmer with 3MOM

Hypothesis:
Smaller small ice
crystals and more liquid
phases



bias		20220101 / 20220330		
nat_530a15_Mo6rc9 / nat_530a15_mo6		00z	12z	All
Arctic All CLIM	TD	-0.034	-0.026	-0.031
	TT	-0.015	-0.012	-0.015
Boreal CLIM	TD	0.0069	0.0048	0.0039
	TT	0.00035	0.00082	0.0017
Canada	TD	0.0015	-0.002	0.0054
	TT	-0.002	0.001	-0.00037
East CLIM	TD	0.0022	0.0021	0.0045
	TT	-0.0018	-0.01	-0.0063
Great Lakes CLIM	TD	0.0011	0.0023	0.0
	TT	-0.026	-0.014	-0.02
Mt West CLIM	TD	-0.037	-0.023	-0.031
	TT	-0.023	-0.021	-0.023
North Plains CLIM	TD	-0.0014	0.0	0.0
	TT	0.0	0.0	0.0
Pacific North West CLIM	TD	0.022	0.011	0.021
	TT	-0.027	-0.024	-0.031
Prairie CLIM	TD	0.0	0.00022	-0.00055
	TT	-0.0061	0.0	-0.0034
United States of America East	TD	0.0011	0.0018	0.0014
	TT	-0.013	-0.0087	-0.01
United States of America West	TD	-0.0055	-0.0044	0.00053
	TT	-0.019	-0.0091	-0.011
full_domain	TD	0.0011	0.0017	0.0033
	TT	-0.0096	-0.0086	-0.0099

TT-bias (full domain)



'nil'

P3_v5.5.0-rc9

P3 v5.3.6 2MOM-noLF

EMET (H22), P0 & UV 3MOM-LF*

bias		20220101 / 20220330		
nat_530a15_Mo6rc9 / nat_530a15_mo6		00Z	12Z	All
Arctic All CLIM	P0	0.012	0.012	0.012
Boreal CLIM	P0	0.027	0.028	0.028
Canada	P0	0.026	0.028	0.027
East CLIM	P0	0.031	0.035	0.033
Great Lakes CLIM	P0	0.018	0.024	0.021
Mt West CLIM	P0	0.013	0.022	0.014
North Plains CLIM	P0	0.027	0.031	0.029
Pacific North West CLIM	P0	-0.02	-0.018	-0.016
Prairie CLIM	P0	0.018	0.011	0.017
United States of America East	P0	0.01	0.01	0.015
United States of America West	P0	-0.029	-0.027	-0.028
full_domain	P0	0.025	0.027	0.026

rmse		20220101 / 20220330		
nat_530a15_Mo6rc9 / nat_530a15_mo6		00Z	12Z	All
Arctic All CLIM	P0	0.0017	0.003	0.0029
Boreal CLIM	P0	0.0073	0.0074	0.0075
Canada	P0	0.0061	0.0075	0.007
East CLIM	P0	0.0033	0.0059	0.0074
Great Lakes CLIM	P0	0.00072	0.0037	0.0031
Mt West CLIM	P0	-0.0006	0.0023	0.003
North Plains CLIM	P0	0.0026	0.0014	0.0034
Pacific North West CLIM	P0	-0.0017	-0.00088	-0.0019
Prairie CLIM	P0	0.0087	0.0067	0.0068
United States of America East	P0	0.0018	0.0019	0.0015
United States of America West	P0	-0.0049	-0.0037	-0.0054
full_domain	P0	0.0049	0.0061	0.006

bias		20220101 / 20220330		
nat_530a15_Mo6rc9 / nat_530a15_mo6		00Z	12Z	All
Arctic All CLIM	UV	0.0026	0.0026	0.003
Boreal CLIM	UV	0.0	0.0018	0.0
Canada	UV	4.2e-05	-0.0011	0.00015
East CLIM	UV	0.0	-0.0001	0.0
Great Lakes CLIM	UV	0.00094	-0.0016	0.0
Mt West CLIM	UV	-0.0009	0.0	0.0
North Plains CLIM	UV	-0.006	-0.0025	-0.0057
Pacific North West CLIM	UV	-0.0045	0.0	-0.0037
Prairie CLIM	UV	0.0013	0.0	0.0
United States of America East	UV	0.0016	0.0011	0.00081
United States of America West	UV	0.0017	0.0	0.0
full_domain	UV	0.00032	0.0	0.0003

rmse		20220101 / 20220330		
nat_530a15_Mo6rc9 / nat_530a15_mo6		00Z	12Z	All
Arctic All CLIM	UV	0.0	-1.2e-05	7e-05
Boreal CLIM	UV	-0.0018	0.0	-0.0011
Canada	UV	-0.0021	-0.0025	-0.0022
East CLIM	UV	-0.00081	-0.0011	-0.0006
Great Lakes CLIM	UV	0.0	-0.0013	0.0
Mt West CLIM	UV	-0.0002	-0.001	-0.00027
North Plains CLIM	UV	-0.00097	-0.0026	-0.0034
Pacific North West CLIM	UV	-0.0063	-0.0048	-0.0041
Prairie CLIM	UV	-0.00092	-0.00074	-0.0006
United States of America East	UV	-0.00038	-0.00097	-0.00086
United States of America West	UV	-0.0012	-0.00044	-0.0019
full_domain	UV	-0.0011	-0.0015	-0.0016

'nil'

P3_v5.5.0-rc9

P3 v5.3.6 2MOM-noLF

EMET (H22), TCC 3MOM-LF*

fbi_clrfew 0 - 2		20220101 / 20220330		
		00z	12z	All
Arctic All CLIM	NT	0.05	0.027	0.044
	TCC	-0.0068	-0.0045	-0.0062
Boreal CLIM	NT	0.0	5.8e-05	0.01
	TCC	0.0	-0.0061	0.0
Canada	NT	0.018	0.019	0.02
	TCC	0.0	-0.0049	-0.0019
East CLIM	NT	0.0	0.0037	0.003
	TCC	-0.00015	9.4e-05	0.0
Great Lakes CLIM	NT	0.0	-0.0021	-0.0037
	TCC	0.0043	0.013	0.0058
Mt West CLIM	NT	0.0091	0.0	-0.00014
	TCC	-0.0047	-0.0045	-0.0029
North Plains CLIM	NT	0.0013	0.005	0.0074
	TCC	0.0	0.0	-0.0082
Pacific North West CLIM	NT	0.0	0.0	0.0078
	TCC	-0.0053	-0.0013	-0.0053
Prairie CLIM	NT	0.0	0.0	0.0
	TCC	-0.00068	-0.0011	-0.002
United States of America East	NT	-0.00059	0.0034	1.2e-06
	TCC	0.0	0.0	0.0
United States of America West	NT	-0.0049	0.0	0.00092
	TCC	-0.0036	-0.0034	-0.0038
full_domain	NT	0.0026	0.0041	0.0042
	TCC	0.0	-0.0021	-0.0012

fbi_sct 3 - 4		20220101 / 20220330		
		00z	12z	All
Arctic All CLIM	NT	-0.02	-0.014	-0.024
	TCC	0.0	0.0	0.0044
Boreal CLIM	NT	0.0075	0.0	0.016
	TCC	0.0	0.0	-0.0031
Canada	NT	0.0081	-0.0022	0.006
	TCC	0.0011	0.003	0.0
East CLIM	NT	0.0	0.0054	-0.015
	TCC	0.0	-0.0044	-0.0018
Great Lakes CLIM	NT	0.05	0.0	0.027
	TCC	0.0067	0.0094	0.0058
Mt West CLIM	NT	-0.11	-0.03	-0.093
	TCC	-0.013	0.017	0.0017
North Plains CLIM	NT	0.0	0.0	0.0
	TCC	0.0	0.0	0.0
Pacific North West CLIM	NT	0.0	0.0	0.0
	TCC	0.0	0.0	0.0
Prairie CLIM	NT	0.033	0.0054	-0.013
	TCC	0.0	0.0072	0.0
United States of America East	NT	0.0	-0.0045	-0.016
	TCC	0.0035	0.0	0.00012
United States of America West	NT	0.0	0.0	-0.027
	TCC	0.0	0.0021	0.009
full_domain	NT	-0.002	-0.0079	-0.014
	TCC	0.00045	-0.00088	0.0016

fbi_bkn_ovc 5 - 8		20220101 / 20220330		
		00z	12z	All
Arctic All CLIM	NT	0.01	0.01	0.013
	TCC	-0.002	0.0	0.0
Boreal CLIM	NT	0.01	0.0042	0.0088
	TCC	-0.0028	0.00058	-0.001
Canada	NT	0.0074	0.0079	0.0079
	TCC	-0.00098	0.0	-0.00061
East CLIM	NT	0.0	0.0	-6e-05
	TCC	0.0	0.0	0.0
Great Lakes CLIM	NT	0.0	0.00033	0.0
	TCC	0.0015	0.0	0.0036
Mt West CLIM	NT	-0.005	-0.0058	-0.007
	TCC	-0.004	-0.0028	0.0
North Plains CLIM	NT	0.0	0.0052	0.0048
	TCC	-0.0035	0.0	-0.0055
Pacific North West CLIM	NT	0.0	0.0	0.0034
	TCC	-0.00041	-0.0029	-0.002
Prairie CLIM	NT	-0.0045	0.0	-0.0025
	TCC	0.0	0.0	-0.00093
United States of America East	NT	0.0	1.6e-05	-0.0013
	TCC	0.0	0.0	0.0
United States of America West	NT	0.0	-0.0064	0.0
	TCC	-0.0039	0.0	-0.0037
full_domain	NT	0.0011	0.00078	0.0012
	TCC	0.0	-0.0014	0.0

Not sure how to interpret those scores.

Seems a bit better than 2MOM-noLF.

'nil'

P3_v5.5.0-rc9

P3 v5.3.6 2MOM-noLF

EMET (H22), PR6 & PR24 3MOM-LF*

dpmean

dpmean		20220101 / 20220330		
nat_530a15_Mo6rc9 / nat_530a15_mo6		00z	12z	All
Arctic All CLIM	PR24	-0.00077	0.00086	0.00064
	PR6	0.00068	-0.00087	1.9e-05
Boreal CLIM	PR24	-0.007	0.0018	-0.0027
	PR6	-0.0017	0.001	0.0007
Canada	PR24	0.00029	0.00078	0.00074
	PR6	0.00055	-0.0003	0.00047
East CLIM	PR24	-0.025	-0.013	-0.017
	PR6	-0.0028	-0.00088	-0.0011
Great Lakes CLIM	PR24	0.0011	0.013	0.0085
	PR6	0.0002	0.0	0.00016
Mt West CLIM	PR24	0.025	0.0042	0.0013
	PR6	0.0051	0.0012	0.0025
North Plains CLIM	PR24	0.00077	0.0017	0.0014
	PR6	0.003	0.00038	0.0021
Pacific North West CLIM	PR24	-0.023	0.014	0.025
	PR6	-0.013	-0.0067	-0.0096
Prairie CLIM	PR24	-0.064	-0.08	-0.077
	PR6	-0.0067	-0.0012	0.00052
United States of America East	PR24	-0.027	-0.019	-0.021
	PR6	-0.0026	-0.00064	-0.0012
United States of America West	PR24	-0.025	0.0092	0.0093
	PR6	0.0	0.0	0.0
full_domain	PR24	-0.0053	0.00053	-0.0053
	PR6	0.00036	0.00052	0.00032

dpmean

ets

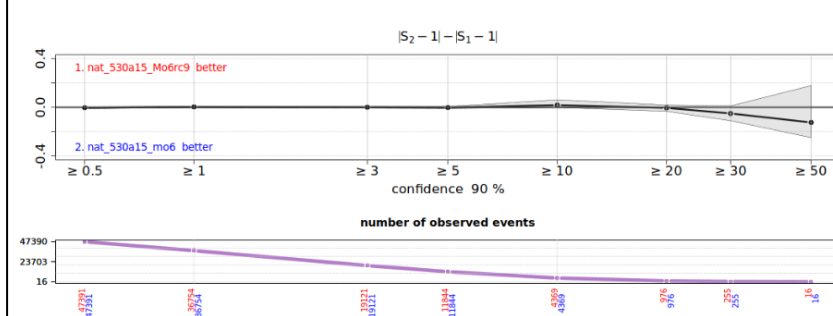
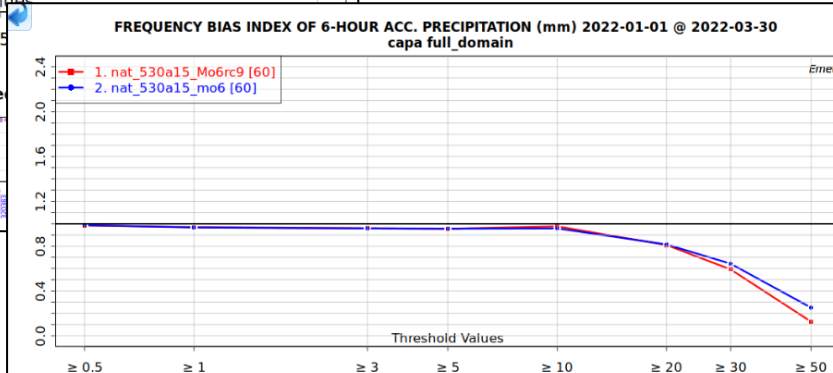
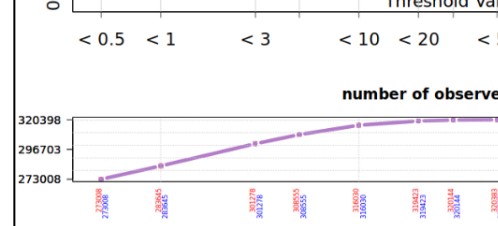
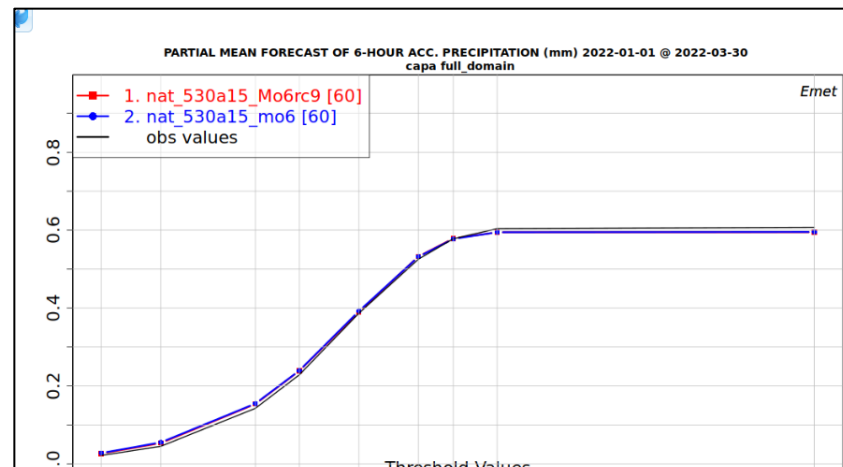
ets		20220101 / 20220330		
nat_530a15_Mo6rc9 / nat_530a15_mo6		00z	12z	All
Arctic All CLIM	PR24	-0.031	0.0	-0.011
	PR6	0.0	0.0	-0.0018
Boreal CLIM	PR24	0.013	0.0024	0.0
	PR6	-0.0075	-0.0015	0.0014
Canada	PR24	0.003	-0.0038	-0.014
	PR6	0.0	0.0044	-0.0016
East CLIM	PR24	-0.0015	-0.0053	0.0
	PR6	0.0	-0.00077	-0.001
Great Lakes CLIM	PR24	-0.0046	-0.002	0.0
	PR6	-0.0019	0.0	-0.0014
Mt West CLIM	PR24	0.0	0.011	0.0099
	PR6	0.0	0.0	0.0
North Plains CLIM	PR24	0.0	0.0	0.0
	PR6	-0.0037	0.0029	0.0
Pacific North West CLIM	PR24	-0.0033	0.0	0.0
	PR6	0.0	0.0	0.0
Prairie CLIM	PR24	0.0	-0.0001	-0.0025
	PR6	-0.0038	-0.0074	-0.0084
United States of America East	PR24	-0.0018	-0.0031	-0.0017
	PR6	0.0	-0.0056	-0.004
United States of America West	PR24	-0.0026	0.0	0.0
	PR6	0.012	-0.002	0.0039
full_domain	PR24	0.0	0.0	0.0
	PR6	0.0	0.0	0.0

ets

fbi

fbi		20220101 / 20220330		
nat_530a15_Mo6rc9 / nat_530a15_mo6		00z	12z	All
Arctic All CLIM	PR24	0.0	0.0	0.0
	PR6	0.0076	-0.048	-0.027
Boreal CLIM	PR24	0.0	-0.022	-0.024
	PR6	-0.018	-0.02	0.0021
Canada	PR24	0.015	0.0013	-0.012
	PR6	0.0	0.0055	0.0024
East CLIM	PR24	-0.012	-0.0072	-0.024
	PR6	-0.005	-0.019	-0.02
Great Lakes CLIM	PR24	-0.01	0.0085	0.0
	PR6	-0.0094	-0.0019	0.0
Mt West CLIM	PR24	-0.016	-0.0069	0.0067
	PR6	0.0083	0.0	0.0044
North Plains CLIM	PR24	0.0	0.0	0.034
	PR6	0.024	0.0	0.014
Pacific North West CLIM	PR24	-0.0016	0.0081	0.013
	PR6	-0.022	-0.0032	-0.017
Prairie CLIM	PR24	-0.04	-0.012	-0.023
	PR6	0.0	-0.0025	-0.0053
United States of America East	PR24	-0.013	-0.0077	-0.012
	PR6	-0.0022	-0.019	-0.02
United States of America West	PR24	-0.0035	0.0035	0.0079
	PR6	-0.0084	0.0	0.0
full_domain	PR24	-0.0076	0.0061	0.0
	PR6	0.0	0.0043	0.0

fbi



pod

pod		20220101 / 20220330		
nat_530a15_Mo6rc9 / nat_530a15_mo6		00z	12z	All
Arctic All CLIM	PR24	0.012	0.0	0.0
	PR6	0.0	-0.018	0.0
Boreal CLIM	PR24	0.0	0.0	0.0
	PR6	-0.01	-0.0024	-0.0014
Canada	PR24	0.0063	0.0	-0.014
	PR6	0.0	0.0058	0.0
East CLIM	PR24	-0.0014	-0.008	-0.0088
	PR6	-0.0014	-0.0085	-0.0068
Great Lakes CLIM	PR24	0.018	-0.0025	0.0
	PR6	0.0	0.0	0.0
Mt West CLIM	PR24	-0.0061	0.012	0.012
	PR6	-0.008	0.0	-0.0025
North Plains CLIM	PR24	-0.018	0.0	-0.0077
	PR6	-0.011	0.0	0.0
Pacific North West CLIM	PR24	0.0023	0.0	0.003
	PR6	0.003	0.016	0.0083
Prairie CLIM	PR24	0.0	-0.0027	-0.012
	PR6	-0.0062	-0.007	-0.0039
United States of America East	PR24	0.0	-0.0083	-0.0093
	PR6	-0.0041	-0.0067	-0.0059
United States of America West	PR24	0.0096	0.019	0.0
	PR6	0.0	0.0	0.0
full_domain	PR24	0.0	-0.0072	-0.0046
	PR6	0.0	0.0	0.0

pod

far

far		20220101 / 20220330		
nat_530a15_Mo6rc9 / nat_530a15_mo6		00z	12z	All
Arctic All CLIM	PR24	-0.03	0.0	-0.012
	PR6	0.0	-0.0028	-0.0015
Boreal CLIM	PR24	0.013	0.0	0.0
	PR6	0.0012	0.016	0.011
Canada	PR24	0.0	-0.0056	-0.0064
	PR6	-0.0048	-0.017	-0.0037
East CLIM	PR24	0.00097	-0.0089	0.0037
	PR6	0.0	-0.0064	0.0016
Great Lakes CLIM	PR24	0.0056	-0.0044	0.0
	PR6	0.0	-0.0014	-0.0015
Mt West CLIM	PR24	0.0	0.0	0.0
	PR6	0.0	0.0	-0.011
North Plains CLIM	PR24	0.0	0.01	0.0
	PR6	0.0	0.0036	0.0
Prairie CLIM	PR24	-0.0045	-0.0084	-0.0044
	PR6	-0.0063	-0.0041	-0.0057
United States of America East	PR24	0.0014	-0.0061	0.0012
	PR6	0.0	-0.0053	-0.0033
United States of America West	PR24	-0.0024	-0.0057	-0.0028
	PR6	0.017	-0.0019	0.0041
full_domain	PR24	0.0044	0.0	0.0
	PR6	-0.0007	-0.0017	-0.0018

far

H22: winter 2022 – precipitation phase

1.2MOM-noLF

2.2MOM-LF

3.3MOM-LF

Note: control uses pcp_type='bourge' → which is Bourgouin diagnostic precipitation types

All other simulations used pcp_type='nil' → P3 determines the precipitation types

Results that will come later:

- Results with the same suite Bourgouin vs. P3 will come later. Should not be very different then two suites (1 with P3 and 1 with Bourgouin)
- A21 and P22 precipitation phase scores.
- 2D maps of some cases.

Emet – precipitation phases

2MOM-noLF

PR>0.01

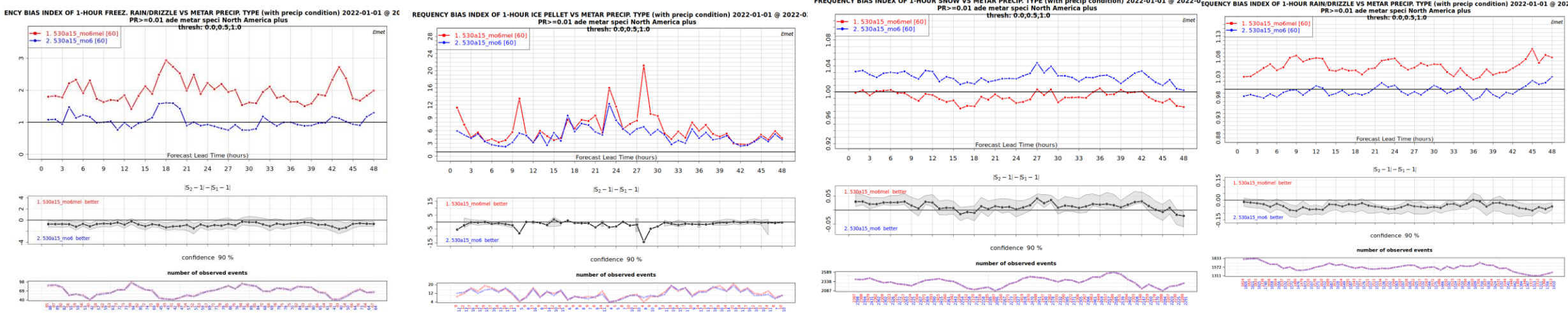
Freezing rain

Ice pellets

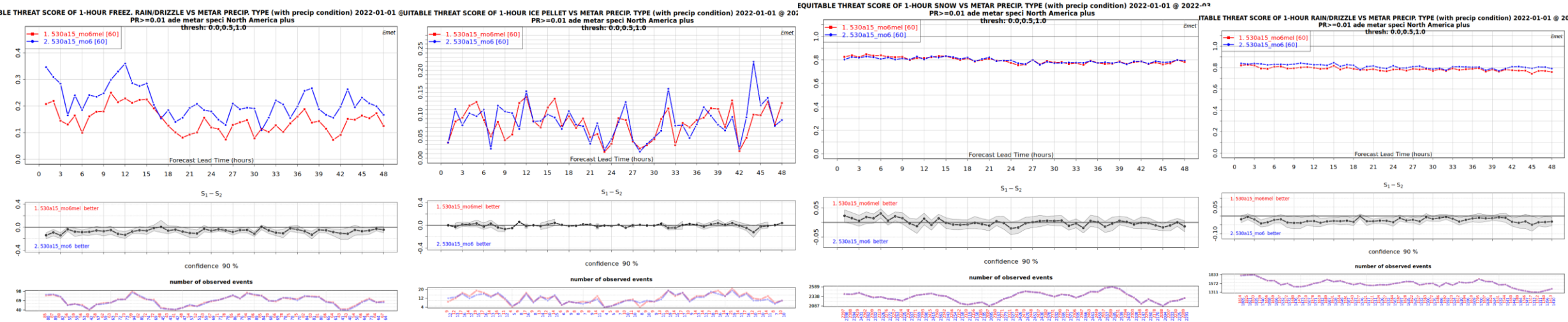
Snow

Rain

FBI



ETS



Emet – precipitation phases

2MOM-noLF

PR>0.01

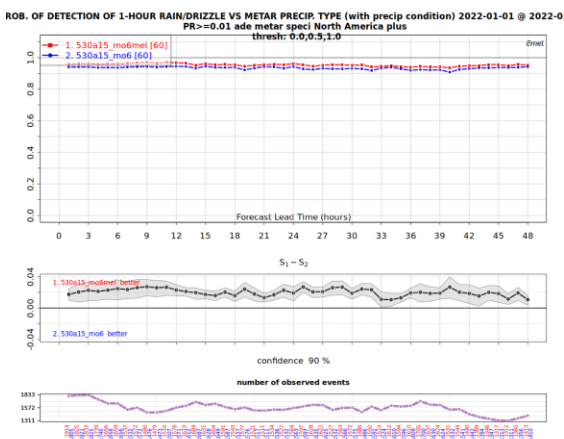
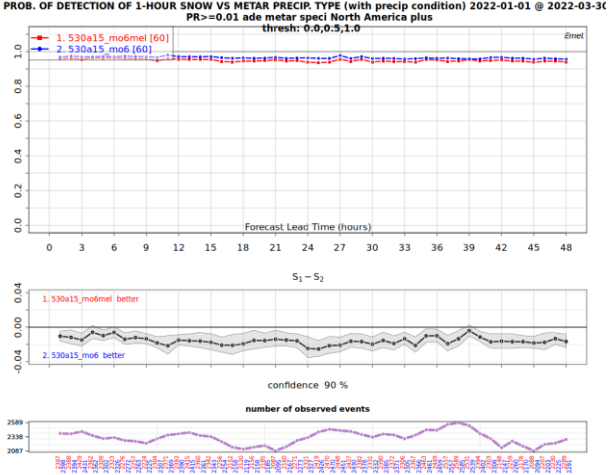
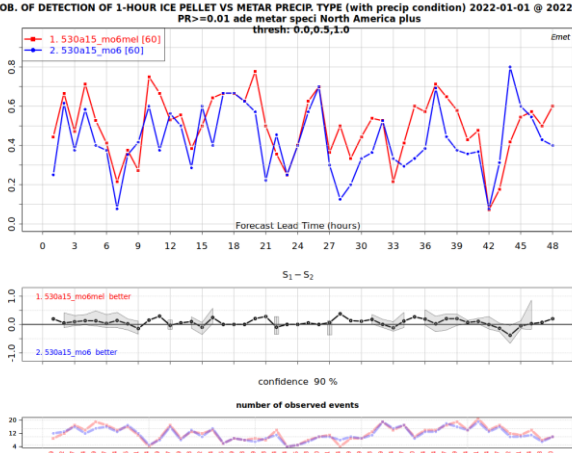
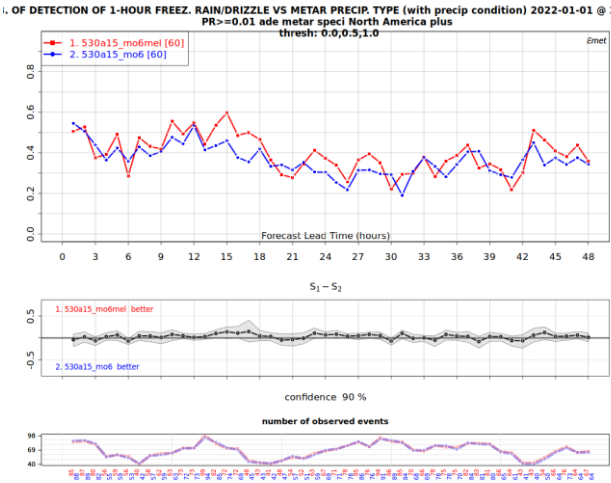
Freezing rain

Ice pellets

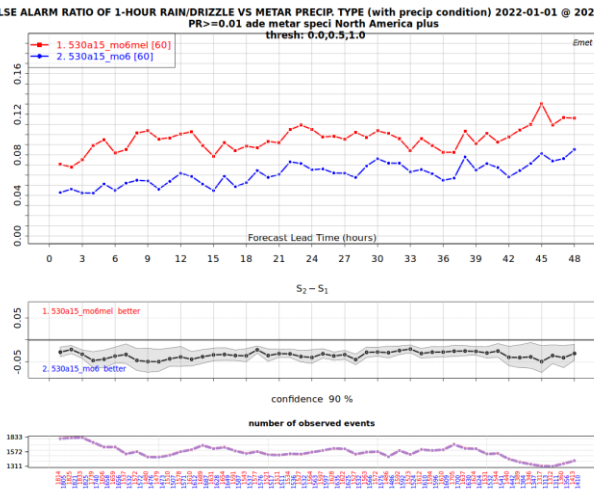
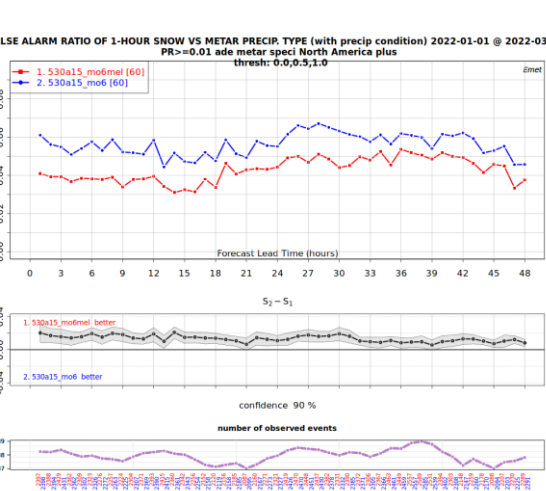
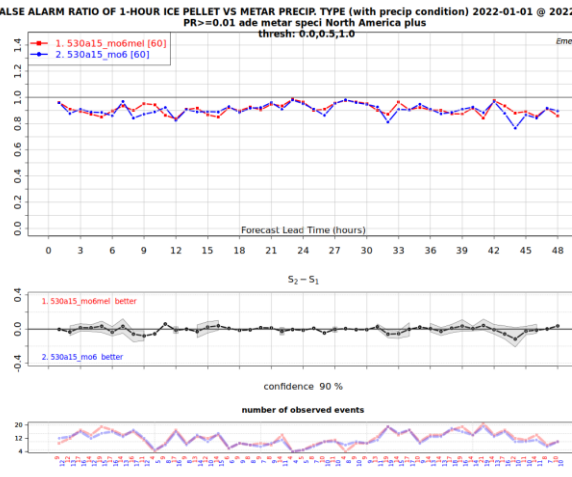
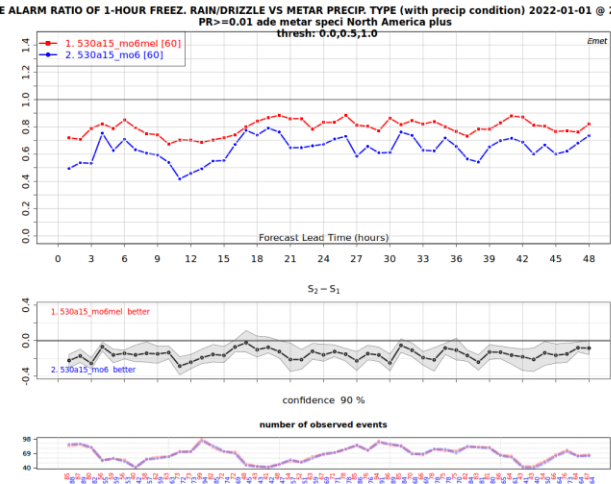
Snow

Rain

POD



FAR



Emet – precipitation phases

2MOM-noLF

PR>0.1

Freezing rain

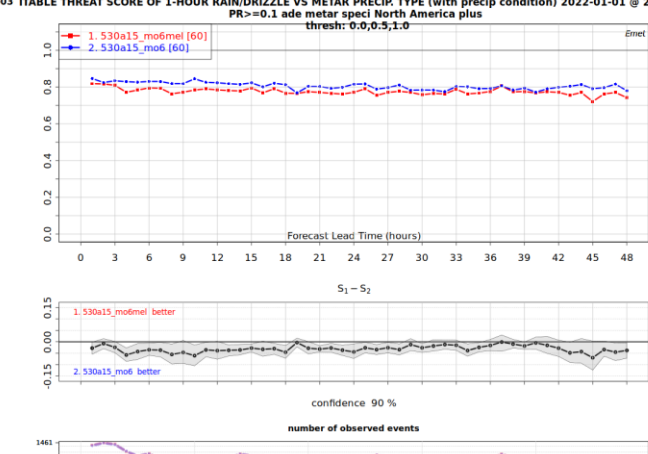
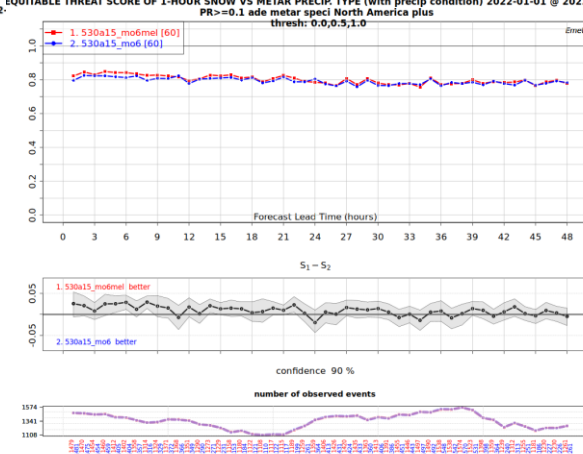
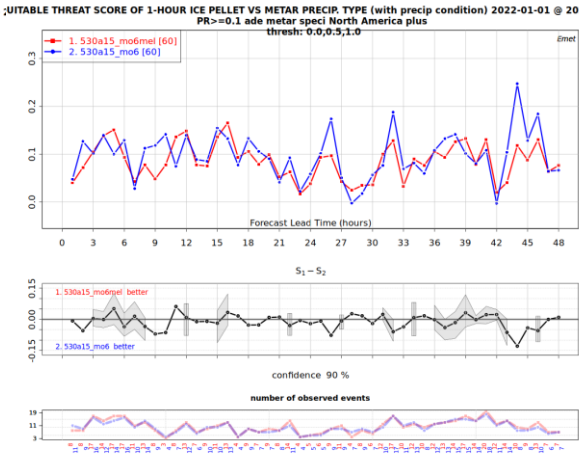
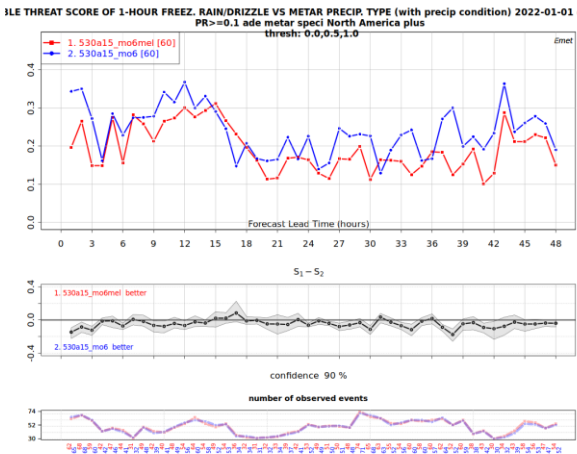
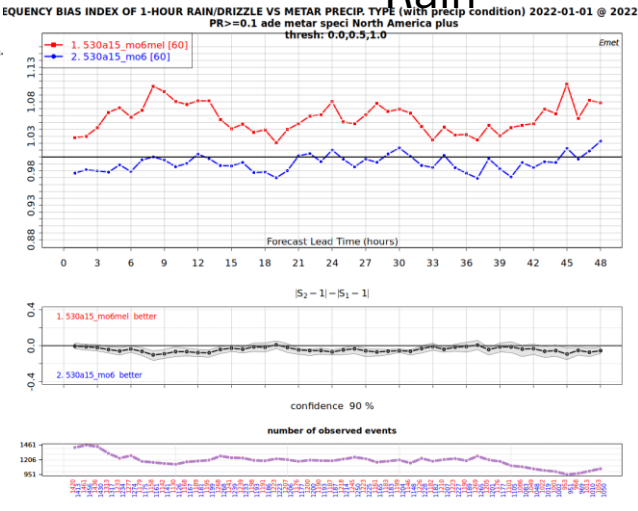
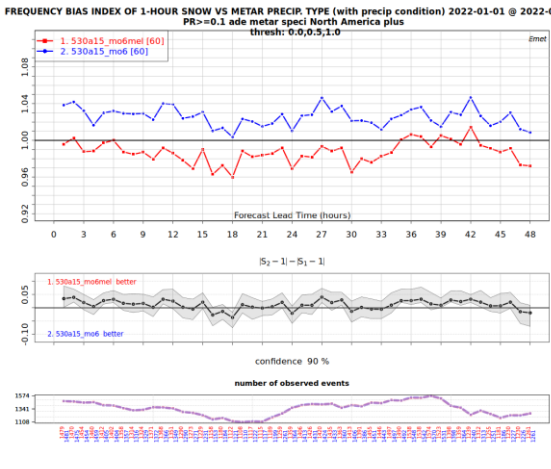
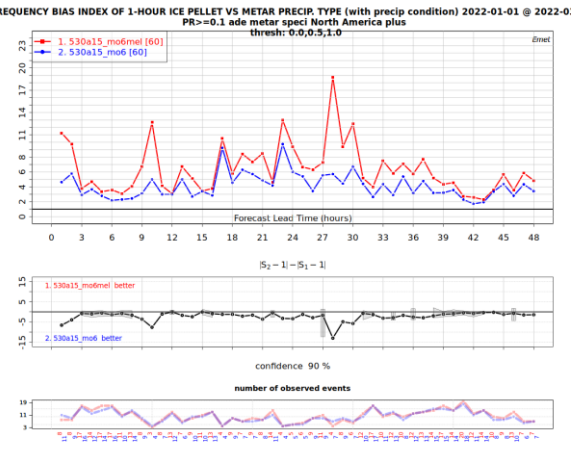
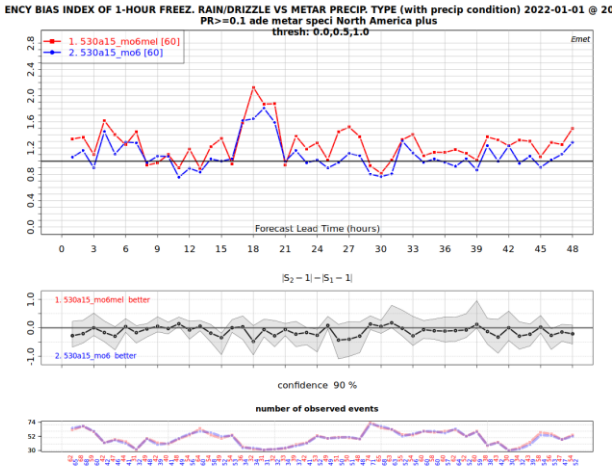
Ice pellets

Snow

Rain

FBI

ETS



Emet – precipitation phases

2MOM-noLF

PR>0.1

Freezing rain

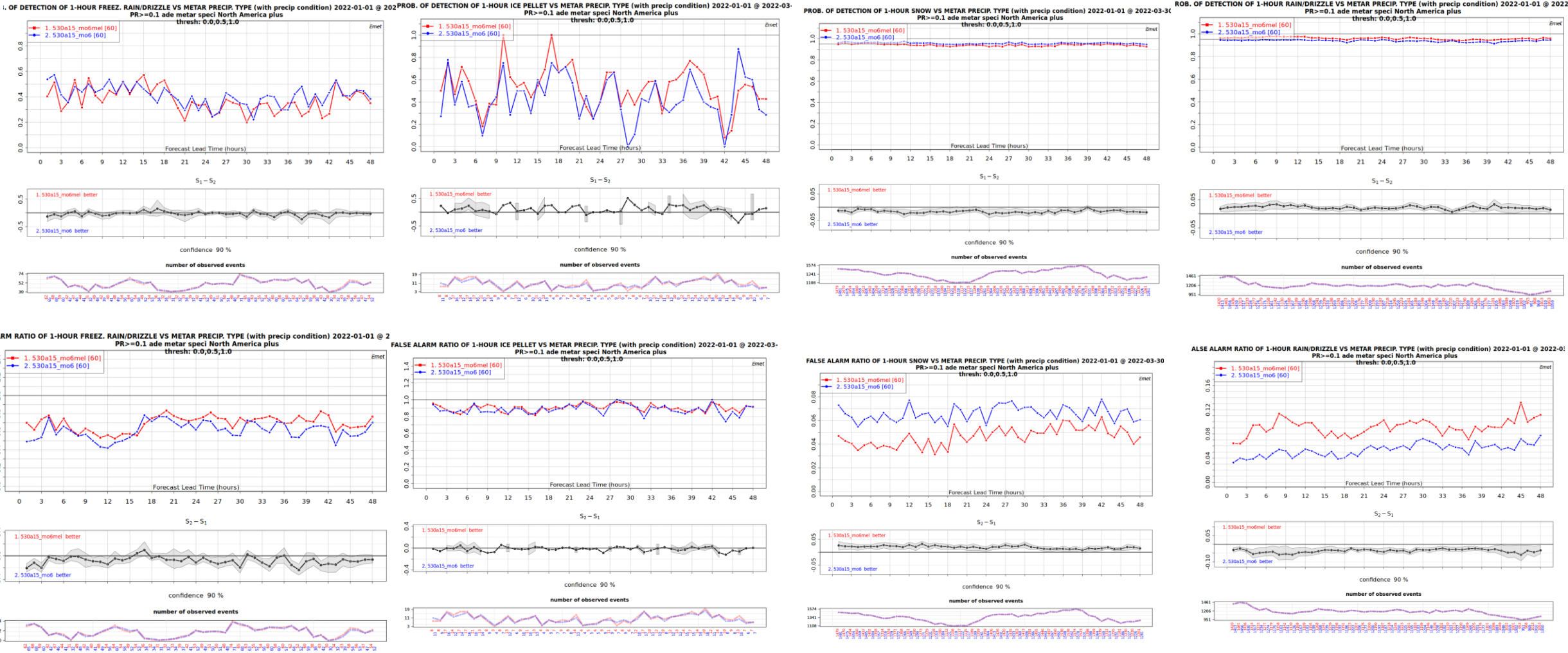
Ice pellets

Snow

Rain

POD

FAR



Emet – precipitation phases

2MOM-noLF

PR>1

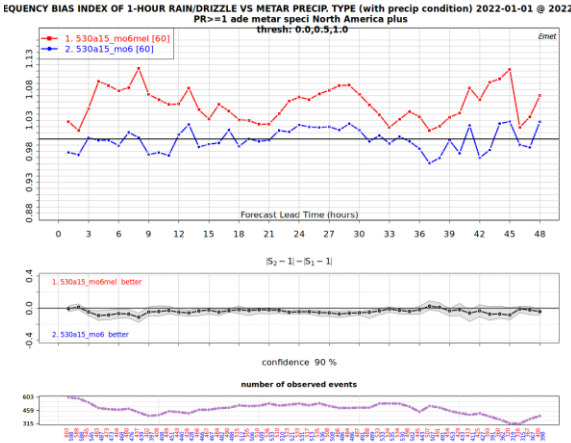
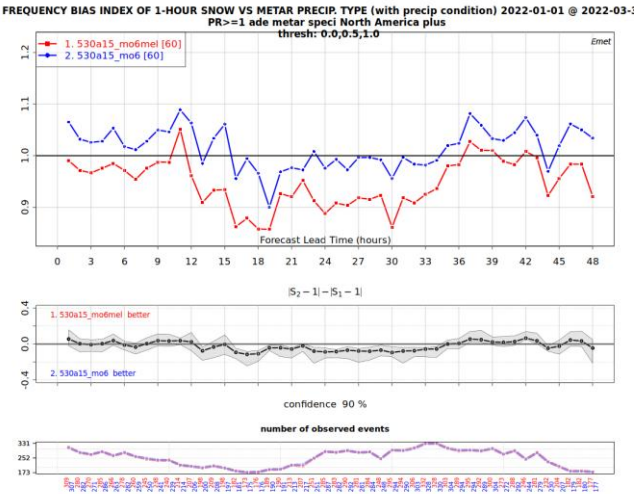
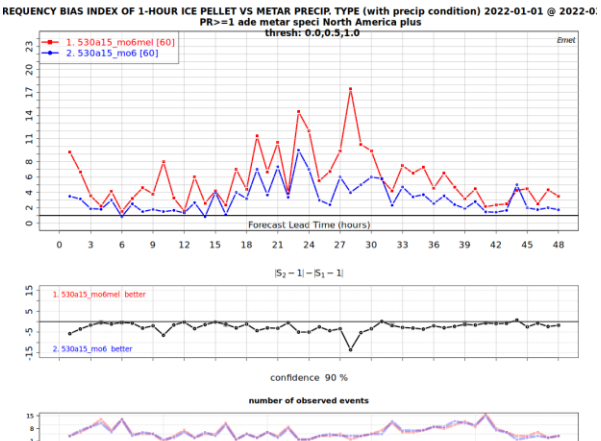
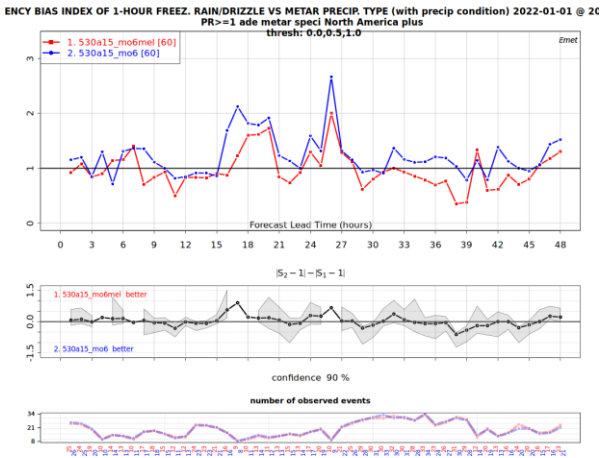
Freezing rain

Ice pellets

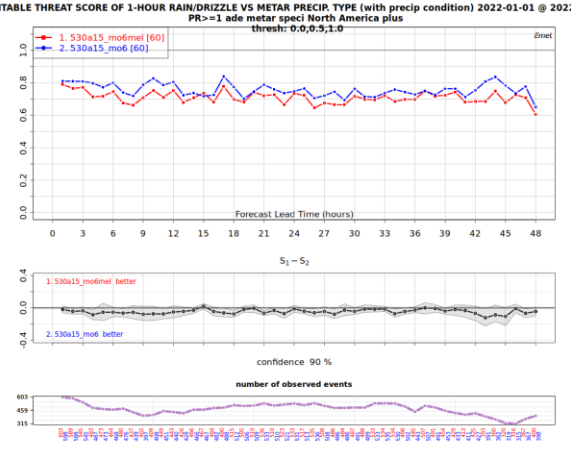
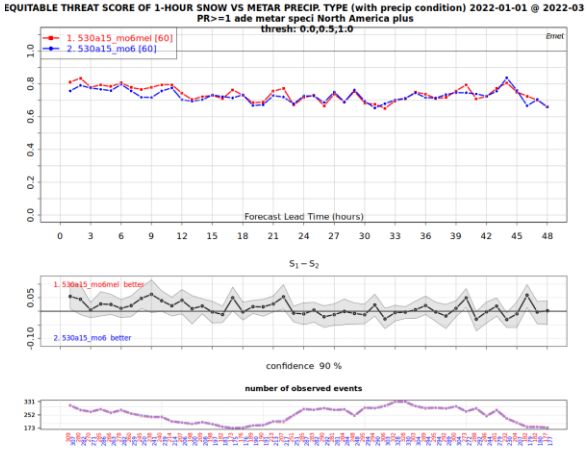
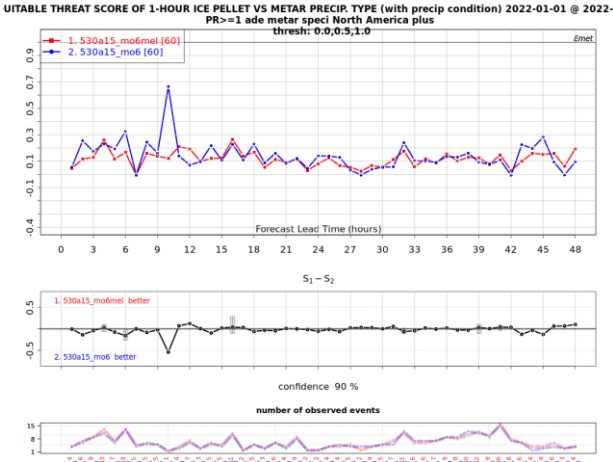
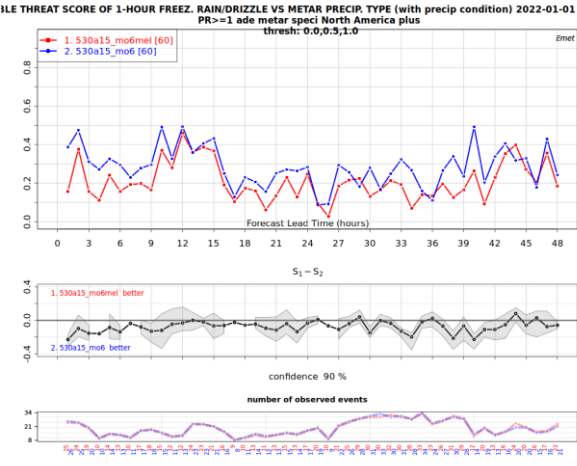
Snow

Rain

FBI



ETS



Emet – precipitation phases 2MOM-noLF

PR>1

Freezing rain

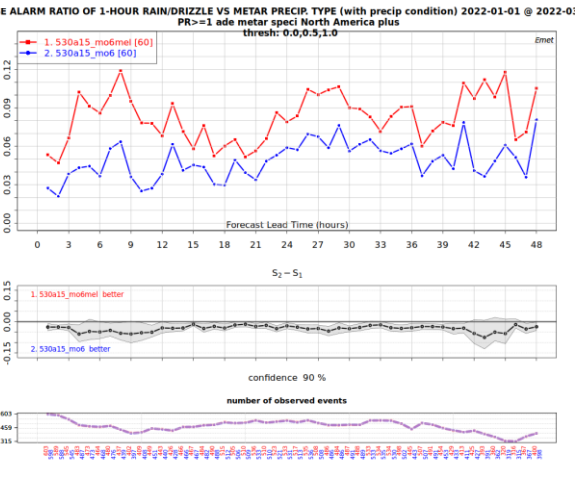
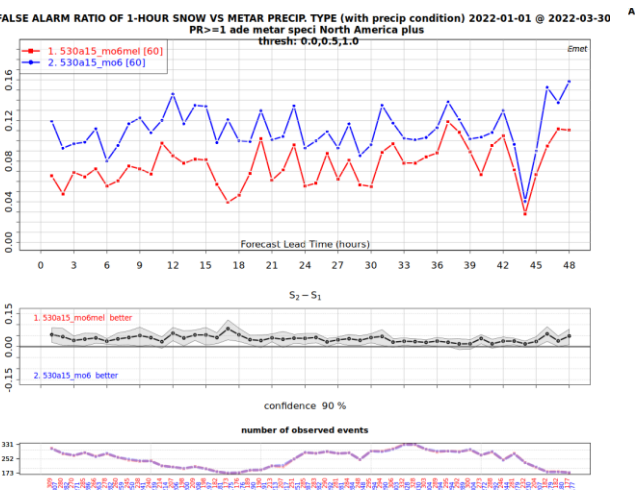
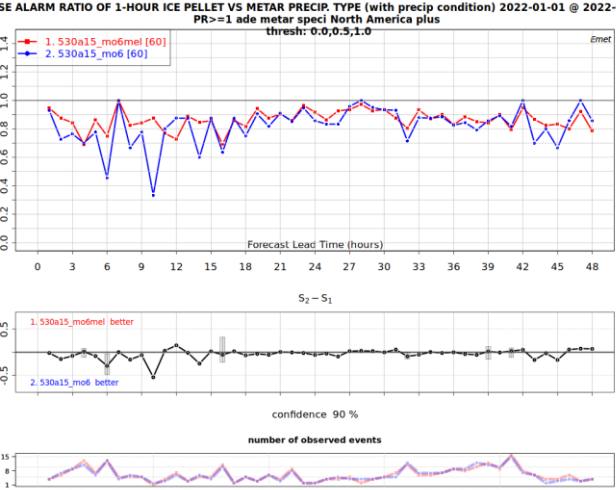
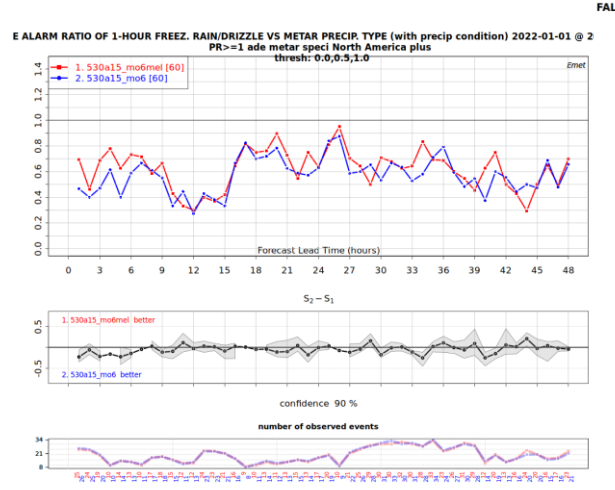
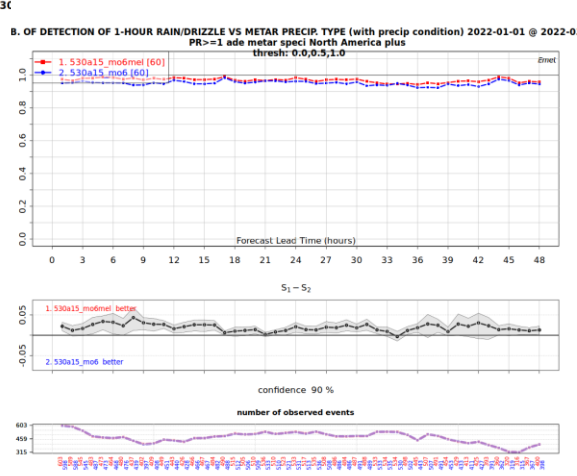
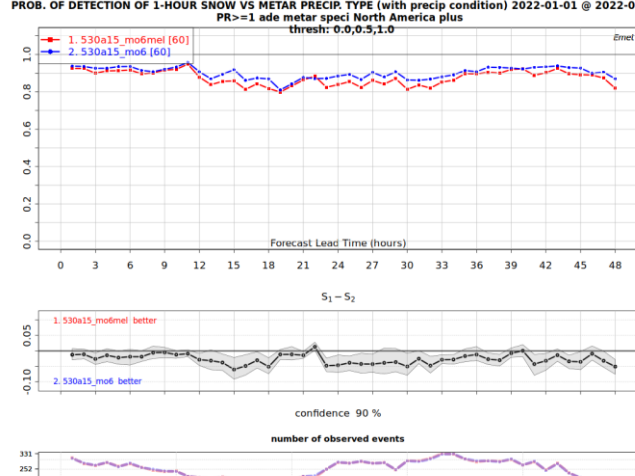
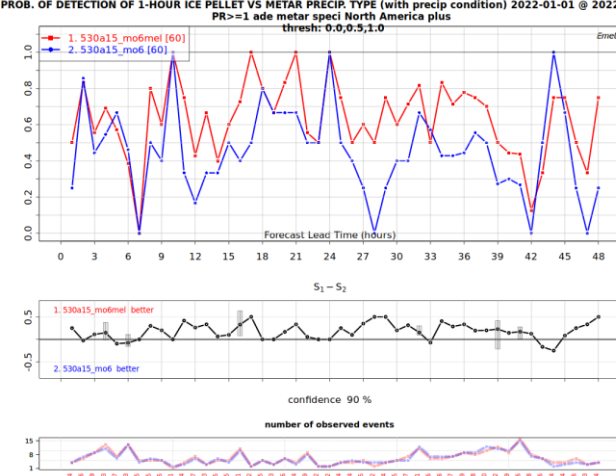
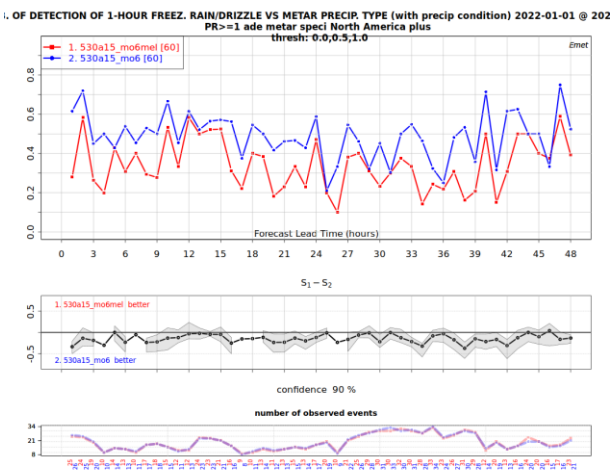
Ice pellets

Snow

Rain

POD

FAR



Emet – precipitation phases

2MOM-noLF

PR>2

Freezing rain

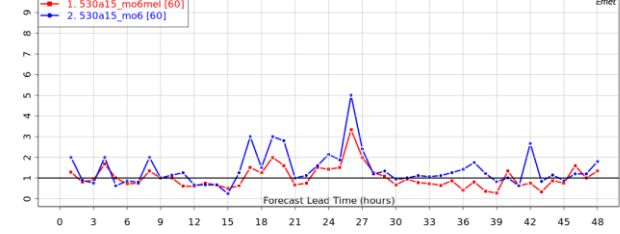
Ice pellets

Snow

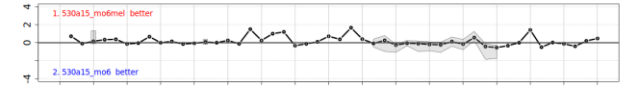
Rain

FBI

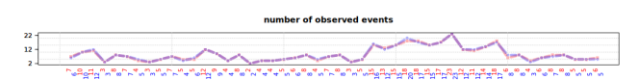
ENCY BIAS INDEX OF 1-HOUR FREEZ. RAIN/DRIZZLE VS METAR PRECIP. TYPE (with precip condition) 2022-01-01 @ 2°C
PR>=2 ade metar speci North America plus
thresh: 0.0,0.5,1.0



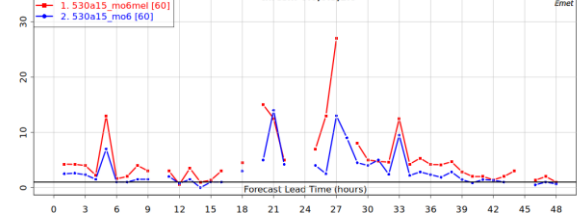
$$|S_2 - I| - |S_1 - I|$$



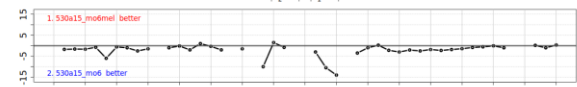
confidence 90 %



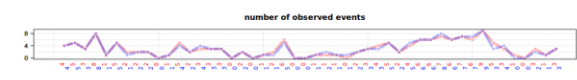
REQUENCY BIAS INDEX OF 1-HOUR ICE PELLET VS METAR PRECIP. TYPE (with precip condition) 2022-01-01 @ 2022-03-10
PR>=2 ade metar speci North America plus
thresh: 0.0,0.5,1.0



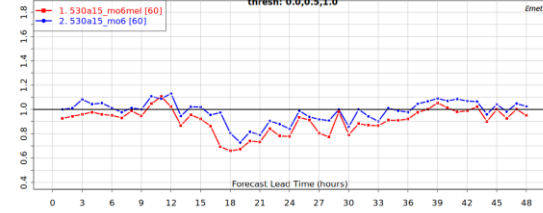
$$|S_2 - I| - |S_1 - I|$$



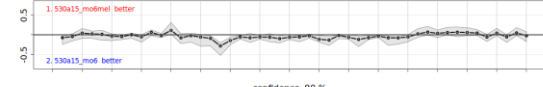
confidence 90 %



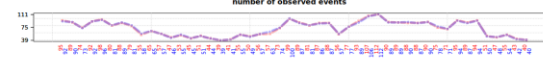
FREQUENCY BIAS INDEX OF 1-HOUR SNOW VS METAR PRECIP. TYPE (with precip condition) 2022-01-01 @ 2022-03-10
PR>=2 ade metar speci North America plus
thresh: 0.0,0.5,1.0



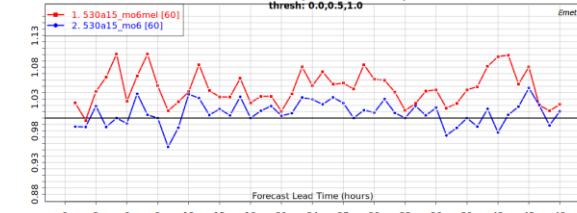
$$|S_2 - I| - |S_1 - I|$$



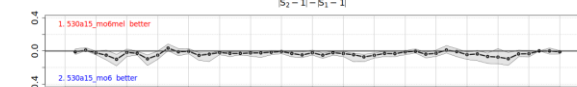
confidence 90 %



EQUITY BIAS INDEX OF 1-HOUR RAIN/DRIZZLE VS METAR PRECIP. TYPE (with precip condition) 2022-01-01 @ 2022-03-10
PR>=2 ade metar speci North America plus
thresh: 0.0,0.5,1.0



$$|S_2 - I| - |S_1 - I|$$

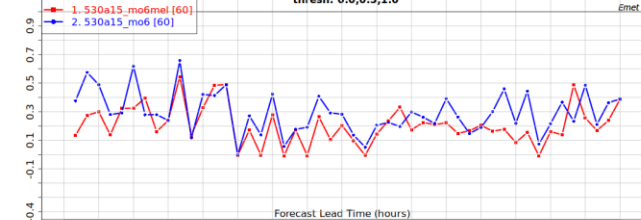


confidence 90 %

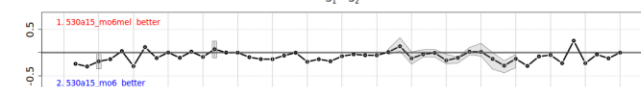


ETS

1LE THREAT SCORE OF 1-HOUR FREEZ. RAIN/DRIZZLE VS METAR PRECIP. TYPE (with precip condition) 2022-01-01 @ 2022-03-10
PR>=2 ade metar speci North America plus
thresh: 0.0,0.5,1.0



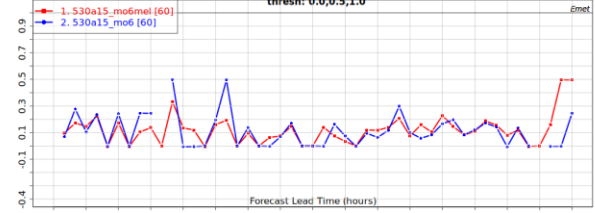
$$S_1 - S_2$$



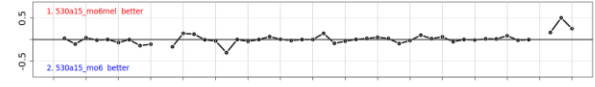
confidence 90 %



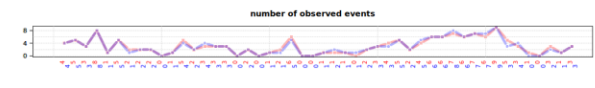
UTABLE THREAT SCORE OF 1-HOUR ICE PELLET VS METAR PRECIP. TYPE (with precip condition) 2022-01-01 @ 2022-03-10
PR>=2 ade metar speci North America plus
thresh: 0.0,0.5,1.0



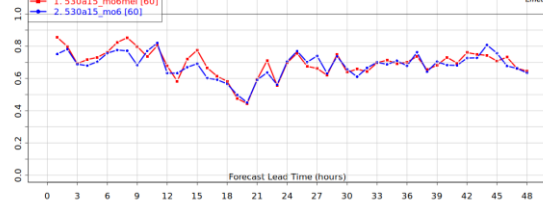
$$S_1 - S_2$$



confidence 90 %



EQUITABLE THREAT SCORE OF 1-HOUR SNOW VS METAR PRECIP. TYPE (with precip condition) 2022-01-01 @ 2022-03-10
PR>=2 ade metar speci North America plus
thresh: 0.0,0.5,1.0



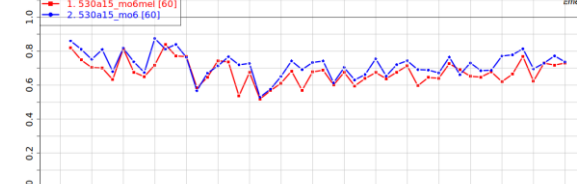
$$S_1 - S_2$$



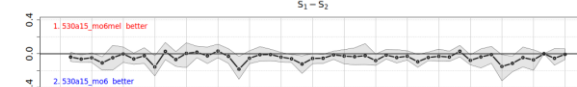
confidence 90 %



UTABLE THREAT SCORE OF 1-HOUR RAIN/DRIZZLE VS METAR PRECIP. TYPE (with precip condition) 2022-01-01 @ 2022-03-10
PR>=2 ade metar speci North America plus
thresh: 0.0,0.5,1.0



$$S_1 - S_2$$



confidence 90 %



Emet – precipitation phases 2MOM-noLF PR>2

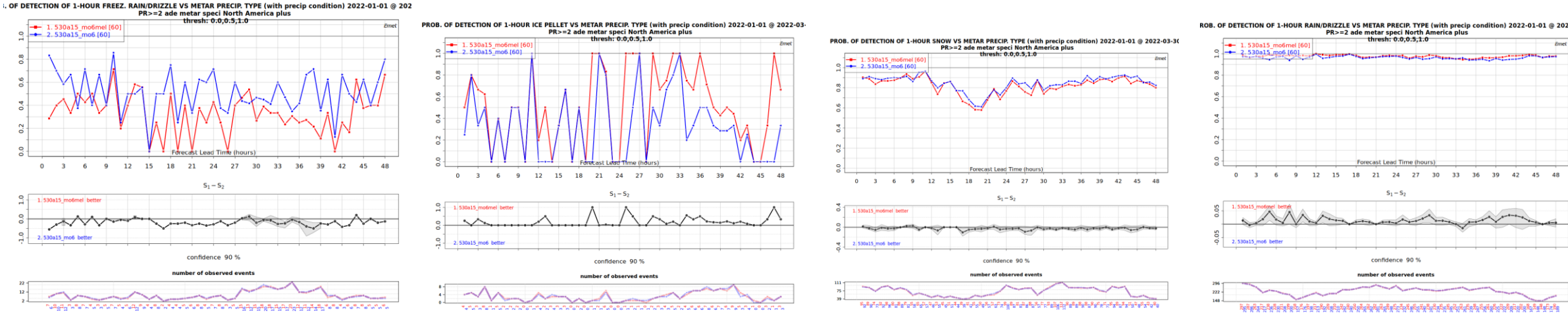
Freezing rain

Ice pellets

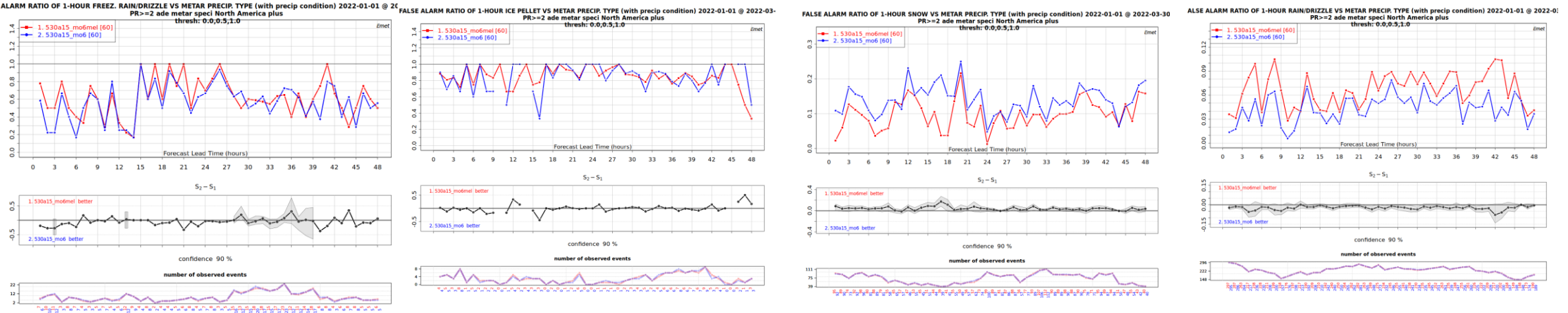
Snow

Rain

POD



FAR



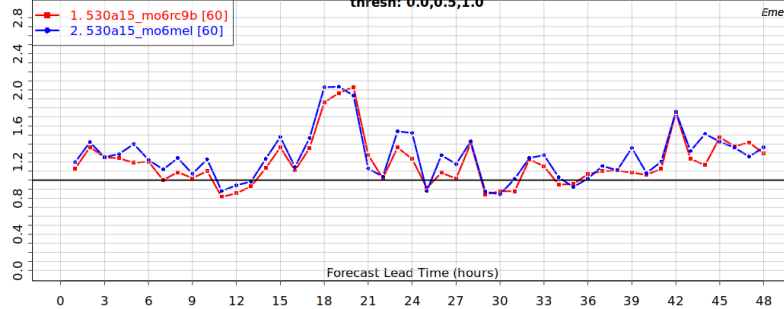
Emet – precipitation phases **2MOM-LF**

LF slightly improves FR FBI, ETS and FAR at small PR thresholds:

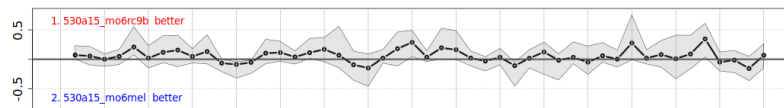
LF degrades SN scores (except POD) at all PR thresholds:

LF improves RN FBI, ETS and FAR at all PR thresholds:

ENCY BIAS INDEX OF 1-HOUR FREEZ. RAIN/DRIZZLE VS METAR PRECIP. TYPE (with precip condition) 2022-01-01 @ 2022-03-01
PR>=0.1 ade metar speci North America plus
thresh: 0.0,0.5,1.0

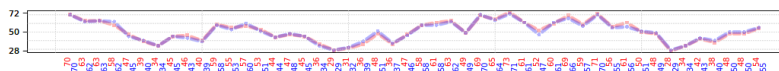


$|S_2 - 1| - |S_1 - 1|$

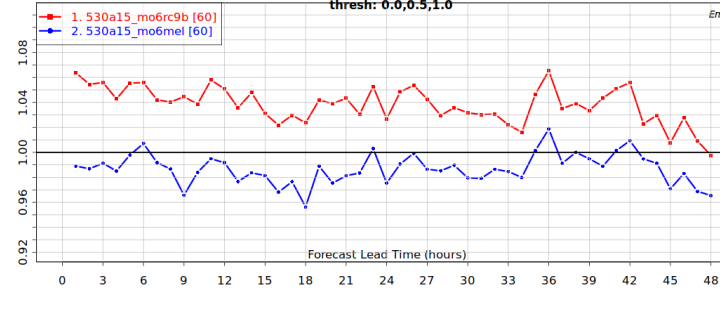


confidence 90 %

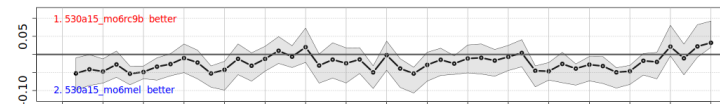
number of observed events



FREQUENCY BIAS INDEX OF 1-HOUR SNOW VS METAR PRECIP. TYPE (with precip condition) 2022-01-01 @ 2022-03-01
PR>=0.1 ade metar speci North America plus
thresh: 0.0,0.5,1.0

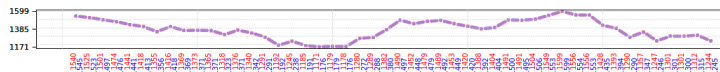


$|S_2 - 1| - |S_1 - 1|$

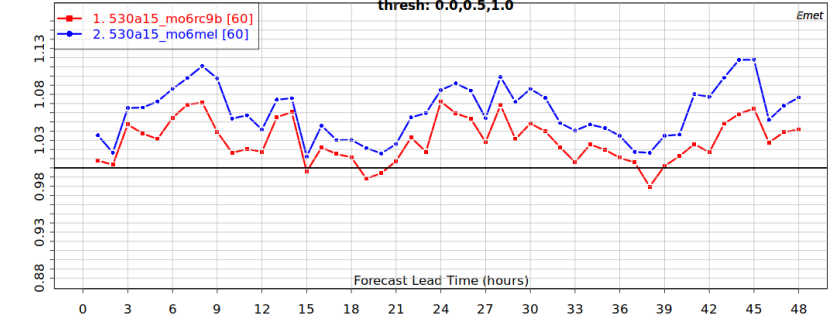


confidence 90 %

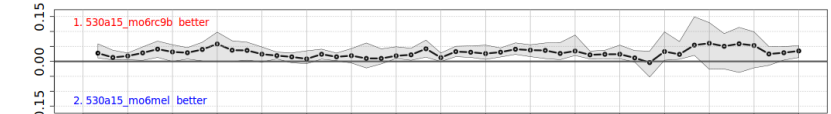
number of observed events



EQUENCY BIAS INDEX OF 1-HOUR RAIN/DRIZZLE VS METAR PRECIP. TYPE (with precip condition) 2022-01-01 @ 2022-03-01
PR>=1 ade metar speci North America plus
thresh: 0.0,0.5,1.0

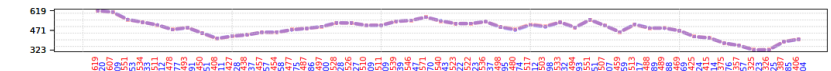


$|S_2 - 1| - |S_1 - 1|$



confidence 90 %

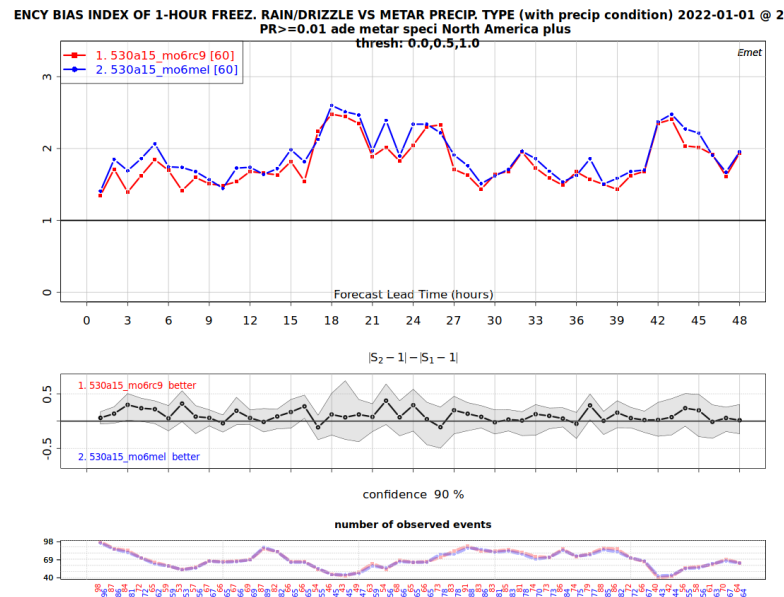
number of observed events



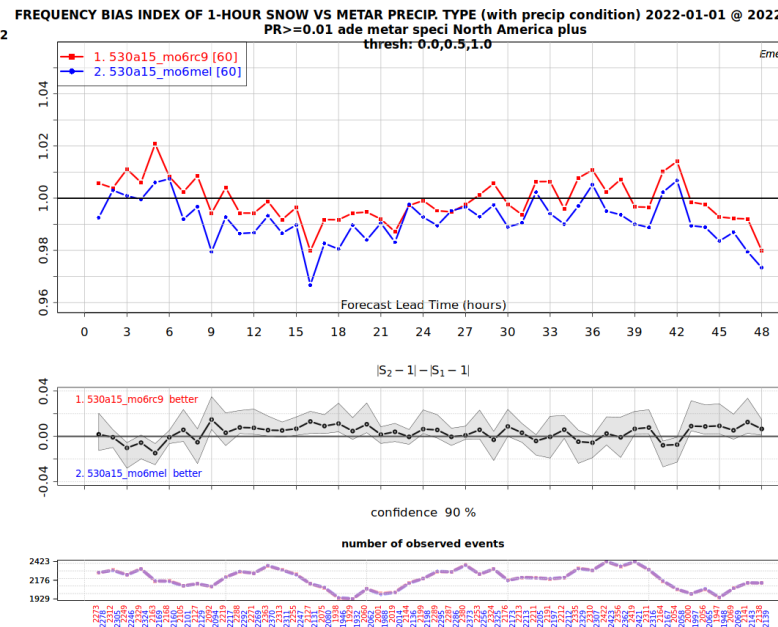
But this is good news (because of obs. undercatch) and it gives closer results to Bourguin...

Emet – precipitation phases 3MOM-LF

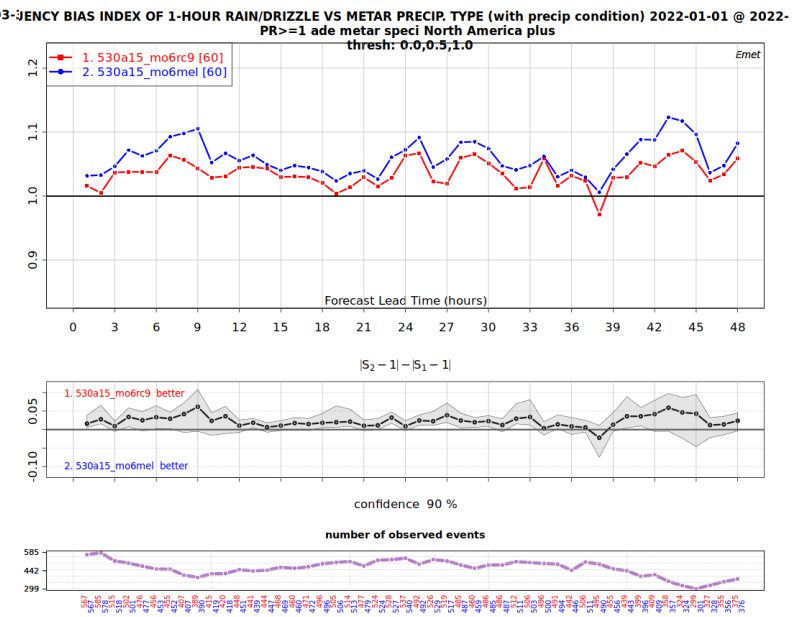
3MOM-LF improves FR FBI, ETS and FAR at small PR thresholds:



3MOM-LF improves SN FBI only at small PR thresholds:



3MOM-LF improves RN FBI, ETS and FAR at all PR thresholds:



Mostly from LF

Spring (S22)

S22: Spring 2022

1.2MOM-noLF

2.2MOM-LF

3.3MOM-LF

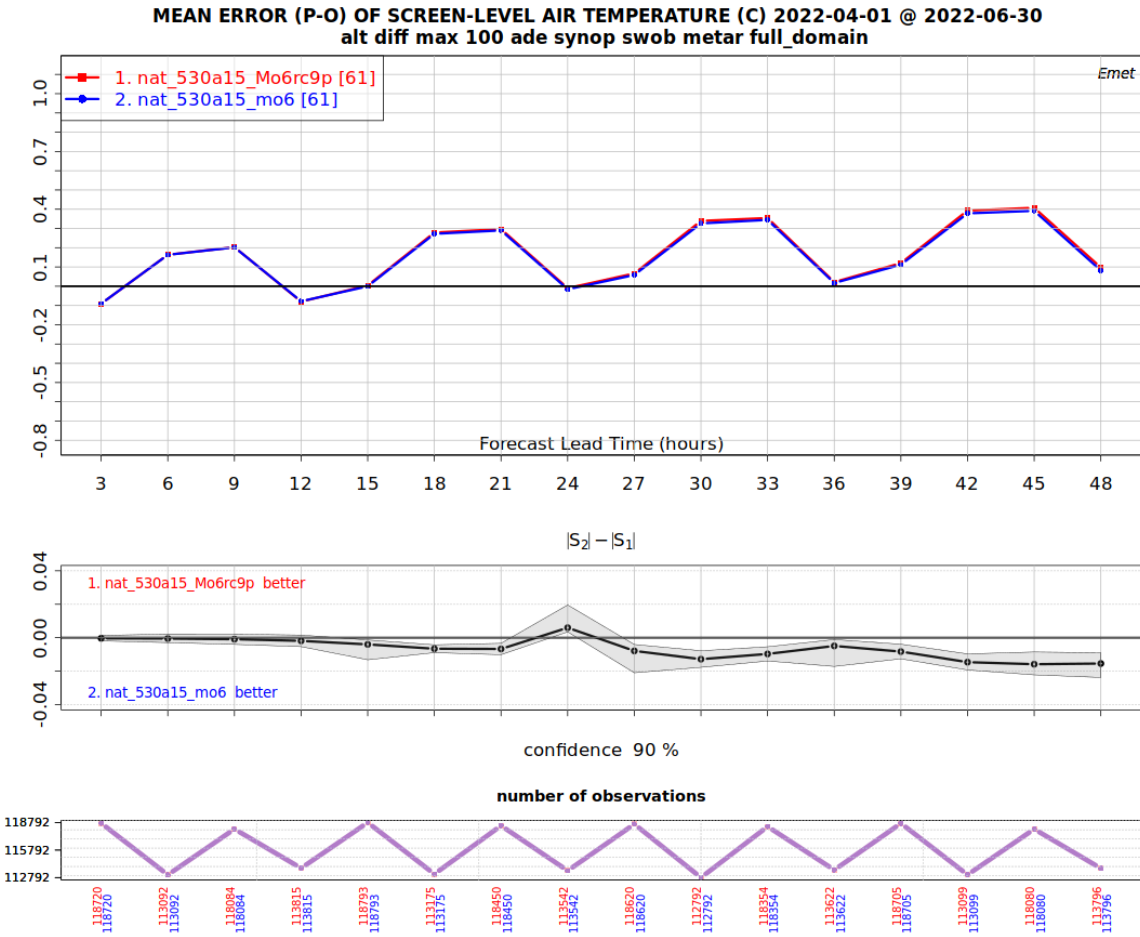
EMET (P22), TT & TD 2MOM-noLF*

P3_v5.5.0-rc9
P3 v5.3.6 2MOM-noLF

Mostly negative effects
everywhere

bias		20220401 / 20220630		
bias nat_530a15_Mo6rc9p / nat_530a15_mo6		00Z	12Z	All
Arctic All CLIM	TD	0.004	0.0048	0.005
	TT	-0.012	-6.4e-05	-4.3e-05
Boreal CLIM	TD	0.0014	-0.001	0.0
	TT	-0.0074	-0.0048	-0.004
Canada	TD	0.0011	0.00087	0.00075
	TT	-0.0092	-0.0072	-0.0046
East CLIM	TD	0.0	0.0015	0.0
	TT	-0.0035	-0.0047	-0.0016
Great Lakes CLIM	TD	0.00074	-0.0056	-0.006
	TT	-0.0059	-0.004	-0.003
Mt West CLIM	TD	-0.0036	-0.0022	-0.00042
	TT	-0.015	-0.0089	0.011
North Plains CLIM	TD	0.0	0.00064	0.0
	TT	-0.0087	-0.011	-0.0086
Pacific North West CLIM	TD	-0.0037	-0.00023	-0.0014
	TT	-0.0043	-0.011	-0.0018
Prairie CLIM	TD	-0.0013	-0.0056	-0.0013
	TT	-0.0099	-1.3e-05	-0.0029
United States of America East	TD	-0.0012	-0.00063	-0.0024
	TT	-0.0096	-0.0052	-0.0054
United States of America West	TD	-0.0012	5.8e-05	-0.0005
	TT	-0.011	-0.009	-0.0042
full_domain	TD	0.00035	-0.0009	-3.1e-05
	TT	-0.012	-0.0072	-0.0063

TT-bias (full domain)



EMET (P22), TT & TD 2MOM-noLF*

P3_v5.5.0-rc9
P3 v5.3.6 2MOM-noLF

bias				
bias nat_530a15_Mo6rc9p / nat_530a15_mo6		20220401 / 20220630		
		00z	12z	All
Arctic All CLIM	TD	0.004	0.0048	0.005
	TT	-0.012	-6.4e-05	-4.3e-05
Boreal CLIM	TD	0.0014	-0.001	0.0
	TT	-0.0074	-0.0048	-0.004
Canada	TD	0.0011	0.00087	0.00075
	TT	-0.0092	-0.0072	-0.0046
East CLIM	TD	0.0	0.0015	0.0
	TT	-0.0035	-0.0047	-0.0016
Great Lakes CLIM	TD	0.00074	-0.0056	-0.006
	TT	-0.0059	-0.004	-0.003
Mt West CLIM	TD	-0.0036	-0.0022	-0.00042
	TT	-0.015	-0.0089	0.011
North Plains CLIM	TD	0.0	0.00064	0.0
	TT	-0.0087	-0.011	-0.0086
Pacific North West CLIM	TD	-0.0037	-0.00023	-0.0014
	TT	-0.0043	-0.011	-0.0018
Prairie CLIM	TD	-0.0013	-0.0056	-0.0013
	TT	-0.0099	-1.3e-05	-0.0029
United States of America East	TD	-0.0012	-0.00063	-0.0024
	TT	-0.0096	-0.0052	-0.0054
United States of America West	TD	-0.0012	5.8e-05	-0.0005
	TT	-0.011	-0.009	-0.0042
full_domain	TD	0.00035	-0.0009	-3.1e-05
	TT	-0.012	-0.0072	-0.0063

rmse				
rmse nat_530a15_Mo6rc9p / nat_530a15_mo6		20220401 / 20220630		
		00z	12z	All
Arctic All CLIM	TD	-0.0044	-0.0039	-0.0043
	TT	-0.0057	-0.0051	-0.0077
Boreal CLIM	TD	-0.00013	-0.0028	-0.0016
	TT	-0.0072	-0.009	-0.01
Canada	TD	-0.00075	0.0	0.0
	TT	-0.0053	-0.0057	-0.0055
East CLIM	TD	0.0015	0.00085	0.00042
	TT	0.0	-0.00073	-0.00075
Great Lakes CLIM	TD	-6.2e-05	0.0	-3.5e-05
	TT	-0.0018	-0.0052	-0.0043
Mt West CLIM	TD	0.0004	0.0	0.0006
	TT	0.0	-2.9e-05	0.00035
North Plains CLIM	TD	0.0	0.0008	0.00092
	TT	-0.0039	0.0	-0.0014
Pacific North West CLIM	TD	0.0	0.002	0.0
	TT	-0.0013	-0.00054	0.0
Prairie CLIM	TD	0.0	0.0	0.0
	TT	-0.0063	-0.0098	-0.0071
United States of America East	TD	0.00049	0.0	0.0
	TT	-0.0018	-0.0065	-0.0051
United States of America West	TD	0.0	0.0	0.0
	TT	0.0	-0.0012	-0.0011
full_domain	TD	0.0	-0.00072	0.0
	TT	-0.0038	-0.0054	-0.0057

stdev				
stdev nat_530a15_Mo6rc9p / nat_530a15_mo6		20220401 / 20220630		
		00z	12z	All
Arctic All CLIM	TD	-0.0048	-0.0059	-0.0054
	TT	-0.0041	-0.0042	-0.0076
Boreal CLIM	TD	-9.5e-05	-0.0012	-0.0013
	TT	-0.0076	-0.011	-0.0096
Canada	TD	-0.00097	-0.0003	-0.00069
	TT	-0.0049	-0.0049	-0.0053
East CLIM	TD	0.00076	0.0	0.00044
	TT	-0.0005	-0.00032	-0.00042
Great Lakes CLIM	TD	-0.0005	-0.00054	0.0004
	TT	0.0	-0.0051	-0.0039
Mt West CLIM	TD	0.00077	0.0	0.00052
	TT	0.0	0.0021	-0.0014
North Plains CLIM	TD	-8.9e-05	0.0013	0.0
	TT	-0.0029	0.0	-0.00042
Pacific North West CLIM	TD	0.0	0.0023	4.1e-06
	TT	-0.00053	0.0015	-0.00053
Prairie CLIM	TD	0.0	0.0	0.0
	TT	0.0	-0.0093	-0.006
United States of America East	TD	0.00035	0.0	0.0
	TT	-0.0015	-0.0056	-0.0043
United States of America West	TD	0.0	0.0	-0.00073
	TT	0.0	0.00061	-0.0012
full_domain	TD	0.0	-0.00029	0.0
	TT	-0.0023	-0.0041	-0.0047

EMET (P22), P0 & UV 2MOM-noLF*

bias < >

bias nat_530a15_Mo6rc9p / nat_530a15_mo6		20220401 / 20220630		
		00z	12z	All
Arctic All CLIM	P0	-0.015	-0.0098	-0.016
Boreal CLIM	P0	0.019	0.023	0.021
Canada	P0	0.0037	0.019	0.018
East CLIM	P0	0.00047	0.021	0.0089
Great Lakes CLIM	P0	-0.0055	-0.0024	-0.0073
Mt West CLIM	P0	-0.018	-0.013	-0.018
North Plains CLIM	P0	0.0037	-0.0078	0.00065
Pacific North West CLIM	P0	-0.0052	-0.0051	-0.0077
Prairie CLIM	P0	-0.0012	0.00017	-0.0041
United States of America East	P0	-0.0082	-0.011	-0.0099
United States of America West	P0	-0.0054	-0.0073	-0.0064
full_domain	P0	0.001	0.011	0.007

rmse < >

rmse nat_530a15_Mo6rc9p / nat_530a15_mo6		20220401 / 20220630		
		00z	12z	All
Arctic All CLIM	P0	-0.0018	-0.00071	-0.0014
Boreal CLIM	P0	4.8e-05	0.0012	0.0014
Canada	P0	0.00015	0.00011	0.00024
East CLIM	P0	0.0	-0.0011	0.0
Great Lakes CLIM	P0	-0.0017	0.0	-0.00043
Mt West CLIM	P0	-0.0064	-0.0024	-0.0051
North Plains CLIM	P0	7.5e-05	0.0	-0.0016
Pacific North West CLIM	P0	0.0	-0.0025	-0.00087
Prairie CLIM	P0	0.0	0.002	0.0
United States of America East	P0	-0.0016	0.0	-0.0015
United States of America West	P0	0.0	0.0	0.0
full_domain	P0	-0.00021	-0.00027	0.00042

bias < >

bias nat_530a15_Mo6rc9p / nat_530a15_mo6		20220401 / 20220630		
		00z	12z	All
Arctic All CLIM	UV	0.0047	0.0017	0.0053
Boreal CLIM	UV	0.0012	0.00072	0.0
Canada	UV	0.001	0.0013	0.00012
East CLIM	UV	-0.00017	-8.6e-05	-0.0006
Great Lakes CLIM	UV	0.0017	0.0016	0.0026
Mt West CLIM	UV	0.0031	0.0	0.0019
North Plains CLIM	UV	0.0	0.0028	0.00077
Pacific North West CLIM	UV	-0.0037	-0.0038	-0.0014
Prairie CLIM	UV	0.0	0.0005	4.7e-05
United States of America East	UV	0.0	-0.00088	-0.00097
United States of America West	UV	0.0	-8.5e-05	0.0029
full_domain	UV	-0.00049	0.00042	-0.00052

rmse < >

rmse nat_530a15_Mo6rc9p / nat_530a15_mo6		20220401 / 20220630		
		00z	12z	All
Arctic All CLIM	UV	0.0004	0.0	0.0
Boreal CLIM	UV	0.0014	0.0018	0.0
Canada	UV	0.001	0.00019	0.0
East CLIM	UV	0.0023	0.0	0.0016
Great Lakes CLIM	UV	0.0	0.0	-0.00016
Mt West CLIM	UV	-0.0012	-0.0034	0.0
North Plains CLIM	UV	-0.00059	0.0012	-0.0012
Pacific North West CLIM	UV	-0.0042	0.0	0.0
Prairie CLIM	UV	0.0	0.0	0.0
United States of America East	UV	0.0	0.0	0.0
United States of America West	UV	-0.0015	0.0	0.0
full_domain	UV	0.00046	0.0	-0.00014

EMET (P22), PR6 & PR24 2MOM-noLF*

P3_v5.5.0-rc9
P3 v5.3.6 2MOM-noLF

dpmean

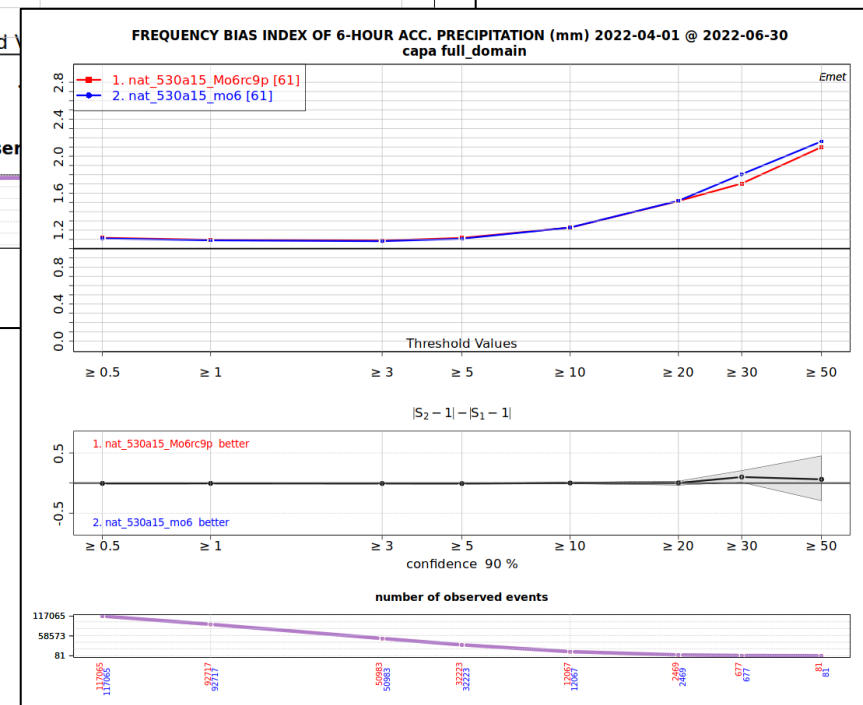
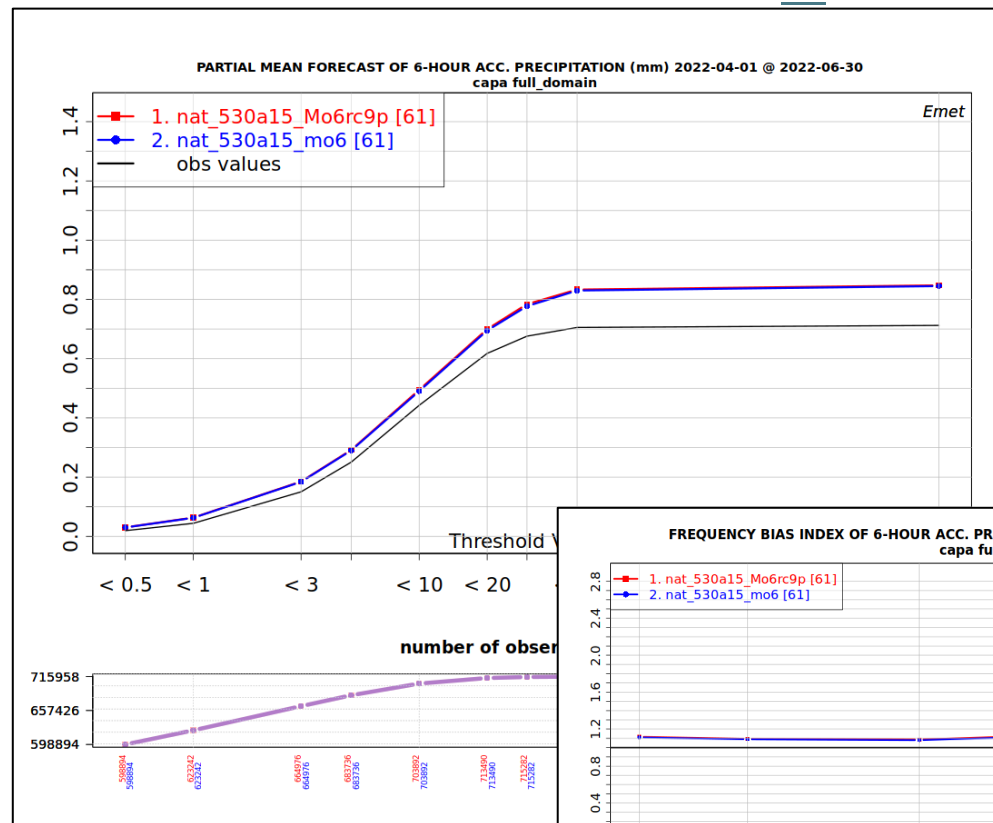
		00z	12z	All
Arctic All CLIM	PR24	0.0054	0.0	0.0
	PR6	-0.0089	-0.0021	-0.0029
Boreal CLIM	PR24	0.0	0.0	-0.015
	PR6	0.0	-0.0038	-0.004
Canada	PR24	0.0	-0.01	-0.0041
	PR6	0.00083	-0.0039	-0.0033
East CLIM	PR24	0.00029	-0.0029	0.0
	PR6	-0.0017	-0.0028	-0.0029
Great Lakes CLIM	PR24	-0.0021	-0.00023	-0.00018
	PR6	-0.0014	0.0	-0.00041
Mt West CLIM	PR24	0.0016	0.0	0.0
	PR6	0.0016	0.00023	0.00077
North Plains CLIM	PR24	0.01	0.0	0.0
	PR6	0.0014	0.0	0.0
Pacific North West CLIM	PR24	-0.03	0.051	0.012
	PR6	0.00092	-0.0058	-0.0044
Prairie CLIM	PR24	0.013	-0.0078	0.0
	PR6	-0.0001	-0.0032	-0.00078
United States of America East	PR24	0.005	-0.0064	0.0
	PR6	-0.0016	-0.00085	-0.0011
United States of America West	PR24	-0.0057	-0.024	-0.02
	PR6	0.0	0.0	0.00062
full_domain	PR24	-0.00071	-0.012	-0.0082
	PR6	-4.3e-05	-0.0038	-0.002

ETS

		00z	12z	All
Arctic All CLIM	PR24	-0.00075	0.0	0.0
	PR6	0.0095	-0.0016	0.0
Boreal CLIM	PR24	0.0	0.00015	-0.00092
	PR6	-0.0024	0.0018	0.0
Canada	PR24	-0.0033	0.0012	0.00086
	PR6	0.0	0.0011	0.0
East CLIM	PR24	0.0067	0.0	0.0025
	PR6	0.0	0.0	0.0
Great Lakes CLIM	PR24	0.0	0.0	0.0
	PR6	0.0	0.0	0.0
Mt West CLIM	PR24	0.0043	0.0	-0.0025
	PR6	-0.00094	0.0012	0.00094
North Plains CLIM	PR24	-0.0075	0.0	0.0
	PR6	0.0	0.0011	-0.00093
Pacific North West CLIM	PR24	0.0032	-0.0037	0.0
	PR6	0.0022	0.004	0.0
Prairie CLIM	PR24	0.0	0.0015	0.0
	PR6	0.0	0.0019	0.0
United States of America East	PR24	0.0	0.0013	0.0018
	PR6	0.0	0.0	0.0
United States of America West	PR24	0.0066	0.0012	0.0
	PR6	0.0024	0.0	-0.0019
full_domain	PR24	0.0015	0.00074	0.00056
	PR6	0.0	0.0017	0.0

FBI

		00z	12z	All
Arctic All CLIM	PR24	0.0	0.0	0.0
	PR6	-0.039	-0.0084	-0.0093
Boreal CLIM	PR24	-0.0038	-0.0075	-0.011
	PR6	-0.0042	-0.0039	-0.0046
Canada	PR24	0.0	0.0	0.0
	PR6	0.022	-0.022	-0.0039
East CLIM	PR24	0.0	0.0	0.0
	PR6	-0.0013	0.0057	0.0
Great Lakes CLIM	PR24	-0.0022	0.0	0.0
	PR6	-0.0023	-0.002	-0.0036
Mt West CLIM	PR24	0.014	-0.0096	0.0
	PR6	0.0	-0.0065	0.051
North Plains CLIM	PR24	0.025	0.0	0.0
	PR6	0.014	0.0	0.0057
Pacific North West CLIM	PR24	-0.012	0.012	0.012
	PR6	-0.017	-0.0031	-0.011
Prairie CLIM	PR24	-0.048	-0.0017	0.0
	PR6	-0.002	-0.0023	-0.0019
United States of America East	PR24	0.0	-0.0043	0.0
	PR6	0.0069	0.0	0.0019
United States of America West	PR24	0.013	-0.014	-0.014
	PR6	0.0	0.0	0.0
full_domain	PR24	-0.0011	-0.0082	-0.0038
	PR6	0.018	-0.032	0.0091



POD

		00z	12z	All
Arctic All CLIM	PR24	0.012	0.0	0.0
	PR6	0.016	-0.0034	0.0027
Boreal CLIM	PR24	0.0	0.0046	0.0036
	PR6	-0.0053	0.0059	-0.0074
Canada	PR24	0.0	0.0019	0.0021
	PR6	0.00036	0.0039	0.00053
East CLIM	PR24	0.0035	0.0	0.0012
	PR6	0.00075	0.0	0.0013
Great Lakes CLIM	PR24	0.0015	0.0	0.0
	PR6	0.0022	0.0	0.00097
Mt West CLIM	PR24	0.0071	0.0029	0.0
	PR6	-0.0018	0.0	0.0
North Plains CLIM	PR24	-0.012	0.0	0.0
	PR6	0.0	0.0084	0.0
Pacific North West CLIM	PR24	0.013	0.0016	0.0065
	PR6	0.0069	0.0061	0.006
Prairie CLIM	PR24	-0.0072	0.0043	0.0
	PR6	0.00095	0.0042	0.004
United States of America East	PR24	0.004	0.0037	0.0033
	PR6	0.0012	-0.0013	0.00038
United States of America West	PR24	0.0	0.0037	0.0013
	PR6	0.0056	0.0	-0.0016
full_domain	PR24	0.0052	0.002	0.0019
	PR6	0.00029	0.0064	0.0

FAR

		00z	12z	All
Arctic All CLIM	PR24	-0.008	0.0	-0.0027
	PR6	-0.011	-0.0023	0.0
Boreal CLIM	PR24	-0.0046	0.0029	-0.0045
	PR6	-0.0023	0.0	-0.0008
Canada	PR24	-0.0042	-0.0031	-0.00029
	PR6	-0.0013	0.0063	0.0
East CLIM	PR24	0.01	0.0	0.0029
	PR6	-0.0009	0.0	-0.00061
Great Lakes CLIM	PR24	0.0	0.0	0.0
	PR6	0.0	-0.0056	0.0
Mt West CLIM	PR24	0.0	-0.0025	-0.0041
	PR6	0.029	0.0017	0.0015
North Plains CLIM	PR24	-0.0064	0.0	0.0
	PR6	0.004	0.0044	-0.0012
Pacific North West CLIM	PR24	0.0066	-0.008	-0.0052
	PR6	0.0	0.0	0.0
Prairie CLIM	PR24	0.0	-0.0047	0.0
	PR6	0.0	0.0025	0.0
United States of America East	PR24	0.0041	0.00092	0.0016
	PR6	0.0	-0.00045	-0.00042
United States of America West	PR24	0.011	-0.00092	-0.0031
	PR6	0.0037	0.0	-0.00076
full_domain	PR24	0.0	-0.00094	-0.00022
	PR6	-0.00099	0.0021	-0.00056

S22: Spring 2022

1.2MOM-noLF

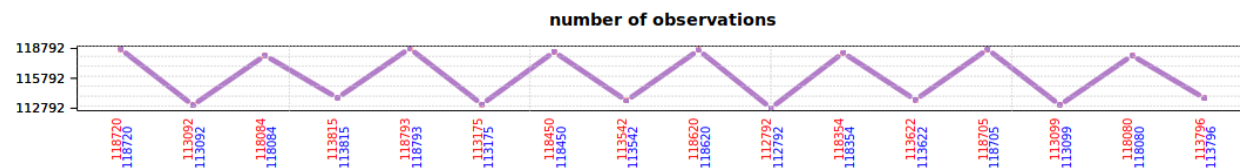
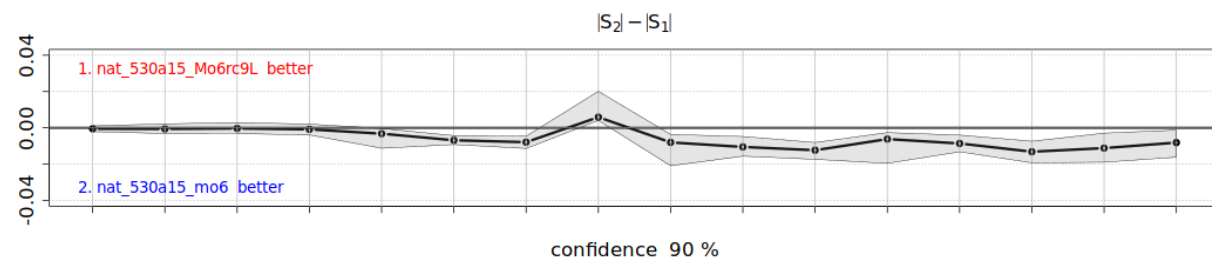
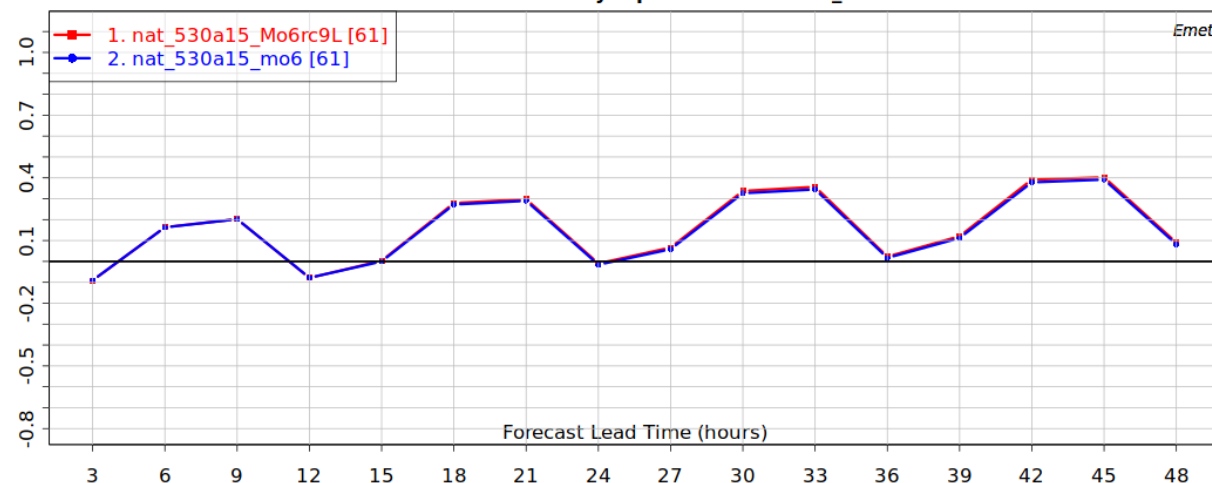
2.2MOM-LF

3.3MOM-LF

Small negative impact

bias		20220401 / 20220630		
nat_530a15_Mo6rc9L / nat_530a15_mo6		00z	12z	All
Arctic All CLIM	TD	-0.00034	0.0023	0.00017
	TT	-0.0069	-0.0004	-0.00024
Boreal CLIM	TD	-0.00066	-0.0014	0.0
	TT	-0.0075	-0.0046	-0.0068
Canada	TD	-5.5e-05	-0.00047	-0.00049
	TT	-0.0097	-0.0071	-0.0055
East CLIM	TD	0.0	0.0	0.0
	TT	-0.0034	-0.0034	-0.00057
Great Lakes CLIM	TD	0.002	-0.0018	-0.0038
	TT	-0.01	-0.0012	-0.0034
Mt West CLIM	TD	-0.0031	-0.0026	0.00022
	TT	-0.016	-0.01	0.011
North Plains CLIM	TD	0.0	0.00065	0.0
	TT	-0.0035	-0.008	-0.0059
Pacific North West CLIM	TD	-0.0038	-0.00064	-0.0018
	TT	-0.011	-0.013	-0.0036
Prairie CLIM	TD	0.0	-0.0052	-0.0017
	TT	0.0	0.00076	0.0
United States of America East	TD	0.00014	0.0011	-0.00089
	TT	-0.0041	-0.0037	-0.0014
United States of America West	TD	-0.0011	-0.00026	0.00078
	TT	-0.012	-0.013	-0.0054
full_domain	TD	-0.0003	-0.0015	-0.00028
	TT	-0.011	-0.0077	-0.0056

MEAN ERROR (P-O) OF SCREEN-LEVEL AIR TEMPERATURE (C) 2022-04-01 @ 2022-06-30
alt diff max 100 ade synop swob metar full domain



EMET (P22), TT & TD 2MOM-LF

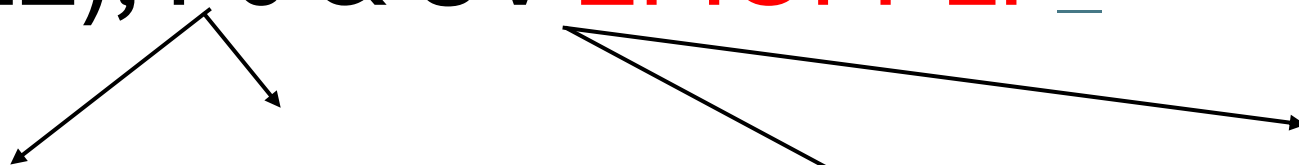
P3_v5.5.0-rc9
P3 v5.3.6 2MOM-noLF

bias				
bias nat_530a15_Mo6rc9L / nat_530a15_mo6		20220401 / 20220630		
		00z	12z	All
Arctic All CLIM	TD	-0.00034	0.0023	0.00017
	TT	-0.0069	-0.0004	-0.00024
Boreal CLIM	TD	-0.00066	-0.0014	0.0
	TT	-0.0075	-0.0046	-0.0068
Canada	TD	-5.5e-05	-0.00047	-0.00049
	TT	-0.0097	-0.0071	-0.0055
East CLIM	TD	0.0	0.0	0.0
	TT	-0.0034	-0.0034	-0.00057
Great Lakes CLIM	TD	0.002	-0.0018	-0.0038
	TT	-0.01	-0.0012	-0.0034
Mt West CLIM	TD	-0.0031	-0.0026	0.00022
	TT	-0.016	-0.01	0.011
North Plains CLIM	TD	0.0	0.00065	0.0
	TT	-0.0035	-0.008	-0.0059
Pacific North West CLIM	TD	-0.0038	-0.00064	-0.0018
	TT	-0.011	-0.013	-0.0036
Prairie CLIM	TD	0.0	-0.0052	-0.0017
	TT	0.0	0.00076	0.0
United States of America East	TD	0.00014	0.0011	-0.00089
	TT	-0.0041	-0.0037	-0.0014
United States of America West	TD	-0.0011	-0.00026	0.00078
	TT	-0.012	-0.013	-0.0054
full_domain	TD	-0.0003	-0.0015	-0.00028
	TT	-0.011	-0.0077	-0.0056

rmse				
rmse nat_530a15_Mo6rc9L / nat_530a15_mo6		20220401 / 20220630		
		00z	12z	All
Arctic All CLIM	TD	-0.0038	-0.0021	-0.0034
	TT	-0.0048	-0.0041	-0.0071
Boreal CLIM	TD	0.0	-0.0026	-0.0009
	TT	-0.0052	-0.0096	-0.009
Canada	TD	0.0	-0.00083	0.0
	TT	-0.0052	-0.004	-0.0052
East CLIM	TD	0.0022	0.0	0.00054
	TT	-5.1e-05	0.0	0.0
Great Lakes CLIM	TD	0.0	0.0	-0.00065
	TT	-0.0021	-0.0039	-0.0032
Mt West CLIM	TD	0.00052	0.0	0.0
	TT	-0.002	0.0	0.0
North Plains CLIM	TD	0.0	0.002	0.0
	TT	-0.00054	0.00027	0.0014
Pacific North West CLIM	TD	0.00047	0.0012	0.00099
	TT	-0.0033	-0.00069	-0.0016
Prairie CLIM	TD	0.0	0.00035	0.00044
	TT	-0.00081	-0.003	-0.002
United States of America East	TD	0.0014	0.00052	0.00031
	TT	-0.0015	-0.0037	-0.0027
United States of America West	TD	0.0	0.0	0.0
	TT	-0.0039	-0.002	-0.0023
full_domain	TD	0.0	-3.5e-05	0.0
	TT	-0.0028	-0.0049	-0.004

stdev				
stdev nat_530a15_Mo6rc9L / nat_530a15_mo6		20220401 / 20220630		
		00z	12z	All
Arctic All CLIM	TD	-0.0041	-0.0031	-0.0045
	TT	-0.0037	-0.0042	-0.0065
Boreal CLIM	TD	0.0	-0.0013	-0.00085
	TT	-0.0051	-0.01	-0.009
Canada	TD	0.0	-0.0013	-0.00064
	TT	-0.0047	-0.0028	-0.0048
East CLIM	TD	0.002	0.0	0.00056
	TT	-0.00028	0.0	0.0
Great Lakes CLIM	TD	0.0	0.0	0.00072
	TT	-0.0015	-0.0043	-0.0035
Mt West CLIM	TD	-0.00095	0.003	7.5e-05
	TT	0.0016	0.0	-0.0029
North Plains CLIM	TD	0.0	0.0025	0.0
	TT	-0.00059	0.0013	0.00074
Pacific North West CLIM	TD	0.00024	0.0	0.00096
	TT	-0.0024	0.0	-0.0016
Prairie CLIM	TD	0.0016	0.0015	0.0018
	TT	-0.0022	-0.0033	-0.0019
United States of America East	TD	0.0015	0.0	0.00069
	TT	-0.0011	-0.0031	-0.0021
United States of America West	TD	0.0	0.0	0.0
	TT	-0.0013	-0.00035	-0.0022
full_domain	TD	0.0	-5.4e-05	0.0
	TT	-0.00075	-0.0022	-0.0028

EMET (P22), P0 & UV 2MOM-LF*



bias				
bias nat_530a15_Mo6rc9L / nat_530a15_mo6		20220401 / 20220630		
		00Z	12z	All
Arctic All CLIM	P0	-0.015	-0.01	-0.016
Boreal CLIM	P0	0.017	0.024	0.021
Canada	P0	0.0035	0.019	0.018
East CLIM	P0	0.0031	0.018	0.011
Great Lakes CLIM	P0	-0.0069	0.0034	-0.005
Mt West CLIM	P0	-0.02	-0.016	-0.021
North Plains CLIM	P0	0.0036	-0.0061	-2.6e-05
Pacific North West CLIM	P0	-0.007	-0.0065	-0.01
Prairie CLIM	P0	-0.0025	-0.0013	-0.0016
United States of America East	P0	-0.01	-0.0046	-0.0084
United States of America West	P0	-0.0068	-0.0071	-0.0088
full_domain	P0	0.0016	0.011	0.0075

rmse				
rmse nat_530a15_Mo6rc9L / nat_530a15_mo6		20220401 / 20220630		
		00Z	12Z	All
Arctic All CLIM	P0	-0.0012	-0.0019	-0.001
Boreal CLIM	P0	0.00075	0.0019	0.0023
Canada	P0	0.00072	0.0014	0.001
East CLIM	P0	0.0	0.0013	0.0
Great Lakes CLIM	P0	-0.0029	-0.0036	0.0
Mt West CLIM	P0	-0.0047	-0.0028	-0.0055
North Plains CLIM	P0	0.0019	0.0	0.0021
Pacific North West CLIM	P0	-0.00028	-0.00024	-0.00017
Prairie CLIM	P0	0.0088	0.0	0.0048
United States of America East	P0	0.0	0.00032	0.0
United States of America West	P0	0.0	0.0	0.0
full_domain	P0	0.00056	0.0008	0.00079

bias				
bias nat_530a15_Mo6rc9L / nat_530a15_mo6		20220401 / 20220630		
		00Z	12Z	All
Arctic All CLIM	UV	0.0067	7.3e-05	0.0046
Boreal CLIM	UV	0.0	-0.002	-0.00048
Canada	UV	-2.9e-05	-0.0015	-0.00054
East CLIM	UV	4.7e-05	0.0	0.00012
Great Lakes CLIM	UV	0.0041	0.0	0.0028
Mt West CLIM	UV	0.0	0.0	0.0
North Plains CLIM	UV	0.00086	0.0052	0.0
Pacific North West CLIM	UV	-0.00067	-0.005	-0.0027
Prairie CLIM	UV	0.0017	0.00012	0.0013
United States of America East	UV	0.0013	0.0	-0.00025
United States of America West	UV	0.0021	-0.00081	0.0
full_domain	UV	0.00016	-0.00044	-0.00015

rmse				
rmse nat_530a15_Mo6rc9L / nat_530a15_mo6		20220401 / 20220630		
		00Z	12z	All
Arctic All CLIM	UV	-0.00036	0.0	0.0
Boreal CLIM	UV	0.0013	0.0	-0.00084
Canada	UV	0.0	-0.00023	0.0
East CLIM	UV	0.0	-0.0007	0.0
Great Lakes CLIM	UV	-0.00014	0.0029	0.0008
Mt West CLIM	UV	0.0	-0.0034	0.0
North Plains CLIM	UV	-0.0044	0.0041	0.0022
Pacific North West CLIM	UV	-0.0021	-0.0049	-0.0021
Prairie CLIM	UV	-0.0023	0.0	0.0
United States of America East	UV	0.0	0.0	-0.00014
United States of America West	UV	-0.0025	-0.00093	-0.0015
full_domain	UV	7.1e-05	0.0	0.0

EMET (P22), PR6 & PR24 2MOM-LF*

P3_v5.5.0-rc9
P3 v5.3.6 2MOM-noLF

ets < >

dpmean

	00z	12z	All
Arctic All CLIM	PR24 -0.0017	-0.0031	-0.0061
	PR6 -0.0013	-0.0012	-0.0013
Boreal CLIM	PR24 0.0	0.0	0.0
	PR6 -0.0027	-9.2e-05	-0.0021
Canada	PR24 -0.0038	-0.0034	-0.0019
	PR6 -0.0039	-0.0027	-0.0071
East CLIM	PR24 -0.0098	0.0015	-0.0016
	PR6 -0.0011	-5.9e-05	-0.0024
Great Lakes CLIM	PR24 -0.0033	-0.0046	-0.0033
	PR6 -0.0022	-0.0069	-0.0011
Mt West CLIM	PR24 -0.0051	-0.0019	-0.0063
	PR6 0.001	0.0062	0.0048
North Plains CLIM	PR24 0.02	0.0	0.0041
	PR6 0.0047	-0.0011	0.0072
Pacific North West CLIM	PR24 -0.026	-0.043	0.012
	PR6 -0.0001	-0.0063	-0.0035
Prairie CLIM	PR24 -0.0041	-0.0011	-0.0083
	PR6 -0.0023	0.0	-0.0018
United States of America East	PR24 0.0	-0.0026	-0.0019
	PR6 -0.0011	-0.0092	-0.0087
United States of America West	PR24 -0.0084	-0.021	-0.017
	PR6 0.0032	-0.0015	-0.0038
full_domain	PR24 -0.0039	-0.0041	-0.004
	PR6 -0.0013	-0.0004	-0.0008

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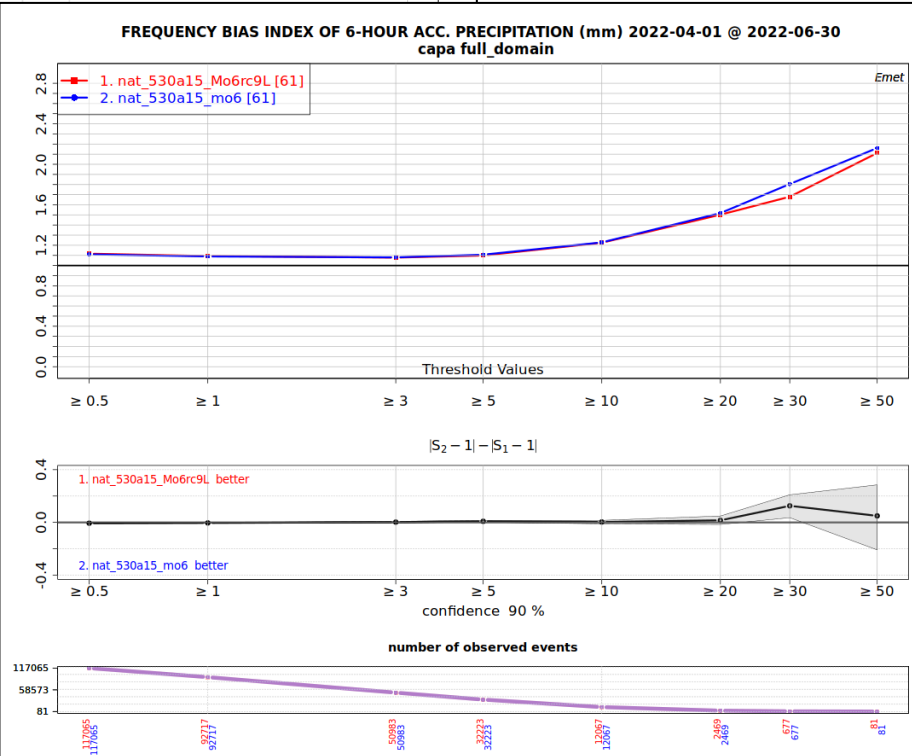
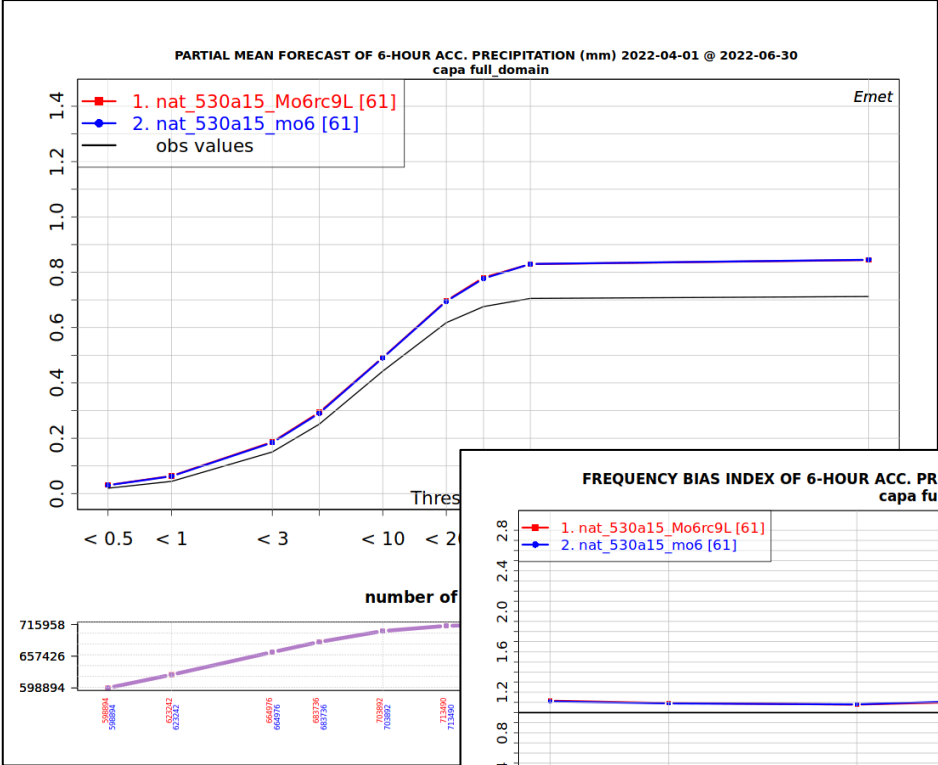
ETS

	00z	12z	All
Arctic All CLIM	PR24 0.012	0.0	-0.0029
	PR6 0.0066	-0.0018	0.0028
Boreal CLIM	PR24 -0.0025	0.0	-0.003
	PR6 -0.0015	0.0	-0.0096
Canada	PR24 0.0038	0.0	0.001
	PR6 -0.0039	0.0093	0.0
East CLIM	PR24 -0.00076	0.0	-0.0011
	PR6 0.0	0.00078	0.0
Great Lakes CLIM	PR24 0.0	0.0	0.0
	PR6 -0.0022	0.009	0.0065
Mt West CLIM	PR24 0.00053	0.00086	0.00072
	PR6 0.0024	0.0019	0.0013
North Plains CLIM	PR24 0.0039	0.0	0.0019
	PR6 -0.0023	0.0044	0.0
Pacific North West CLIM	PR24 0.0024	0.002	-0.0026
	PR6 0.0034	0.0042	0.0026
Prairie CLIM	PR24 0.0	0.0034	0.0013
	PR6 0.0	0.0	-0.00052
United States of America East	PR24 -5e-05	0.00059	0.00043
	PR6 -0.0012	0.0	0.0
United States of America West	PR24 0.0036	0.001	0.00061
	PR6 0.004	0.0	0.0
full_domain	PR24 0.0024	0.00067	0.0014
	PR6 -0.001	0.004	0.0

fbi < >

FBI

	00z	12z	All
Arctic All CLIM	PR24 0.0	-0.0042	-0.0042
	PR6 -0.0068	-0.0049	-0.0052
Boreal CLIM	PR24 0.0	0.0	0.0
	PR6 0.015	-0.0073	0.0
Canada	PR24 0.014	0.0	0.002
	PR6 0.027	0.0	-0.00093
East CLIM	PR24 0.0	0.0036	0.0018
	PR6 0.026	0.0014	-0.00081
Great Lakes CLIM	PR24 0.0	0.0	0.0
	PR6 -0.0074	0.0	-0.031
Mt West CLIM	PR24 0.00091	0.0013	-0.00095
	PR6 0.0	0.0058	0.0055
North Plains CLIM	PR24 0.062	0.0	0.0
	PR6 0.014	0.0	0.0042
Pacific North West CLIM	PR24 -0.0057	0.012	0.0076
	PR6 0.0	-0.0017	-0.007
Prairie CLIM	PR24 0.019	0.0	-0.0012
	PR6 0.0	0.0	0.0
United States of America East	PR24 0.0	0.002	0.0
	PR6 -0.00045	-0.0013	-0.0018
United States of America West	PR24 -0.0058	-0.0096	-0.0057
	PR6 0.0088	-0.002	0.11
full_domain	PR24 -0.0023	-0.0013	-0.0016
	PR6 0.023	0.0	0.015



pod < >

POD

	00z	12z	All
Arctic All CLIM	PR24 0.019	0.0025	0.002
	PR6 0.012	0.0	0.0015
Boreal CLIM	PR24 -0.0048	0.0	0.0
	PR6 -0.014	0.0	-0.015
Canada	PR24 0.0015	0.0	0.0
	PR6 -0.0088	0.0036	0.00047
East CLIM	PR24 -0.0019	0.0	-0.0041
	PR6 0.00097	0.0011	0.00054
Great Lakes CLIM	PR24 0.0013	-0.0028	0.0037
	PR6 -0.006	0.037	0.02
Mt West CLIM	PR24 0.0029	0.002	0.005
	PR6 0.0	0.00092	0.0
North Plains CLIM	PR24 -0.0074	0.0	0.0
	PR6 -0.015	0.0	0.0
Pacific North West CLIM	PR24 0.01	0.0089	0.0041
	PR6 0.0051	0.0042	0.0045
Prairie CLIM	PR24 0.0014	0.0046	0.0037
	PR6 0.0	0.0	-0.00082
United States of America East	PR24 -0.0015	0.0	-0.0026
	PR6 -0.00054	0.0	0.00039
United States of America West	PR24 0.0024	0.0035	0.0028
	PR6 0.0068	0.0	-0.0029
full_domain	PR24 -0.00062	0.00027	0.0017
	PR6 -0.0061	0.0	0.00069

far < >

FAR

	00z	12z	All
Arctic All CLIM	PR24 0.0021	-0.0021	-0.0023
	PR6 0.0083	-0.0025	0.0023
Boreal CLIM	PR24 0.0	-0.005	-0.0054
	PR6 -0.0017	0.0	0.0
Canada	PR24 0.0036	0.0	0.002
	PR6 -0.0047	0.018	0.00058
East CLIM	PR24 0.0	0.0017	0.00078
	PR6 0.0	0.00066	-0.00065
Great Lakes CLIM	PR24 0.0	0.0	0.0
	PR6 -0.003	0.0098	0.0045
Mt West CLIM	PR24 0.0	0.0	0.0
	PR6 0.0058	0.0052	0.004
North Plains CLIM	PR24 0.012	0.001	0.0029
	PR6 0.0034	0.0018	0.0
Pacific North West CLIM	PR24 0.0031	-0.0033	-0.0041
	PR6 0.0033	0.0	0.0
Prairie CLIM	PR24 0.0	0.0039	0.0026
	PR6 0.0	0.0	0.0
United States of America East	PR24 0.0006	0.00088	0.00059
	PR6 0.0	0.00069	0.00041
United States of America West	PR24 0.0087	-0.00057	-0.00089
	PR6 0.0044	0.0	0.0
full_domain	PR24 0.00084	0.00065	0.00086
	PR6 -0.00063	0.0054	0.0

S22: Spring 2022

1.2MOM-noLF

2.2MOM-LF

3.3MOM-LF

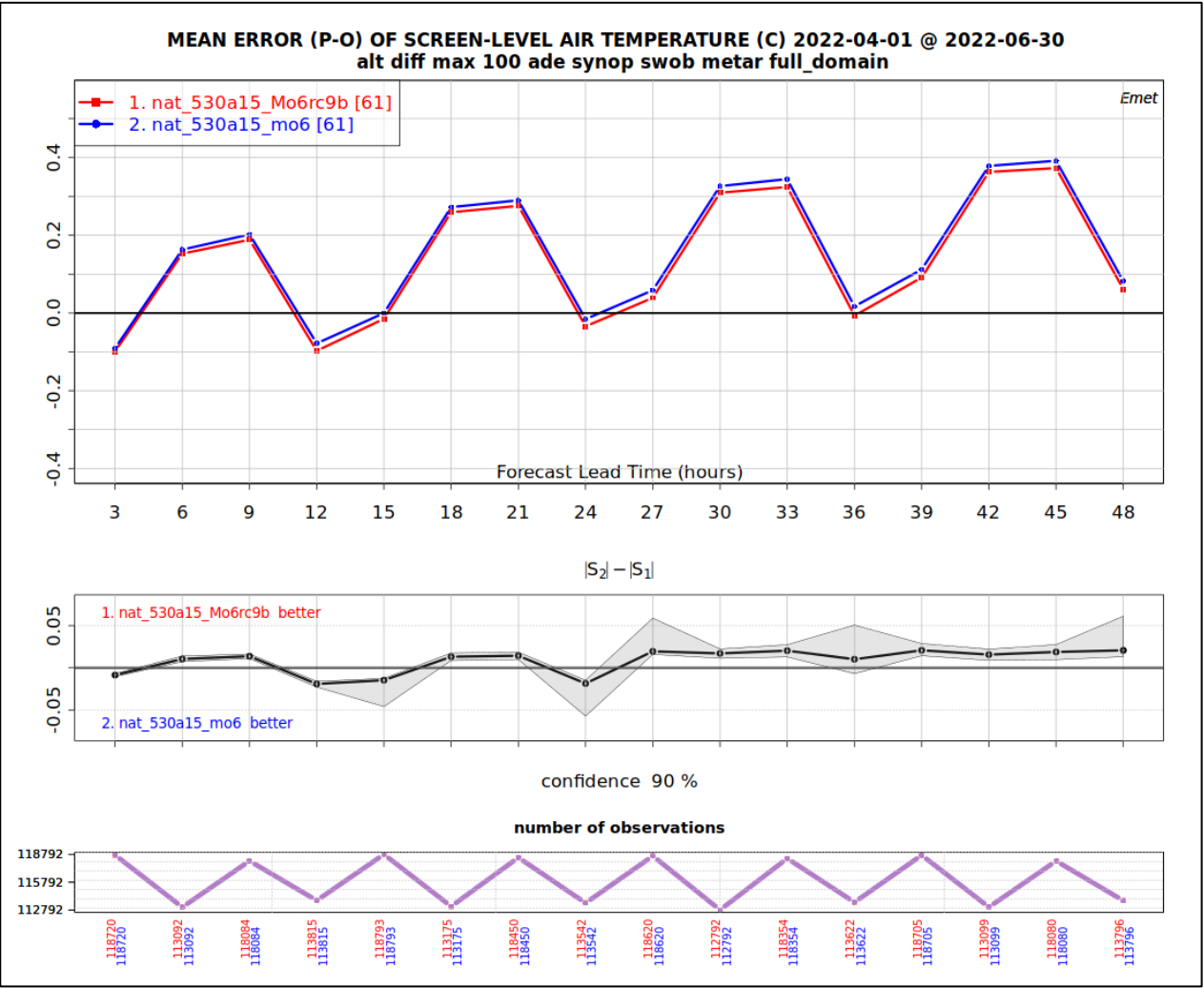
EMET (P22), TT & TD 3MOM-LF*

P3_v5.5.0-rc9
P3 v5.3.6 2MOM-noLF

Mostly positive effects
everywhere, except for
TD

bias nat_530a15_Mo6rc9b / nat_530a15_mo6		20220401 / 20220630		
		00z	12z	All
Arctic All CLIM	TD	-0.01	-0.011	-0.012
	TT	-0.00035	0.0091	0.0068
Boreal CLIM	TD	-0.018	-0.021	-0.023
	TT	-0.0033	-0.00063	0.002
Canada	TD	-0.017	-0.019	-0.022
	TT	0.0083	0.008	0.0083
East CLIM	TD	0.001	0.0	0.00052
	TT	0.0046	0.005	0.0058
Great Lakes CLIM	TD	0.0028	-0.014	-0.006
	TT	-0.01	-0.0015	-0.0042
Mt West CLIM	TD	0.0043	0.0044	0.00057
	TT	-0.0058	-0.0041	-0.021
North Plains CLIM	TD	-0.026	-0.021	-0.027
	TT	0.018	0.021	0.021
Pacific North West CLIM	TD	0.012	0.02	0.021
	TT	0.0077	-0.002	0.012
Prairie CLIM	TD	0.0	-0.0039	-0.003
	TT	0.0038	0.004	0.003
United States of America East	TD	0.00032	-0.0015	0.00024
	TT	0.0067	0.0063	0.0071
United States of America West	TD	0.0053	0.003	-0.0044
	TT	0.0032	0.0099	0.0029
full_domain	TD	-0.0056	-0.012	-0.013
	TT	0.0075	0.011	0.0077

TT-bias (full domain)



EMET (P22), TT & TD 3MOM-LF*

P3_v5.5.0-rc9
P3 v5.3.6 2MOM-noLF

bias nat_530a15_Mo6rc9b / nat_530a15_mo6		20220401 / 20220630		
		00z	12z	All
Arctic All CLIM	TD	-0.01	-0.011	-0.012
	TT	-0.00035	0.0091	0.0068
Boreal CLIM	TD	-0.018	-0.021	-0.023
	TT	-0.0033	-0.00063	0.002
Canada	TD	-0.017	-0.019	-0.022
	TT	0.0083	0.008	0.0083
East CLIM	TD	0.001	0.0	0.00052
	TT	0.0046	0.005	0.0058
Great Lakes CLIM	TD	0.0028	-0.014	-0.006
	TT	-0.01	-0.0015	-0.0042
Mt West CLIM	TD	0.0043	0.0044	0.00057
	TT	-0.0058	-0.0041	-0.021
North Plains CLIM	TD	-0.026	-0.021	-0.027
	TT	0.018	0.021	0.021
Pacific North West CLIM	TD	0.012	0.02	0.021
	TT	0.0077	-0.002	0.012
Prairie CLIM	TD	0.0	-0.0039	-0.003
	TT	0.0038	0.004	0.003
United States of America East	TD	0.00032	-0.0015	0.00024
	TT	0.0067	0.0063	0.0071
United States of America West	TD	0.0053	0.003	-0.0044
	TT	0.0032	0.0099	0.0029
full_domain	TD	-0.0056	-0.012	-0.013
	TT	0.0075	0.011	0.0077

rmse nat_530a15_Mo6rc9b / nat_530a15_mo6		20220401 / 20220630		
		00z	12z	All
Arctic All CLIM	TD	-0.0035	-0.0019	-0.0022
	TT	-0.0079	-0.0066	-0.0083
Boreal CLIM	TD	-0.0073	-0.0032	-0.007
	TT	0.0	-0.0031	0.0
Canada	TD	-0.006	-0.0031	-0.005
	TT	-0.001	0.0	0.0
East CLIM	TD	0.0014	0.0027	0.0014
	TT	0.0049	0.0071	0.0059
Great Lakes CLIM	TD	0.0	0.0044	0.0034
	TT	0.0029	0.0019	0.0021
Mt West CLIM	TD	0.0	0.0	-0.00049
	TT	0.0024	0.0052	0.004
North Plains CLIM	TD	-0.0032	0.0	0.0
	TT	0.0041	0.0035	0.0
Pacific North West CLIM	TD	0.0025	0.01	0.0044
	TT	0.0024	0.0027	0.0018
Prairie CLIM	TD	0.0	0.0023	0.0
	TT	0.0	0.0027	0.0
United States of America East	TD	0.0	0.0051	0.0015
	TT	0.0021	0.0038	0.003
United States of America West	TD	0.00072	0.0	0.0
	TT	0.00092	0.0089	0.0051
full_domain	TD	-0.0017	0.0	-0.00012
	TT	0.002	0.0016	0.0017

stdev nat_530a15_Mo6rc9b / nat_530a15_mo6		20220401 / 20220630		
		00z	12z	All
Arctic All CLIM	TD	-0.0032	-0.0011	-0.0017
	TT	-0.0084	-0.0079	-0.01
Boreal CLIM	TD	-0.0033	0.0	-0.0019
	TT	0.00082	-0.0024	-0.0026
Canada	TD	-0.0012	0.0	-0.00038
	TT	-0.0021	-0.00095	-0.0015
East CLIM	TD	0.0019	0.0027	0.0013
	TT	0.0018	0.0062	0.0044
Great Lakes CLIM	TD	-0.00088	0.0061	0.0038
	TT	0.0033	0.0021	0.0022
Mt West CLIM	TD	0.00057	0.0	0.00015
	TT	0.0056	0.0084	0.0078
North Plains CLIM	TD	0.0	0.0	0.0
	TT	0.0023	0.0	0.0
Pacific North West CLIM	TD	0.0	0.0065	0.0
	TT	-0.00023	0.0034	0.0015
Prairie CLIM	TD	0.0	0.0027	0.0013
	TT	0.0	0.0022	0.0
United States of America East	TD	0.0	0.0053	0.0018
	TT	0.0016	0.0	0.0024
United States of America West	TD	0.0	0.0	0.0
	TT	0.002	0.0068	0.0052
full_domain	TD	1.3e-05	0.0	0.001
	TT	0.002	0.0014	0.00048

EMET (P22), P0 & UV 3MOM-LF*

bias		20220401 / 20220630		
nat_530a15_Mo6rc9b / nat_530a15_mo6		00z	12z	All
Arctic All CLIM	P0	-0.0089	-0.013	-0.015
Boreal CLIM	P0	0.0057	0.015	0.013
Canada	P0	-0.00076	0.01	0.011
East CLIM	P0	0.0032	0.011	0.0095
Great Lakes CLIM	P0	0.003	-0.0042	0.0035
Mt West CLIM	P0	-0.0025	-0.002	-0.00038
North Plains CLIM	P0	-0.0016	0.0034	0.0
Pacific North West CLIM	P0	-0.0067	-0.0053	-0.0056
Prairie CLIM	P0	-0.0016	0.00071	0.004
United States of America East	P0	-0.00014	0.0031	-0.00052
United States of America West	P0	0.0	0.0	0.0026
full_domain	P0	-3.6e-05	0.0035	0.0038

rmse		20220401 / 20220630		
nat_530a15_Mo6rc9b / nat_530a15_mo6		00z	12z	All
Arctic All CLIM	P0	0.0018	0.0	0.00027
Boreal CLIM	P0	0.0	0.0037	0.002
Canada	P0	0.0014	0.0024	0.0017
East CLIM	P0	0.0	0.00046	0.0022
Great Lakes CLIM	P0	0.0	0.0	0.0
Mt West CLIM	P0	0.0	0.0015	0.002
North Plains CLIM	P0	0.0	0.0055	0.0017
Pacific North West CLIM	P0	0.00091	0.0	0.0013
Prairie CLIM	P0	0.0026	0.0091	0.0096
United States of America East	P0	0.0	0.0017	0.0008
United States of America West	P0	0.002	0.0	0.0023
full_domain	P0	0.00047	0.0024	0.002

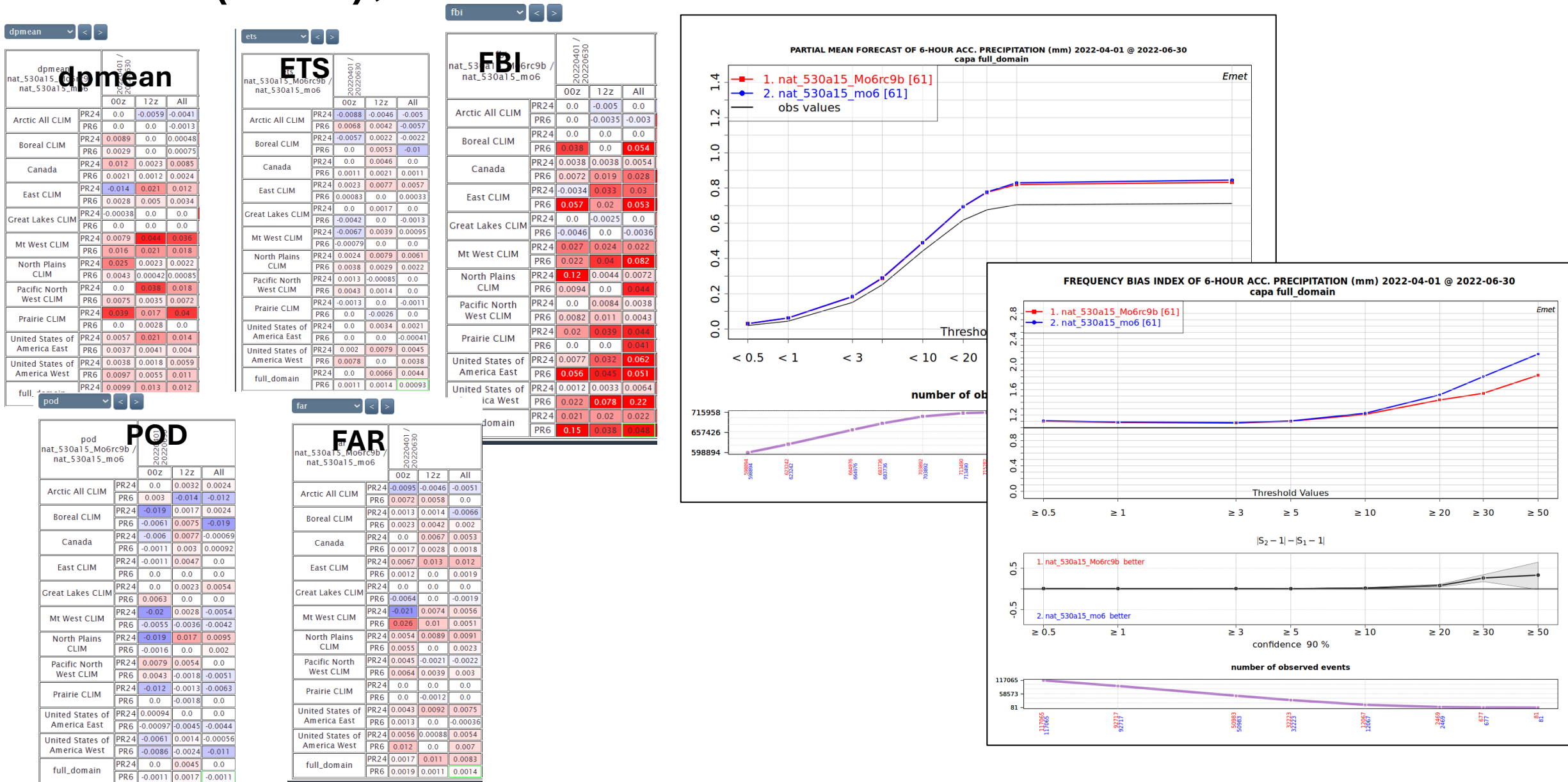
bias		20220401 / 20220630		
nat_530a15_Mo6rc9b / nat_530a15_mo6		00z	12z	All
Arctic All CLIM	UV	0.0029	-0.002	0.0022
Boreal CLIM	UV	-0.00061	-5.7e-05	-0.0011
Canada	UV	0.00038	-0.0009	-0.0018
East CLIM	UV	0.0041	0.0033	0.0038
Great Lakes CLIM	UV	0.0054	0.002	0.005
Mt West CLIM	UV	0.0053	0.0071	0.0053
North Plains CLIM	UV	0.00078	0.00039	0.00023
Pacific North West CLIM	UV	-0.011	-0.013	-0.016
Prairie CLIM	UV	-0.0019	0.0	0.0032
United States of America East	UV	0.0034	0.002	0.0026
United States of America West	UV	0.0024	0.0037	0.005
full_domain	UV	-0.00022	0.00022	0.00016

rmse		20220401 / 20220630		
nat_530a15_Mo6rc9b / nat_530a15_mo6		00z	12z	All
Arctic All CLIM	UV	0.0	-0.0042	-0.0017
Boreal CLIM	UV	0.0017	0.0003	0.0
Canada	UV	5.7e-06	-0.0027	-0.0021
East CLIM	UV	0.0026	-0.00073	0.00058
Great Lakes CLIM	UV	-0.0016	0.0012	-0.00046
Mt West CLIM	UV	0.003	-0.013	0.0
North Plains CLIM	UV	-0.0054	-0.002	-0.0035
Pacific North West CLIM	UV	-0.0041	-0.0037	-0.0052
Prairie CLIM	UV	-5.2e-05	-0.0031	-0.0019
United States of America East	UV	-0.0016	0.0	-0.00098
United States of America West	UV	-0.0039	-0.0053	-0.0038
full_domain	UV	-0.002	-0.00095	-0.0017

EMET (P22), PR6 & PR24 3MOM-LF*

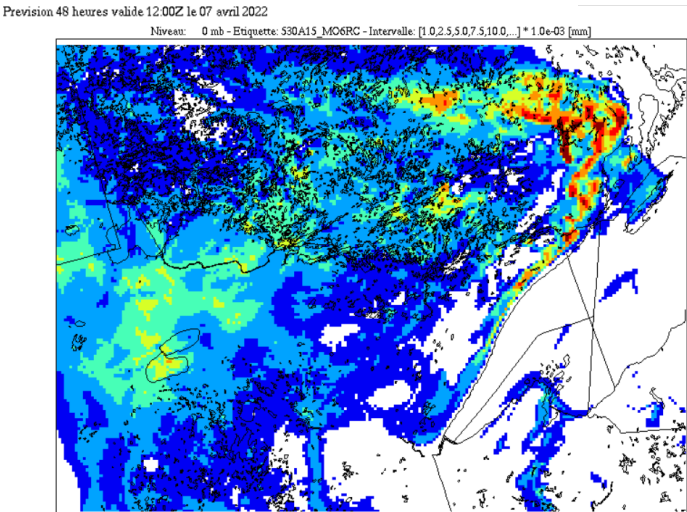
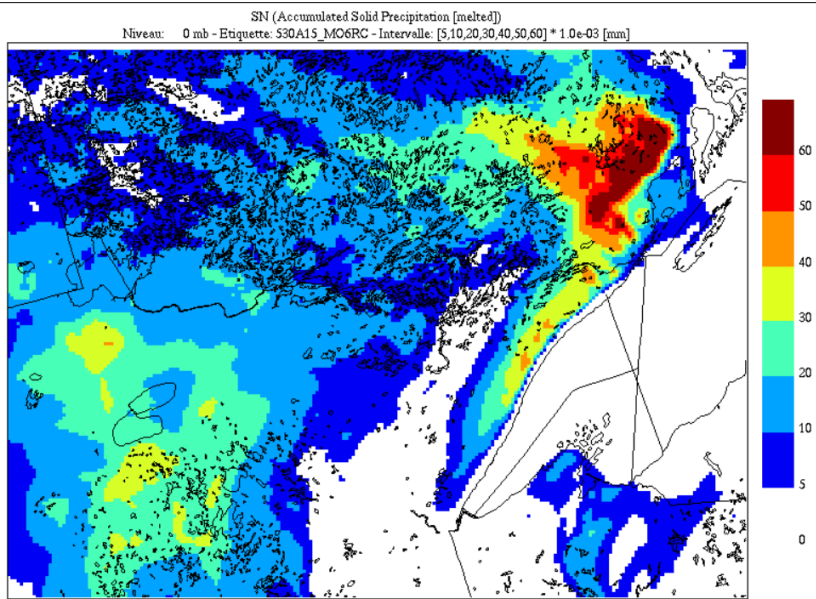
P3_v5.5.0-rc9

P3 v5.3.6 2MOM-noLF



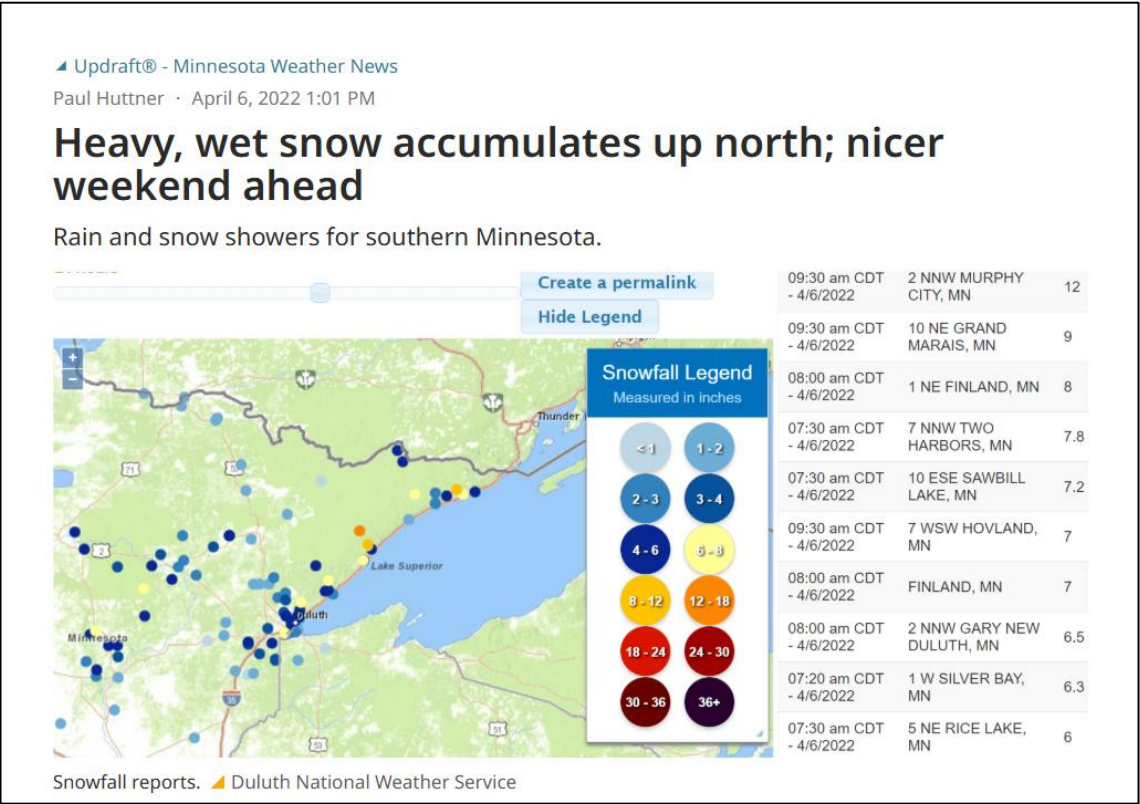
Other advantage during cold seasons of LF: wet snow

Total snow



Wet snow given in
P3 with LF

Up to 35 mm of wet (heavy) snow observed
On April 6, 2022



Summer (E22)

Summer (E22)

1.2MOM-noLF

2.2MOM-LF

3.3MOM-LF

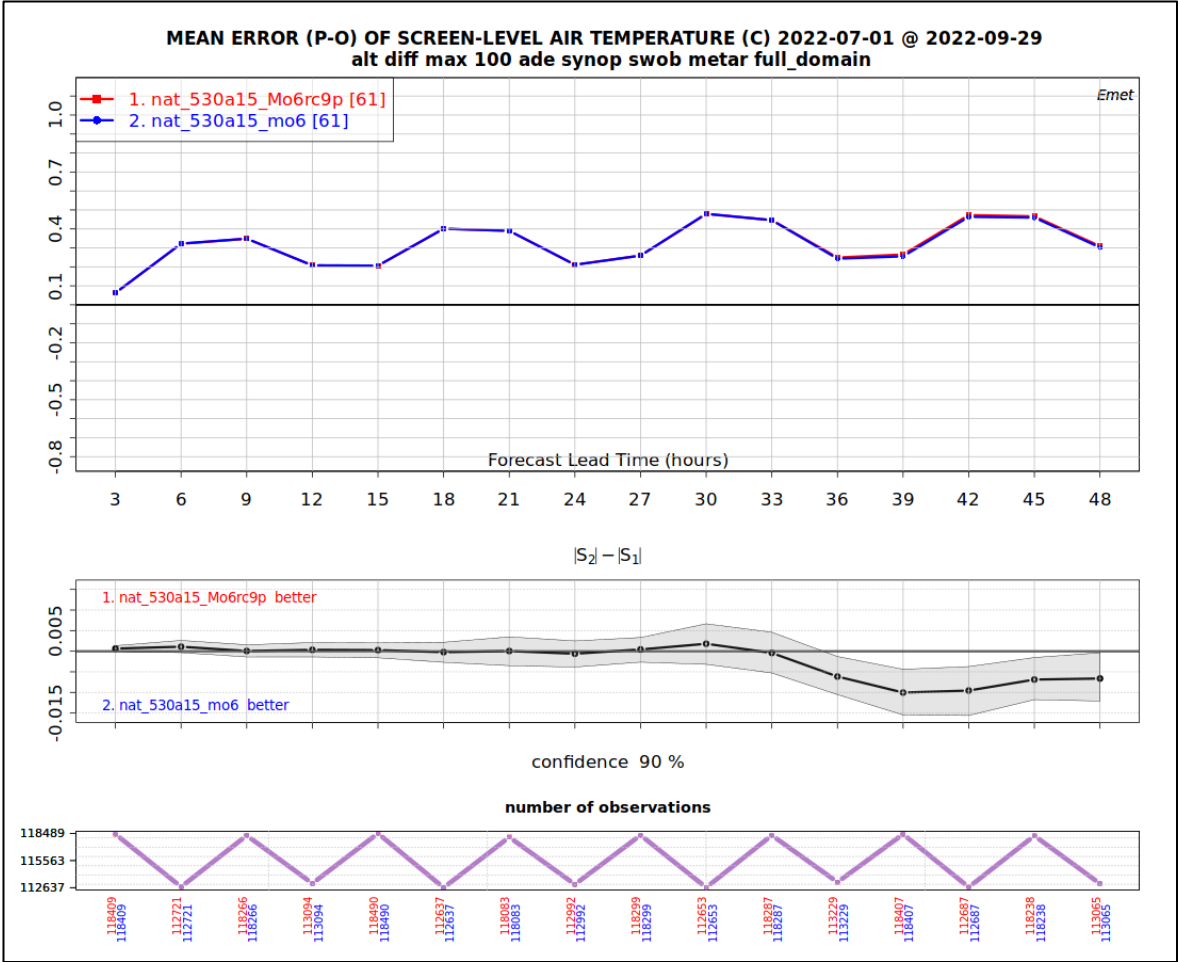
EMET (E22), TT & TD 2MOM-noLF*

P3_v5.5.0-rc9
P3 v5.3.6 2MOM-noLF

Pretty neutral

bias		20220701 / 20220929		
nat_530a15_Mo6rc9p / nat_530a15_mo6		00z	12z	All
Arctic All CLIM	TD	0.00098	-0.0014	-0.00041
	TT	0.0094	0.0098	-9.2e-05
Boreal CLIM	TD	0.00057	0.0	0.0
	TT	-0.0035	-0.0055	-0.0041
Canada	TD	-8.5e-05	0.0	0.0
	TT	-0.0031	-0.0034	-0.0025
East CLIM	TD	0.0	0.0	0.0
	TT	0.0004	0.0	0.0
Great Lakes CLIM	TD	-0.00012	0.0014	5e-05
	TT	0.00036	-0.00094	-0.0018
Mt West CLIM	TD	-0.00028	0.0	0.0015
	TT	0.0	0.0	0.0
North Plains CLIM	TD	0.0	-0.0018	0.0
	TT	-0.0021	-0.0072	-0.0053
Pacific North West CLIM	TD	0.0	0.001	-0.00073
	TT	1.6e-05	0.0	-1.8e-05
Prairie CLIM	TD	3.9e-05	0.0	0.0
	TT	0.0015	-0.012	0.00098
United States of America East	TD	0.00091	0.0	-0.00022
	TT	0.00067	-0.0028	-0.0014
United States of America West	TD	-0.00059	0.0025	0.0
	TT	0.0	-0.00041	-0.0015
full_domain	TD	0.0	0.0	0.0
	TT	0.00011	-0.0037	-0.0025

TT-bias (full domain)



EMET (E22), P0 & UV 2MOM-noLF*

bias < >

bias nat_530a15_Mo6rc9p / nat_530a15_mo6		20220701 / 20220929		
		00z	12z	All
Arctic All CLIM	P0	0.019	0.015	0.017
Boreal CLIM	P0	0.011	0.0088	0.011
Canada	P0	0.0093	0.0072	0.0085
East CLIM	P0	-0.0031	-0.00017	0.002
Great Lakes CLIM	P0	-0.0042	0.0	-0.0019
Mt West CLIM	P0	-0.0024	-0.0024	-0.00062
North Plains CLIM	P0	-0.0017	0.00043	-0.003
Pacific North West CLIM	P0	-0.00028	0.0031	0.0019
Prairie CLIM	P0	0.0	-0.0041	-0.0032
United States of America East	P0	-0.0016	0.0	-0.00082
United States of America West	P0	-0.0027	-0.0064	-0.0041
full_domain	P0	0.0062	0.0048	0.0077

rmse < >

rmse nat_530a15_Mo6rc9p / nat_530a15_mo6		20220701 / 20220929		
		00z	12z	All
Arctic All CLIM	P0	0.011	0.0049	0.0092
Boreal CLIM	P0	0.0033	0.0013	0.0023
Canada	P0	0.0033	0.0022	0.0029
East CLIM	P0	0.0	0.0	0.0029
Great Lakes CLIM	P0	0.0	0.0029	0.0
Mt West CLIM	P0	0.0	-0.0013	-0.00051
North Plains CLIM	P0	0.0	-0.00013	0.0
Pacific North West CLIM	P0	0.0	-0.00031	0.0
Prairie CLIM	P0	0.0	0.0	0.0
United States of America East	P0	0.0	0.0039	0.0027
United States of America West	P0	-0.0034	0.0	-0.0028
full_domain	P0	0.0022	0.0025	0.0024

bias < >

bias nat_530a15_Mo6rc9p / nat_530a15_mo6		20220701 / 20220929		
		00z	12z	All
Arctic All CLIM	UV	0.0	0.0046	0.0029
Boreal CLIM	UV	0.0	0.0024	0.0
Canada	UV	-0.00028	0.0025	0.0013
East CLIM	UV	0.0	0.0011	0.00057
Great Lakes CLIM	UV	-0.00017	0.0	0.0
Mt West CLIM	UV	0.0	0.0	0.0
North Plains CLIM	UV	-0.00022	-0.0017	0.0
Pacific North West CLIM	UV	-0.00079	0.0	0.0
Prairie CLIM	UV	0.0	0.0	0.0
United States of America East	UV	-0.00015	0.0	0.00091
United States of America West	UV	0.0018	0.00046	0.0016
full_domain	UV	0.00095	0.00078	0.00075

rmse < >

rmse nat_530a15_Mo6rc9p / nat_530a15_mo6		20220701 / 20220929		
		00z	12z	All
Arctic All CLIM	UV	0.0037	0.0068	0.0063
Boreal CLIM	UV	-0.002	0.0	-0.0013
Canada	UV	0.00013	0.0	0.0011
East CLIM	UV	0.0	0.0	0.0
Great Lakes CLIM	UV	0.0015	0.0	0.0
Mt West CLIM	UV	-0.0012	-0.00017	0.00036
North Plains CLIM	UV	-0.0002	0.0	0.00041
Pacific North West CLIM	UV	0.0	0.0	-0.00086
Prairie CLIM	UV	0.0	0.0027	5.6e-05
United States of America East	UV	0.0	0.0012	0.0011
United States of America West	UV	0.0015	0.0023	0.0
full_domain	UV	0.0004	0.0	0.0

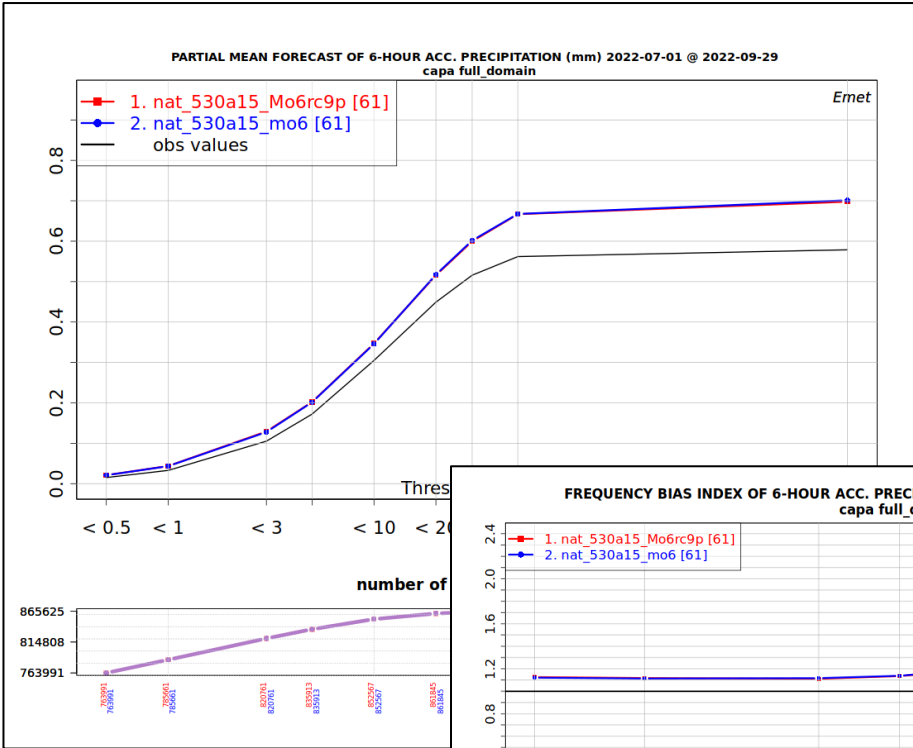
EMET (E22), PR6 & PR24 2MOM-noLF*

P3_v5.5.0-rc9
P3 v5.3.6 2MOM-noLF

dpmean				
dpmean				
nat_530a15_Mo6rc9p / nat_530a15_mo6				
	00z	12z	All	
Arctic All CLIM	PR24 -0.011	-0.00082	-0.0064	
	PR6 -0.001	-0.0034	-0.0027	
Boreal CLIM	PR24 -0.0029	0.003	0.011	
	PR6 0.0	0.00028	0.0017	
Canada	PR24 -0.0019	0.0069	-0.0062	
	PR6 -0.00022	0.0014	0.00048	
East CLIM	PR24 0.0	-0.0031	0.0032	
	PR6 0.0	0.0	0.0	
Great Lakes CLIM	PR24 -0.01	0.0072	0.002	
	PR6 0.0	0.0	0.0	
Mt West CLIM	PR24 -0.00074	0.0	0.0	
	PR6 0.0	0.00039	4.6e-05	
North Plains CLIM	PR24 0.0	0.0	0.0	
	PR6 0.0022	0.0	0.002	
Pacific North West CLIM	PR24 -0.0024	-0.0054	-0.0056	
	PR6 -0.0004	-0.0066	-0.0043	
Prairie CLIM	PR24 0.02	0.011	0.0078	
	PR6 -0.00019	0.0	-0.00013	
United States of America East	PR24 0.013	0.00097	0.0035	
	PR6 0.0	-0.0033	0.0	
United States of America West	PR24 -0.0074	0.00012	-0.0034	
	PR6 -0.0007	-0.0029	0.0	
full_domain	PR24 0.003	0.0	0.0018	
	PR6 -0.00021	0.0	-0.00019	

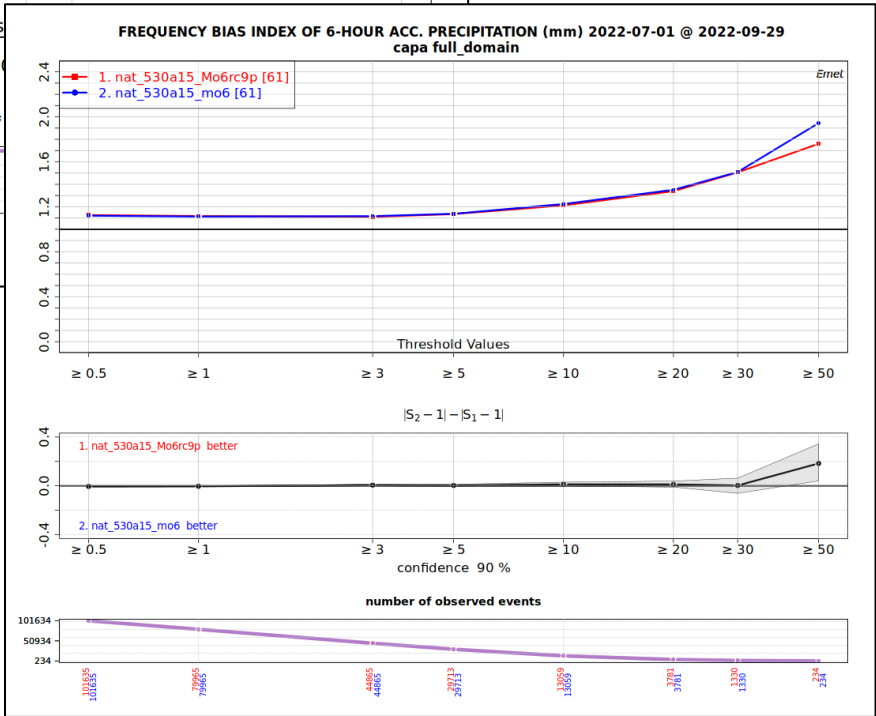
ETS				
ETS				
nat_530a15_Mo6rc9p / nat_530a15_mo6				
	00z	12z	All	
Arctic All CLIM	PR24 0.0	-0.0015	-0.0026	
	PR6 0.0	-0.0035	-0.00093	
Boreal CLIM	PR24 0.0	0.0	0.0	
	PR6 -0.0022	0.0022	-0.0012	
Canada	PR24 0.0015	0.00057	0.0018	
	PR6 0.0026	0.0033	0.0026	
East CLIM	PR24 0.0012	0.0014	0.00052	
	PR6 -0.00097	0.0	0.0	
Great Lakes CLIM	PR24 0.0	0.0051	0.0027	
	PR6 0.0	0.0014	0.0	
Mt West CLIM	PR24 0.0	0.0021	0.0	
	PR6 0.0	0.0049	0.00097	
North Plains CLIM	PR24 -0.00086	0.0022	0.0031	
	PR6 0.0	0.0	0.0013	
Pacific North West CLIM	PR24 -0.0064	-0.004	0.0	
	PR6 -0.015	-0.0046	-0.01	
Prairie CLIM	PR24 0.0044	0.0	0.0014	
	PR6 0.0017	0.0	0.0013	
United States of America East	PR24 0.0	0.0016	0.0012	
	PR6 0.0	0.0	0.00093	
United States of America West	PR24 -0.0014	0.0	0.0	
	PR6 0.0064	0.0076	0.0056	
full_domain	PR24 0.00085	0.0014	0.0017	
	PR6 0.0	0.0013	0.00048	

FBI				
FBI				
nat_530a15_Mo6rc9p / nat_530a15_mo6				
	00z	12z	All	
Arctic All CLIM	PR24 0.0066	-0.0026	-0.0023	
	PR6 0.031	-0.017	-0.0049	
Boreal CLIM	PR24 0.019	0.029	0.035	
	PR6 0.0	0.011	0.00041	
Canada	PR24 -0.0037	0.02	0.0057	
	PR6 0.0017	0.019	0.0031	
East CLIM	PR24 0.0	0.0	0.0	
	PR6 0.0018	0.0	0.0	
Great Lakes CLIM	PR24 0.0	-0.0027	0.0015	
	PR6 0.033	0.0	0.0	
Mt West CLIM	PR24 -0.0041	0.0	0.0	
	PR6 -0.0044	0.0	0.0	
North Plains CLIM	PR24 0.0	0.019	0.0	
	PR6 0.013	0.0	0.0051	
Pacific North West CLIM	PR24 -0.0042	0.004	-0.0078	
	PR6 0.005	-0.023	-0.003	
Prairie CLIM	PR24 0.0058	0.0	0.0	
	PR6 0.0	0.0	-0.0024	
United States of America East	PR24 0.0014	0.0	0.0	
	PR6 0.0	-0.003	0.0	
United States of America West	PR24 -0.012	0.0	-0.0018	
	PR6 0.01	-0.0085	0.0	
full_domain	PR24 0.0	0.0	0.0038	
	PR6 0.0012	-0.0016	0.022	



POD				
POD				
nat_530a15_Mo6rc9p / nat_530a15_mo6				
	00z	12z	All	
Arctic All CLIM	PR24 0.0	0.0	0.0012	
	PR6 0.0094	-0.0024	0.00096	
Boreal CLIM	PR24 0.0	0.0	0.0	
	PR6 -0.0046	0.00099	-0.0032	
Canada	PR24 0.0048	0.0	-8.2e-05	
	PR6 0.0	0.0051	0.00089	
East CLIM	PR24 0.0043	0.0	0.0	
	PR6 -0.0016	0.0	0.0	
Great Lakes CLIM	PR24 -0.0084	0.0075	0.0049	
	PR6 0.0	0.0	0.0	
Mt West CLIM	PR24 0.0	0.0031	0.0021	
	PR6 0.0	0.0063	0.0015	
North Plains CLIM	PR24 0.0052	0.0	0.0025	
	PR6 0.00077	0.0	0.0	
Pacific North West CLIM	PR24 -0.0087	0.0	0.0	
	PR6 -0.012	-0.0035	-0.0023	
Prairie CLIM	PR24 0.0	0.0	0.0	
	PR6 0.0031	0.0	0.0024	
United States of America East	PR24 0.0	0.0022	0.00093	
	PR6 0.0	0.0024	0.0021	
United States of America West	PR24 -0.0041	0.0013	0.0	
	PR6 0.014	0.017	0.014	
full_domain	PR24 0.0026	0.0015	-0.00039	
	PR6 0.0	0.0032	0.00035	

FAR				
FAR				
nat_530a15_Mo6rc9p / nat_530a15_mo6				
	00z	12z	All	
Arctic All CLIM	PR24 -0.0093	0.0037	-0.0025	
	PR6 0.0	-0.006	-0.0026	
Boreal CLIM	PR24 0.0031	0.0	0.0045	
	PR6 -0.003	0.0028	0.0	
Canada	PR24 -0.00095	0.0053	0.0047	
	PR6 0.0047	0.0	0.0049	
East CLIM	PR24 0.002	0.0013	0.0012	
	PR6 0.0	0.0	0.0	
Great Lakes CLIM	PR24 -0.0032	0.0069	0.0028	
	PR6 0.0	0.0015	0.0	
Mt West CLIM	PR24 0.0063	0.00066	0.0	
	PR6 0.0	0.0061	0.00089	
North Plains CLIM	PR24 0.0019	0.004	0.0052	
	PR6 0.0	0.0037	0.0064	
Pacific North West CLIM	PR24 -0.013	-0.0026	-0.0082	
	PR6 -0.025	-0.012	-0.02	
Prairie CLIM	PR24 0.0087	0.0	0.0019	
	PR6 0.0	0.0	-0.00064	
United States of America East	PR24 0.0012	0.002	0.0016	
	PR6 0.0	0.0	0.0011	
United States of America West	PR24 -0.0022	0.0	0.0015	
	PR6 0.014	0.007	0.0086	
full_domain	PR24 0.0018	0.0018	0.0025	
	PR6 0.0	0.0029	0.0021	



Summer (E22)

1.2MOM-noLF

2.2MOM-LF

3.3MOM-LF

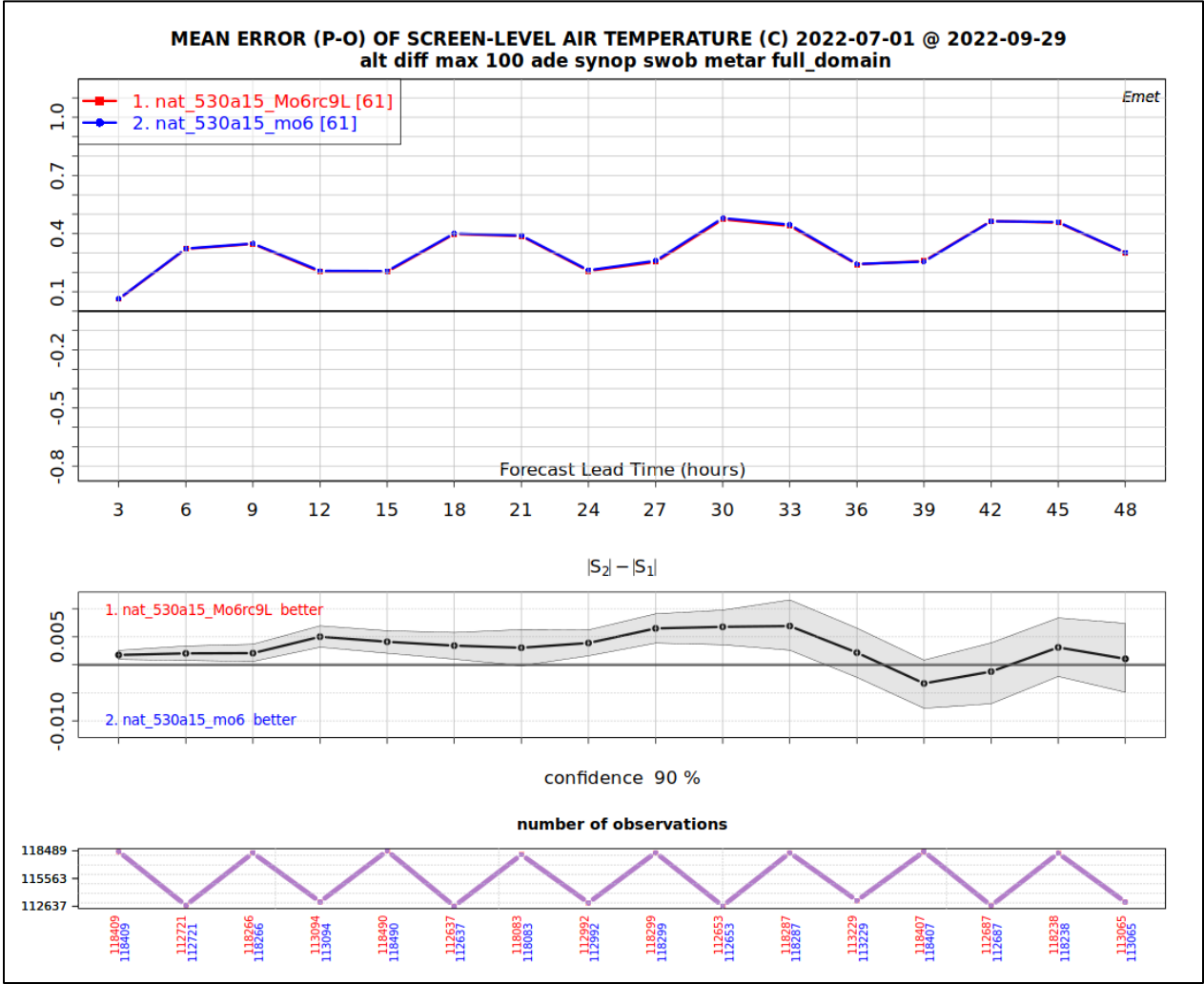
EMET (E22), TT & TD 2MOM-LF*

P3_v5.5.0-rc9
P3 v5.3.6 2MOM-noLF

Small positive or neutral
effects everywhere

bias		< >		
bias		20220701 / 20220929		
nat_530a15_Mo6rc9L / nat_530a15_mo6		00z	12z	All
Arctic All CLIM	TD	0.0	8.4e-05	-0.00079
	TT	0.0053	0.0074	-0.0038
Boreal CLIM	TD	0.002	0.0	0.00051
	TT	-0.00094	0.00019	-0.00037
Canada	TD	0.00054	0.0	-0.001
	TT	0.0019	0.0015	0.0016
East CLIM	TD	-0.00072	-0.0008	0.0
	TT	0.0015	0.0	0.0012
Great Lakes CLIM	TD	-0.00052	0.00063	0.00027
	TT	0.0038	-0.0036	-0.00079
Mt West CLIM	TD	0.0013	0.0	0.0
	TT	0.0024	0.0057	-0.00048
North Plains CLIM	TD	-0.00049	0.0	-0.0046
	TT	0.0018	0.0024	0.0036
Pacific North West CLIM	TD	0.0	0.0	0.00019
	TT	0.0	-4.7e-06	0.0
Prairie CLIM	TD	-0.00065	0.0	0.00022
	TT	0.0007	-0.0018	0.00048
United States of America East	TD	-0.0021	0.0	-0.00061
	TT	0.0035	0.00059	0.0026
United States of America West	TD	0.00018	0.002	0.0
	TT	0.0044	0.003	0.0024
full_domain	TD	0.00028	-0.00059	-0.00035
	TT	0.0039	0.0015	0.0026

TT-bias (full domain)



EMET (E22), P0 & UV 2MOM-LF*

P3_v5.5.0-rc9
P3 v5.3.6 2MOM-noLF

bias		20220701 / 20220929		
nat_530a15_Mo6rc9L / nat_530a15_mo6		00z	12z	All
Arctic All CLIM	P0	0.017	0.015	0.016
Boreal CLIM	P0	0.0089	0.0093	0.0094
Canada	P0	0.0083	0.0068	0.0078
East CLIM	P0	-0.0041	2.4e-05	0.00086
Great Lakes CLIM	P0	-0.00035	0.0	0.0
Mt West CLIM	P0	-0.0023	0.0011	-0.00026
North Plains CLIM	P0	-0.00031	0.0014	-0.0015
Pacific North West CLIM	P0	0.0	0.00092	0.00064
Prairie CLIM	P0	0.0	-0.011	-0.0011
United States of America East	P0	-0.0012	0.0	0.0
United States of America West	P0	0.0	0.0	-0.0022
full_domain	P0	0.005	0.0039	0.0069

rmse		20220701 / 20220929		
nat_530a15_Mo6rc9L / nat_530a15_mo6		00z	12z	All
Arctic All CLIM	P0	0.0078	0.0072	0.0075
Boreal CLIM	P0	0.002	0.002	0.0019
Canada	P0	0.0033	0.0031	0.0039
East CLIM	P0	0.0	0.0027	0.0
Great Lakes CLIM	P0	0.0	0.0	0.0022
Mt West CLIM	P0	0.0	7.1e-05	-0.00035
North Plains CLIM	P0	7.1e-05	0.0052	0.0028
Pacific North West CLIM	P0	-0.00016	-0.00088	-0.00063
Prairie CLIM	P0	-0.00047	0.0	0.0
United States of America East	P0	0.0	0.0027	0.0012
United States of America West	P0	0.0	0.0	-1.7e-05
full_domain	P0	0.0023	0.0033	0.0034

bias		20220701 / 20220929		
nat_530a15_Mo6rc9L / nat_530a15_mo6		00z	12z	All
Arctic All CLIM	UV	0.0023	0.0042	0.0025
Boreal CLIM	UV	0.0	0.0	0.0
Canada	UV	0.0015	0.0012	0.0014
East CLIM	UV	0.0	0.0	0.00033
Great Lakes CLIM	UV	0.0	0.0056	0.0029
Mt West CLIM	UV	0.0	0.00022	-0.0012
North Plains CLIM	UV	0.0	0.0	0.0019
Pacific North West CLIM	UV	0.0	0.0	0.0
Prairie CLIM	UV	0.0	0.0	0.0
United States of America East	UV	0.0024	0.0017	0.0026
United States of America West	UV	7.7e-05	0.0	0.00076
full_domain	UV	0.0019	0.00069	0.0019

rmse		20220701 / 20220929		
nat_530a15_Mo6rc9L / nat_530a15_mo6		00z	12z	All
Arctic All CLIM	UV	0.0017	0.0017	0.0012
Boreal CLIM	UV	0.0	-0.0021	-0.00064
Canada	UV	0.00011	0.00065	-0.00014
East CLIM	UV	0.00024	0.0	0.0
Great Lakes CLIM	UV	0.00011	0.0	6e-05
Mt West CLIM	UV	0.0058	0.0031	0.0017
North Plains CLIM	UV	0.0	0.0011	0.00015
Pacific North West CLIM	UV	0.0003	0.00044	0.00077
Prairie CLIM	UV	-5e-05	0.0	0.0
United States of America East	UV	-0.00087	0.0016	0.00012
United States of America West	UV	0.0044	0.0	0.00084
full_domain	UV	0.0017	0.0	0.00014

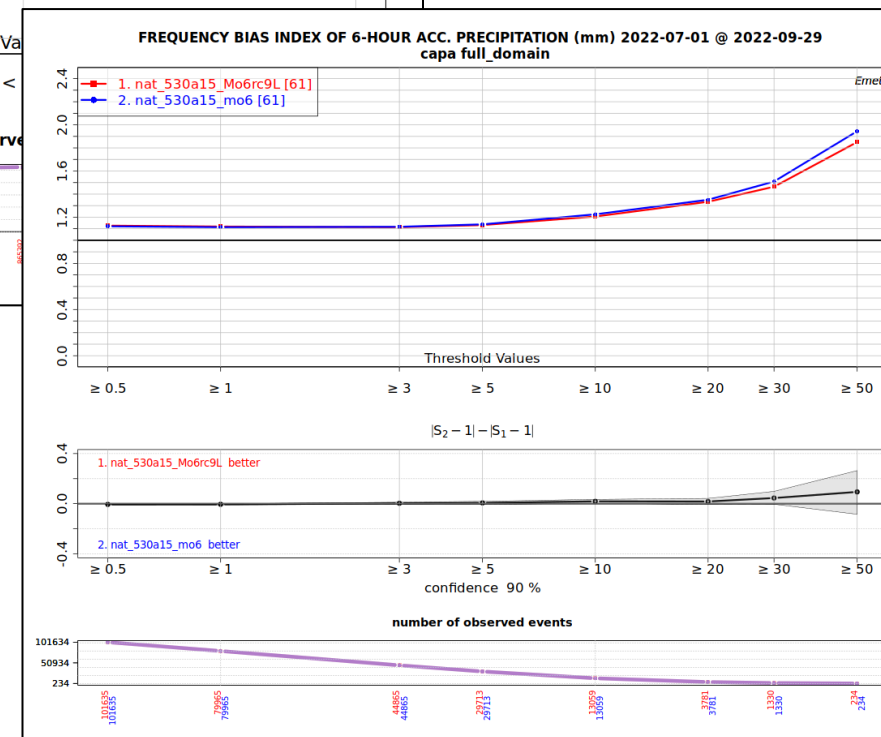
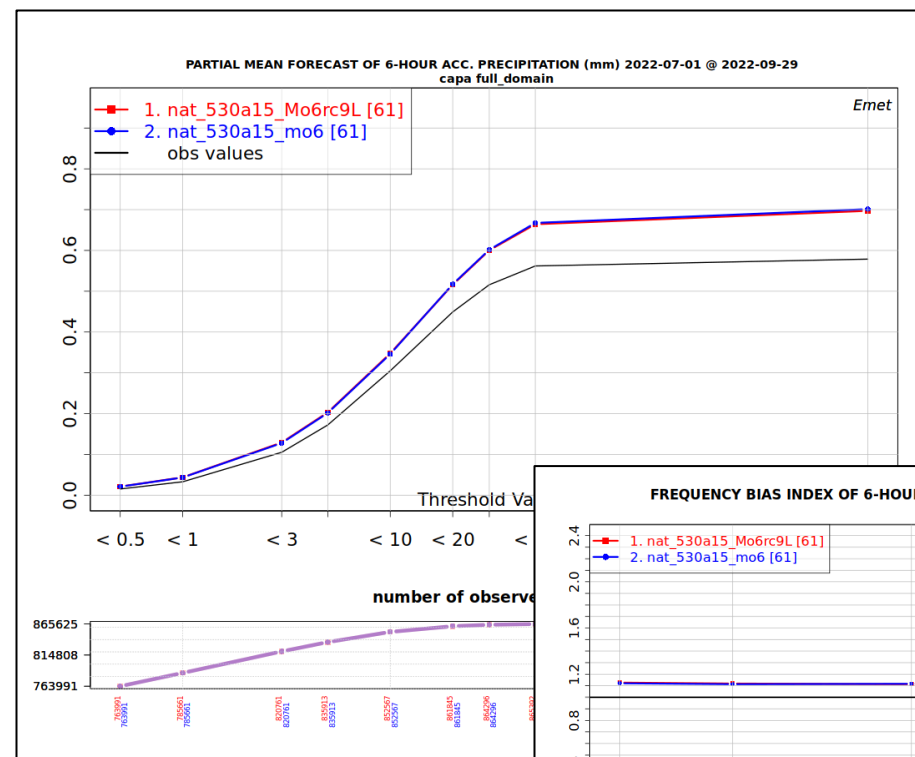
EMET (E22), PR6 & PR24 2MOM-LF*

P3_v5.5.0-rc9
P3 v5.3.6 2MOM-noLF

dpmean				
dpmean				
nat_530a15_Mo6rc9L / nat_530a15_mo6				
		00z	12z	All
Arctic All CLIM	PR24	-0.028	-0.0023	-0.018
	PR6	-0.0058	-0.0038	-0.0052
Boreal CLIM	PR24	0.014	0.014	0.014
	PR6	0.00052	0.0022	0.0023
Canada	PR24	-0.0019	0.0025	0.00092
	PR6	-0.00022	-0.00022	-0.00018
East CLIM	PR24	-0.0026	0.0021	-0.0025
	PR6	0.0	0.0015	0.0009
Great Lakes CLIM	PR24	0.0	0.0	-0.0012
	PR6	-0.0011	0.00051	-0.0018
Mt West CLIM	PR24	-0.002	0.00032	0.0
	PR6	0.0	0.0	0.0
North Plains CLIM	PR24	0.017	0.0	-0.00014
	PR6	0.0029	-0.0027	0.0
Pacific North West CLIM	PR24	-0.0055	-0.00046	-0.0024
	PR6	-0.00035	-0.00088	-0.00099
Prairie CLIM	PR24	0.015	0.0099	0.0
	PR6	-0.00056	0.0	-0.0025
United States of America East	PR24	0.0	0.0054	0.003
	PR6	0.0021	0.0	0.0
United States of America West	PR24	-0.0025	-0.00069	-0.00058
	PR6	-0.0001	0.0	-0.00075
full_domain	PR24	-0.0013	0.0031	0.0028
	PR6	-0.00019	-0.00022	-0.00021

ets				
ets				
nat_530a15_Mo6rc9L / nat_530a15_mo6				
		00z	12z	All
Arctic All CLIM	PR24	0.0	-0.0016	-0.0013
	PR6	0.0061	0.0	0.0043
Boreal CLIM	PR24	0.005	0.009	0.0088
	PR6	0.0012	0.0017	0.004
Canada	PR24	0.002	0.0053	0.0068
	PR6	0.00059	0.005	0.0025
East CLIM	PR24	0.0067	-0.0015	0.0037
	PR6	0.0	0.0	0.0
Great Lakes CLIM	PR24	0.0017	-0.002	-0.0015
	PR6	0.0	0.0	0.0
Mt West CLIM	PR24	0.0008	0.0	0.0
	PR6	-0.0051	0.0015	0.0029
North Plains CLIM	PR24	0.0038	0.0	0.0033
	PR6	0.0	0.01	0.0038
Pacific North West CLIM	PR24	0.01	0.0	-0.00056
	PR6	-0.0065	0.0019	0.0
Prairie CLIM	PR24	-0.0021	0.0	0.0
	PR6	0.0	0.0	0.0
United States of America East	PR24	0.0027	-0.00045	0.0015
	PR6	0.0	0.0	0.0
United States of America West	PR24	0.0	0.0	0.0
	PR6	0.0043	0.0034	0.0038
full_domain	PR24	0.005	0.0012	0.0029
	PR6	0.0	0.00034	0.00027

fbi				
fbi				
nat_530a15_Mo6rc9L / nat_530a15_mo6				
		00z	12z	All
Arctic All CLIM	PR24	0.021	-0.035	-0.035
	PR6	0.032	-0.081	-0.064
Boreal CLIM	PR24	0.016	0.036	0.038
	PR6	0.022	0.023	0.016
Canada	PR24	0.013	0.0095	0.013
	PR6	0.015	0.0081	0.0029
East CLIM	PR24	0.0	0.012	0.0035
	PR6	0.0	0.0	0.0
Great Lakes CLIM	PR24	-0.0067	0.0	-0.0019
	PR6	0.076	0.0	0.043
Mt West CLIM	PR24	0.0	0.0041	0.0
	PR6	-0.0059	0.0	0.0
North Plains CLIM	PR24	0.0	0.0	0.0038
	PR6	0.0035	-0.0034	0.0
Pacific North West CLIM	PR24	0.0043	-0.0022	0.0
	PR6	0.0	-0.0061	-0.004
Prairie CLIM	PR24	-0.026	0.0082	0.0
	PR6	-0.03	-0.0021	-0.061
United States of America East	PR24	0.0	0.008	0.0024
	PR6	0.005	0.0	0.0
United States of America West	PR24	0.055	0.0063	0.031
	PR6	0.014	0.014	0.022
full_domain	PR24	0.0	0.019	0.0063
	PR6	0.012	-0.0018	0.00096



pod				
pod				
nat_530a15_Mo6rc9L / nat_530a15_mo6				
		00z	12z	All
Arctic All CLIM	PR24	0.0046	0.0	0.0017
	PR6	0.014	0.0019	0.01
Boreal CLIM	PR24	-0.0057	-0.0032	-0.0023
	PR6	0.0	0.0026	0.0061
Canada	PR24	0.015	-0.0018	0.007
	PR6	0.0	0.0018	0.0013
East CLIM	PR24	0.014	-6e-05	0.0
	PR6	0.0	0.0	0.0
Great Lakes CLIM	PR24	0.0026	-0.0047	-0.0023
	PR6	-0.0063	0.0	0.0
Mt West CLIM	PR24	0.0016	0.0	0.0
	PR6	-0.013	0.0	-0.0071
North Plains CLIM	PR24	0.0	0.0	0.0
	PR6	0.0	0.022	0.0061
Pacific North West CLIM	PR24	-0.0032	0.0012	-0.00083
	PR6	0.00076	0.0066	0.0045
Prairie CLIM	PR24	0.0	0.0	0.0
	PR6	0.0016	0.0	0.0
United States of America East	PR24	0.0064	-0.0012	-0.0011
	PR6	-0.0029	0.0	0.0
United States of America West	PR24	0.0	0.0	0.0
	PR6	0.0061	0.0061	0.0057
full_domain	PR24	0.0079	-0.00086	0.0017
	PR6	0.0	0.0012	0.00086

far				
far				
nat_530a15_Mo6rc9L / nat_530a15_mo6				
		00z	12z	All
Arctic All CLIM	PR24	0.0	-0.0017	-0.0023
	PR6	0.0	-0.0014	-0.00098
Boreal CLIM	PR24	0.0042	0.016	0.015
	PR6	0.0018	0.0	0.0012
Canada	PR24	0.0077	0.01	0.011
	PR6	0.00078	0.00026	0.0041
East CLIM	PR24	0.0086	-0.00074	0.0051
	PR6	0.0	0.0	0.0
Great Lakes CLIM	PR24	0.0	-0.0028	-0.002
	PR6	0.0029	0.0	-0.001
Mt West CLIM	PR24	-0.0009	0.00071	0.0
	PR6	-0.0071	0.0023	0.0016
North Plains CLIM	PR24	0.01	0.0	0.0066
	PR6	0.0041	0.016	0.0059
Pacific North West CLIM	PR24	0.0037	-0.00054	0.0
	PR6	-0.0026	-0.0028	-0.0011
Prairie CLIM	PR24	-0.0075	0.0027	0.0
	PR6	-0.0011	0.0	-0.001
United States of America East	PR24	0.0035	0.0022	0.0025
	PR6	0.0	0.0	0.0
United States of America West	PR24	0.0	0.0	0.0
	PR6	0.0	0.0054	0.0
full_domain	PR24	0.0068	0.0031	0.0045
	PR6	0.0	0.0024	0.0

Summer (E22)

1.2MOM-noLF

2.2MOM-LF

3.3MOM-LF

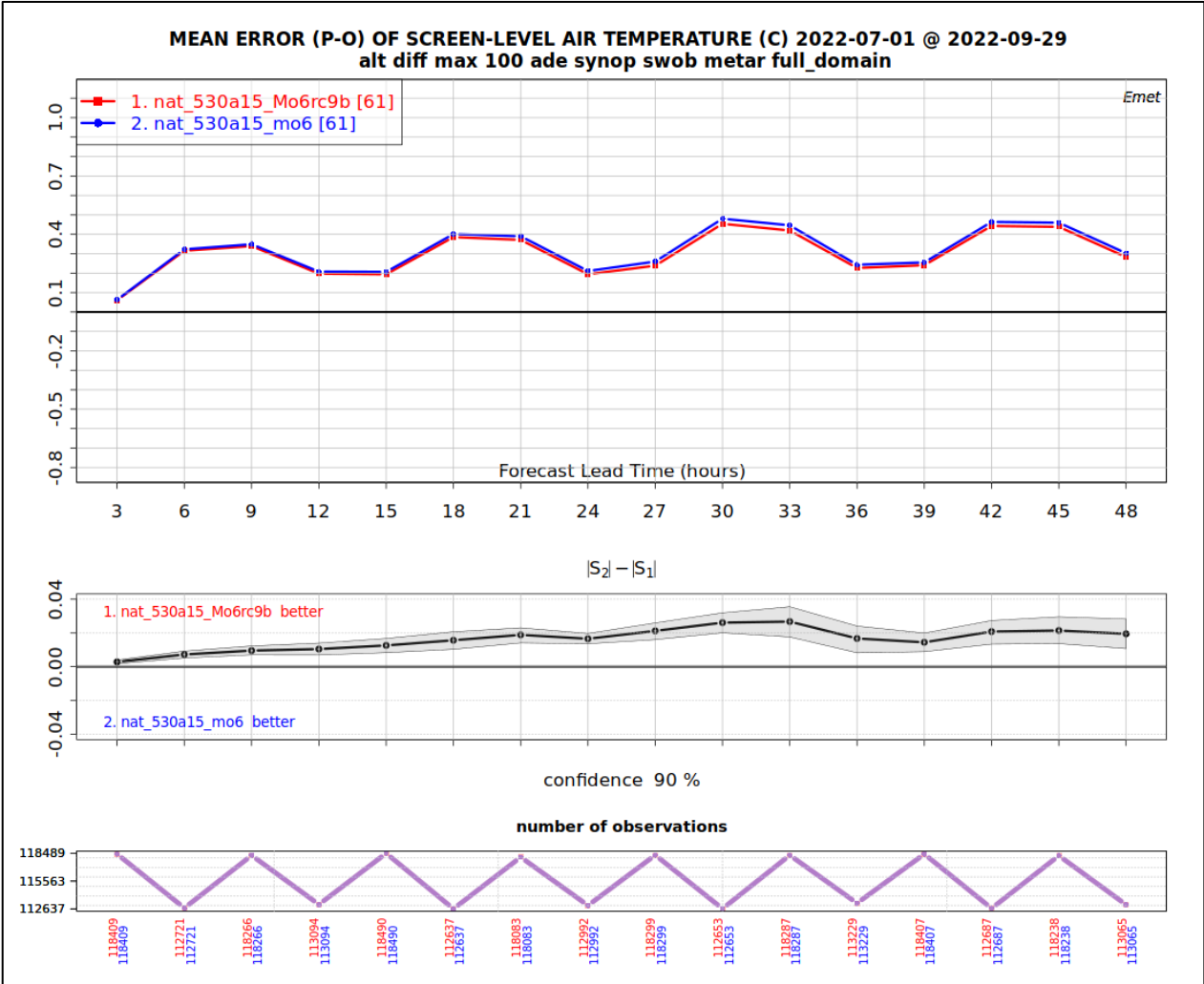
EMET (E22), TT & TD 3MOM-LF*

P3_v5.5.0-rc9
P3 v5.3.6 2MOM-noLF

Positive effects
everywhere

bias		20220701 / 20220929		
nat_530a15_Mo6rc9 / nat_530a15_mo6		00z	12z	All
Arctic All CLIM	TD	0.0037	0.0014	0.00024
	TT	0.0026	-0.0014	0.0
Boreal CLIM	TD	0.0051	0.0032	0.0041
	TT	0.0032	0.0076	0.005
Canada	TD	0.001	0.006	0.0056
	TT	0.0095	0.01	0.011
East CLIM	TD	-0.001	0.002	0.0
	TT	0.0055	0.0088	0.0067
Great Lakes CLIM	TD	0.0045	0.0035	0.0064
	TT	0.011	0.012	0.014
Mt West CLIM	TD	0.0072	0.014	0.011
	TT	0.0033	0.0073	0.0051
North Plains CLIM	TD	0.0041	0.013	0.0076
	TT	0.014	0.016	0.017
Pacific North West CLIM	TD	0.0	0.0	0.0
	TT	-0.00065	-0.0014	-0.0012
Prairie CLIM	TD	0.0	0.0023	0.0
	TT	0.015	0.02	0.015
United States of America East	TD	0.00024	0.0029	0.0017
	TT	0.01	0.017	0.014
United States of America West	TD	0.0051	0.014	0.0092
	TT	0.0069	0.0039	0.0053
full_domain	TD	0.0045	0.0074	0.0055
	TT	0.011	0.014	0.013

TT-bias (full domain)



EMET (E22), P0 & UV 3MOM-LF*

bias

bias nat_530a15_Mo6rc9 / nat_530a15_mo6		20220701 / 20220929		
		00z	12z	All
Arctic All CLIM	P0	0.011	0.0073	0.01
Boreal CLIM	P0	0.00072	0.0021	0.00096
Canada	P0	-0.0034	-0.0014	-0.003
East CLIM	P0	0.0041	-0.00016	0.00051
Great Lakes CLIM	P0	0.0068	0.0068	0.01
Mt West CLIM	P0	-0.0057	0.0026	0.0003
North Plains CLIM	P0	0.0081	-7.1e-06	0.0045
Pacific North West CLIM	P0	0.00052	0.00045	0.00069
Prairie CLIM	P0	0.011	0.011	0.016
United States of America East	P0	0.015	0.023	0.021
United States of America West	P0	0.021	0.013	0.017
full_domain	P0	-0.005	-0.004	-0.0057

rmse

rmse nat_530a15_Mo6rc9 / nat_530a15_mo6		20220701 / 20220929		
		00z	12z	All
Arctic All CLIM	P0	0.012	0.0076	0.011
Boreal CLIM	P0	0.0064	0.0048	0.0063
Canada	P0	0.0094	0.0065	0.0082
East CLIM	P0	0.0076	0.0079	0.011
Great Lakes CLIM	P0	0.018	0.013	0.012
Mt West CLIM	P0	0.0069	0.0057	0.0086
North Plains CLIM	P0	0.0067	0.0076	0.0087
Pacific North West CLIM	P0	7.7e-06	-0.0012	-0.00068
Prairie CLIM	P0	0.0044	0.008	0.011
United States of America East	P0	0.011	0.012	0.014
United States of America West	P0	0.013	0.0087	0.014
full_domain	P0	0.0098	0.0075	0.0091

bias

bias nat_530a15_Mo6rc9 / nat_530a15_mo6		20220701 / 20220929		
		00z	12z	All
Arctic All CLIM	UV	0.0	0.00064	0.002
Boreal CLIM	UV	0.0	-0.00011	0.0013
Canada	UV	0.001	6.4e-05	-0.00012
East CLIM	UV	0.0048	0.0067	0.008
Great Lakes CLIM	UV	0.0047	0.015	0.013
Mt West CLIM	UV	-0.00045	0.00084	0.006
North Plains CLIM	UV	-0.0031	0.00035	-0.00095
Pacific North West CLIM	UV	0.0	0.0	-0.00065
Prairie CLIM	UV	0.012	0.0	0.0046
United States of America East	UV	0.01	0.0094	0.012
United States of America West	UV	-0.004	-0.0022	0.0022
full_domain	UV	0.0044	0.0053	0.0049

rmse

rmse nat_530a15_Mo6rc9 / nat_530a15_mo6		20220701 / 20220929		
		00z	12z	All
Arctic All CLIM	UV	0.0	0.0019	0.0
Boreal CLIM	UV	0.0	0.0	0.0
Canada	UV	-0.0031	0.00059	-0.0011
East CLIM	UV	0.00031	0.0011	0.0018
Great Lakes CLIM	UV	0.00045	0.0048	0.0049
Mt West CLIM	UV	-0.001	0.0031	0.00078
North Plains CLIM	UV	-0.0067	0.001	0.0018
Pacific North West CLIM	UV	-0.0011	0.0	0.0
Prairie CLIM	UV	0.004	0.0034	0.0022
United States of America East	UV	0.0013	0.0037	0.0027
United States of America West	UV	-0.0019	0.0	-0.0015
full_domain	UV	-0.0011	0.0011	0.00086

EMET (E22), PR6 & PR24 3MOM-LF*

P3_v5.5.0-rc9
P3 v5.3.6 2MOM-noLF

dpmean

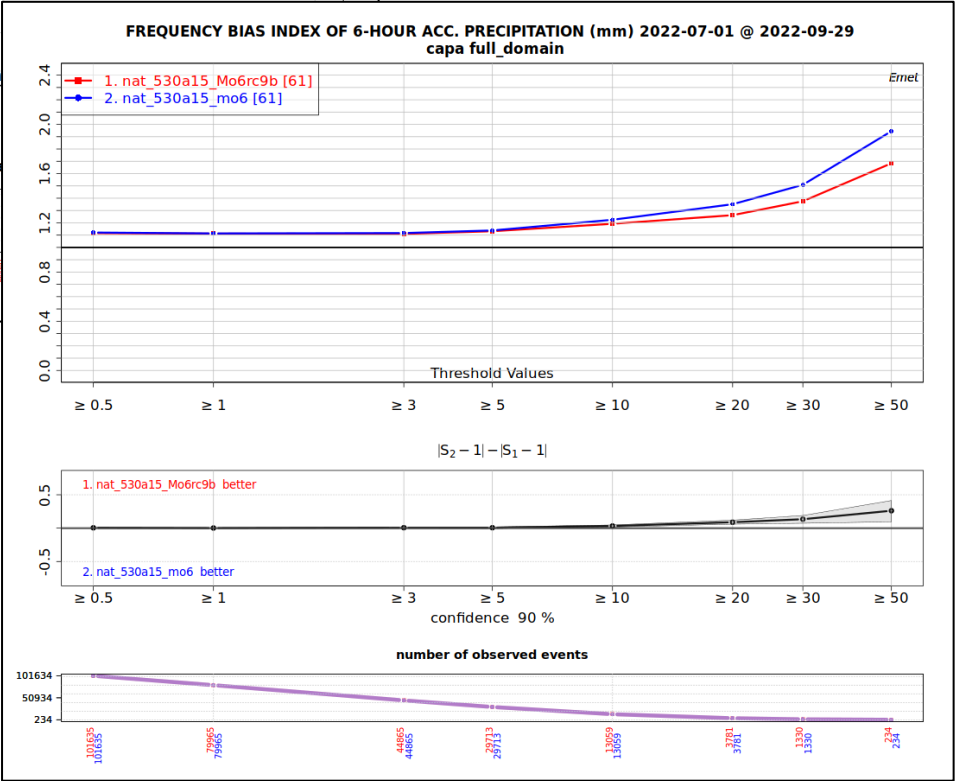
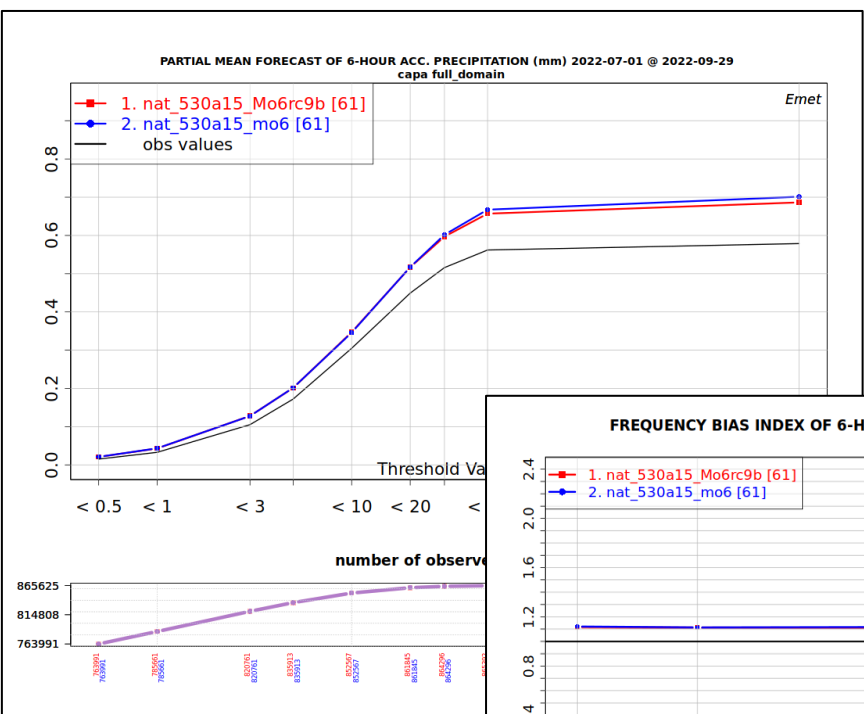
dpmean	nat_530a15_Mo6rc9 / nat_530a15_mo6	20220701 / 20220929	00z	12z	All
Arctic All CLIM	PR24	-0.0071	-0.0043	-0.0099	
	PR6	-0.00078	-0.002	-0.0022	
Boreal CLIM	PR24	0.01	0.017	0.019	
	PR6	0.00058	0.0081	0.0044	
Canada	PR24	-0.00031	0.011	0.004	
	PR6	0.0027	0.0041	0.0037	
East CLIM	PR24	0.0059	0.0088	0.0058	
	PR6	0.006	0.0047	0.005	
Great Lakes CLIM	PR24	0.034	0.0	0.0	
	PR6	-0.0028	0.0	-0.0032	
Mt West CLIM	PR24	0.011	0.019	0.017	
	PR6	0.0032	0.011	0.006	
North Plains CLIM	PR24	0.0	0.0074	0.012	
	PR6	0.0038	0.0	0.0075	
Pacific North West CLIM	PR24	-0.0026	0.0	0.0	
	PR6	-0.00042	0.0	-0.00021	
Prairie CLIM	PR24	-0.001	0.016	0.011	
	PR6	-0.00083	0.0	0.0	
United States of America East	PR24	0.02	0.013	0.018	
	PR6	0.005	0.0	0.0028	
United States of America West	PR24	0.011	0.016	0.015	
	PR6	0.002	0.0017	0.004	
full_domain	PR24	0.011	0.013	0.013	
	PR6	0.0033	0.0031	0.0034	

ets

ets	nat_530a15_Mo6rc9 / nat_530a15_mo6	20220701 / 20220929	00z	12z	All
Arctic All CLIM	PR24	0.0	-0.0025	-0.0014	
	PR6	0.0	0.0	0.0	
Boreal CLIM	PR24	0.0025	-0.00095	0.00089	
	PR6	0.006	0.0014	0.0055	
Canada	PR24	0.0099	0.0034	0.0075	
	PR6	0.001	0.007	0.0064	
East CLIM	PR24	0.0	0.003	0.0023	
	PR6	0.0018	0.0029	0.0035	
Great Lakes CLIM	PR24	0.0	-0.0014	-0.0034	
	PR6	0.0	0.0013	-0.0017	
Mt West CLIM	PR24	-0.0012	0.0026	0.0024	
	PR6	-0.005	0.0063	0.0023	
North Plains CLIM	PR24	0.021	0.012	0.019	
	PR6	0.0021	0.0024	0.0047	
Pacific North West CLIM	PR24	-0.016	-0.0021	-0.0073	
	PR6	-0.011	0.0	0.0014	
Prairie CLIM	PR24	0.0023	0.0071	0.0078	
	PR6	0.0013	0.0034	0.0041	
United States of America East	PR24	0.0	0.002	0.0015	
	PR6	-0.0014	0.0026	0.0	
United States of America West	PR24	0.0086	0.0025	0.004	
	PR6	0.007	0.0049	0.0051	
full_domain	PR24	0.0086	0.0025	0.004	
	PR6	0.007	0.0049	0.0051	

fbi

fbi	nat_530a15_Mo6rc9 / nat_530a15_mo6	20220701 / 20220929	00z	12z	All
Arctic All CLIM	PR24	-0.0092	0.0	0.018	
	PR6	0.037	-0.0038	0.0085	
Boreal CLIM	PR24	0.021	0.053	0.047	
	PR6	0.018	0.056	0.063	
Canada	PR24	0.0074	0.03	0.024	
	PR6	0.071	0.054	0.097	
East CLIM	PR24	0.034	0.013	0.03	
	PR6	0.11	0.0064	0.064	
Great Lakes CLIM	PR24	0.024	-0.019	0.0	
	PR6	0.017	-0.0056	-0.0091	
Mt West CLIM	PR24	0.017	0.043	0.038	
	PR6	0.027	0.033	0.0062	
North Plains CLIM	PR24	0.0	0.015	0.059	
	PR6	0.015	0.0	0.013	
Pacific North West CLIM	PR24	-0.00015	0.013	0.0036	
	PR6	0.0014	0.0	0.066	
Prairie CLIM	PR24	0.0	0.022	0.044	
	PR6	-0.0036	0.0	0.0	
United States of America East	PR24	0.029	0.0098	0.042	
	PR6	0.039	0.0	0.022	
United States of America West	PR24	0.027	0.049	0.036	
	PR6	0.016	0.037	0.042	
full_domain	PR24	0.029	0.048	0.048	
	PR6	0.072	0.028	0.065	



pod

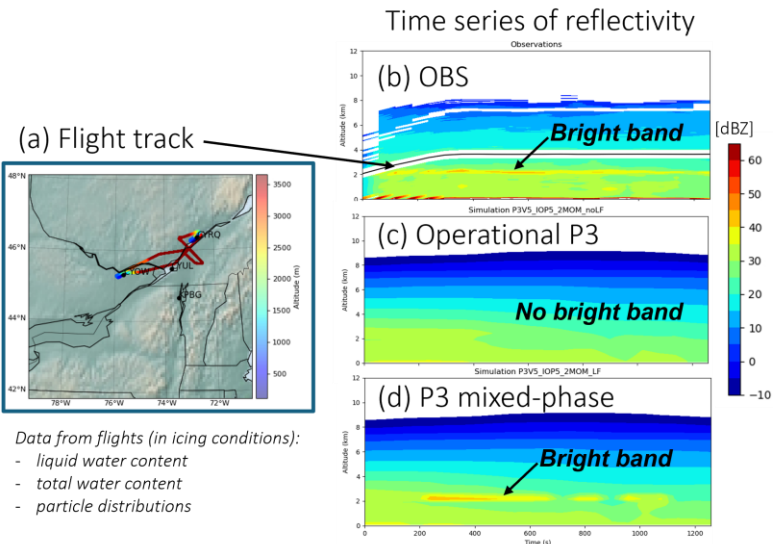
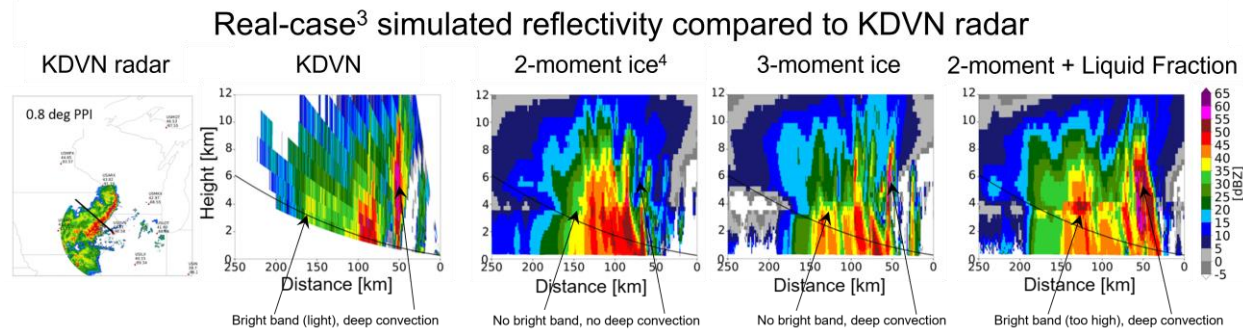
pod	nat_530a15_Mo6rc9 / nat_530a15_mo6	20220701 / 20220929	00z	12z	All
Arctic All CLIM	PR24	-0.0019	0.0	-0.001	
	PR6	0.0	0.0018	0.006	
Boreal CLIM	PR24	0.0022	-0.00095	0.0	
	PR6	0.0079	0.0	0.0054	
Canada	PR24	0.0086	0.0	0.0	
	PR6	0.0015	0.0074	0.00058	
East CLIM	PR24	0.0022	0.0	0.0	
	PR6	0.0	0.0045	0.0026	
Great Lakes CLIM	PR24	0.0	0.0	-0.0055	
	PR6	0.0028	0.0034	0.0013	
Mt West CLIM	PR24	-0.0037	0.0	-0.0017	
	PR6	-0.016	0.0065	-0.0067	
North Plains CLIM	PR24	0.024	0.0078	0.015	
	PR6	0.0032	0.0042	0.0061	
Pacific North West CLIM	PR24	-0.04	-0.0033	-0.017	
	PR6	-0.013	0.0	0.0	
Prairie CLIM	PR24	0.0	-0.0013	0.0012	
	PR6	0.0037	0.0053	0.0028	
United States of America East	PR24	0.0	0.0	0.00052	
	PR6	-0.0039	0.0015	0.0	
United States of America West	PR24	0.0019	0.0	0.0	
	PR6	0.0	0.0085	0.0	
full_domain	PR24	0.0	0.0	0.0	
	PR6	0.0	0.0057	0.0013	

far

far	nat_530a15_Mo6rc9 / nat_530a15_mo6	20220701 / 20220929	00z	12z	All
Arctic All CLIM	PR24	0.0	-0.0024	0.0	
	PR6	0.0	0.0	0.0	
Boreal CLIM	PR24	0.0042	0.017	0.0064	
	PR6	0.0048	0.0013	0.0055	
Canada	PR24	0.013	0.019	0.013	
	PR6	0.0063	0.024	0.016	
East CLIM	PR24	0.0018	0.0065	0.0078	
	PR6	0.004	0.0039	0.0062	
Great Lakes CLIM	PR24	0.0	-0.0026	-0.0038	
	PR6	0.0	0.0	0.0	
Mt West CLIM	PR24	0.0039	0.01	0.0058	
	PR6	-0.0071	0.012	0.0061	
North Plains CLIM	PR24	0.031	0.019	0.029	
	PR6	0.0047	0.007	0.013	
Pacific North West CLIM	PR24	-0.0057	0.005	-0.0031	
	PR6	-0.012	0.0	0.0049	
Prairie CLIM	PR24	0.005	0.01	0.009	
	PR6	0.0	0.0032	0.0033	
United States of America East	PR24	0.0	0.0056	0.0027	
	PR6	-0.0014	0.0038	0.0	
United States of America West	PR24	0.011	0.0075	0.011	
	PR6	0.012	0.0099	0.012	
full_domain	PR24	0.0065	0.0095	0.0073	
	PR6	0.0029	0.011	0.0085	

Other advantages of 3MOM_LF

Better representation of
reflectivity



An improved representation of hail
simulations and properties

The \$1B Calgary hailstorm (June 2020)

Operational
P3 (HRDPS)

P3
3MOM_LF

